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(12) **United States Patent**
Bober et al.

(10) **Patent No.:** **US 12,407,186 B2**

(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **AUXILIARY TABLE DEVICE AND METHOD THEREOF**

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(73) Assignee: **WBTEC, LLC**, Sarasota, FL (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/733,090**

(22) Filed: **Jun. 4, 2024**

(65) **Prior Publication Data**

US 2024/0333032 A1 Oct. 3, 2024

Related U.S. Application Data

(63) Continuation-in-part of application No. 18/342,278, filed on Jun. 27, 2023, now Pat. No. 12,003,122, (Continued)

(30) **Foreign Application Priority Data**

Jul. 16, 2021 (PL) 438515

(51) **Int. Cl.**

H02J 50/00 (2016.01)

A47B 5/04 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **H02J 50/005** (2020.01); **A47B 5/04** (2013.01); **A47B 13/081** (2013.01); **H02J 50/10** (2016.02);

(Continued)

(58) **Field of Classification Search**

CPC A47B 5/02; A47B 5/04; A47B 2200/008

(Continued)

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Primary Examiner — Jose V Chen

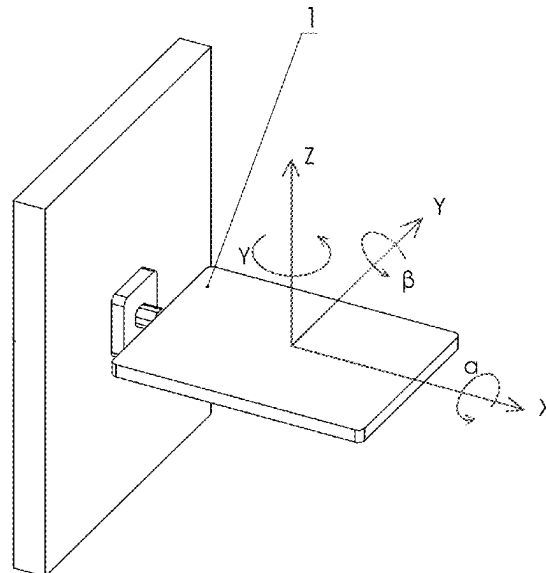
(74) Attorney, Agent, or Firm — The Farrell Law Firm, P.C.

(57)

ABSTRACT

An auxiliary table device is provided comprising a horizontal surface comprising a usable surface; a first surface vertically positioned below the horizontal surface and configured to engage with a first side of a portion of furniture; and a second surface positioned below the horizontal surface and configured to adjustably engage with a second side of the portion of furniture while in a vertical position, wherein at least one edge of the horizontal surface extends beyond an edge of the first or second side of the portion of furniture when the first surface and the second surface are engaged with their respective sides of the furniture.

20 Claims, 66 Drawing Sheets



Related U.S. Application Data

which is a continuation-in-part of application No. 17/850,445, filed on Jun. 27, 2022, now Pat. No. 11,689,068, which is a continuation-in-part of application No. 16/681,487, filed on Nov. 12, 2019, now Pat. No. 11,374,417.

- (60) Provisional application No. 62/758,239, filed on Nov. 9, 2018.

(51) **Int. Cl.**

A47B 13/08 (2006.01)

H02J 50/10 (2016.01)

(52) **U.S. Cl.**

CPC . *A47B 2200/0035* (2013.01); *A47B 2200/008* (2013.01); *A47B 2220/0077* (2013.01)

(58) **Field of Classification Search**

USPC 108/152, 49, 42, 47; 297/153, 145
See application file for complete search history.

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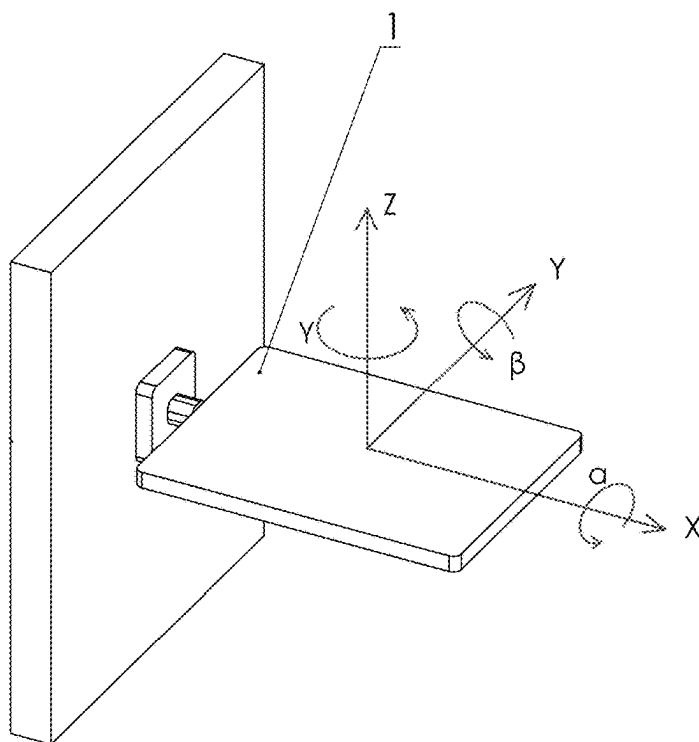


FIG. 1

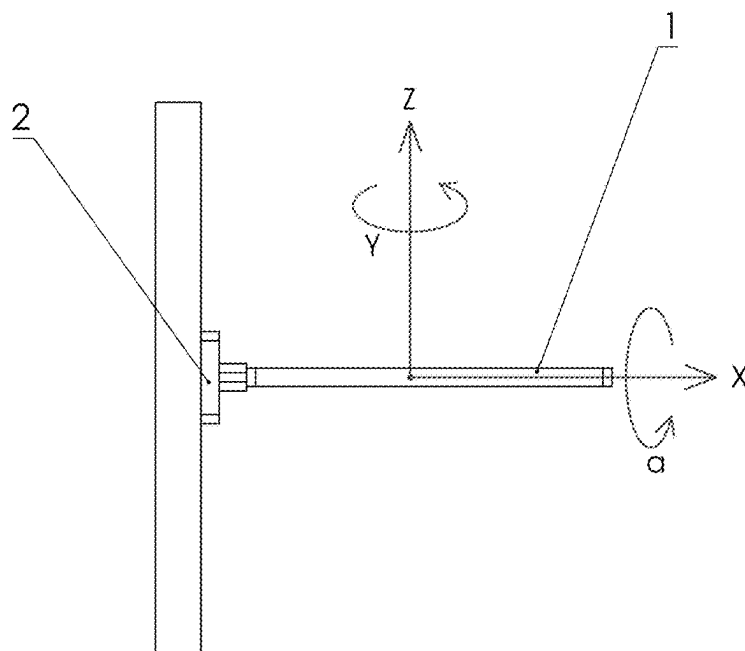


FIG. 2

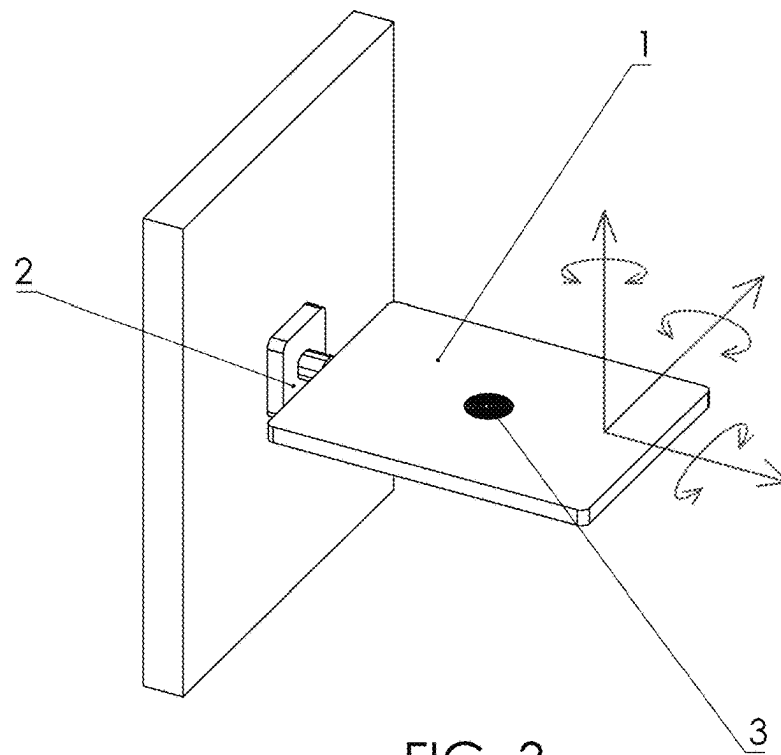


FIG. 3

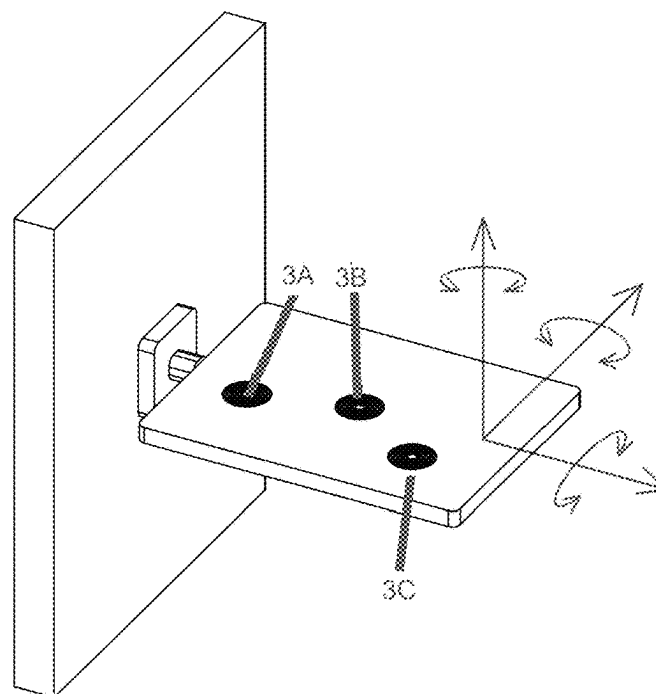


FIG. 4

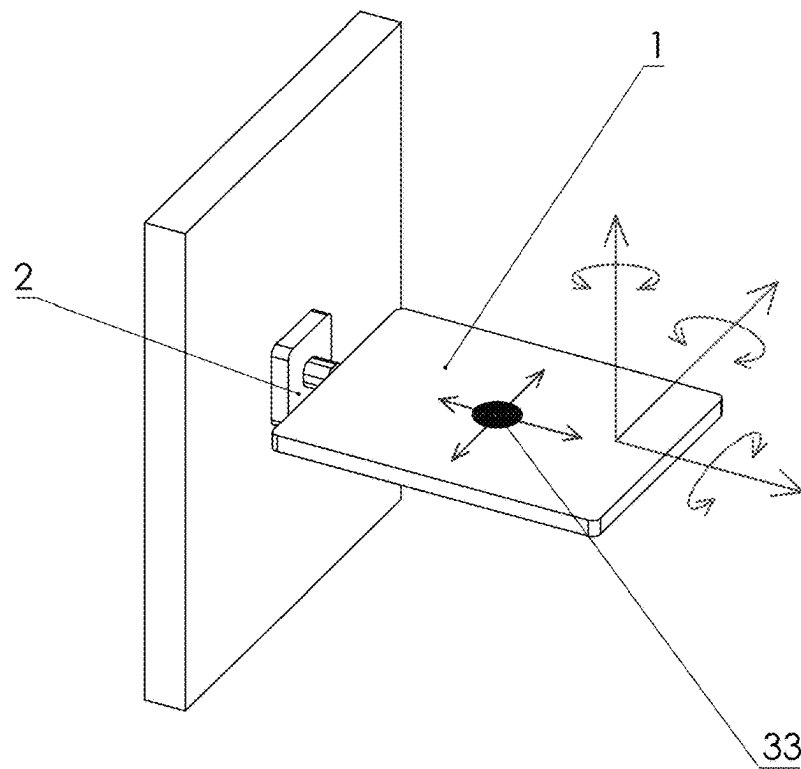


FIG. 5

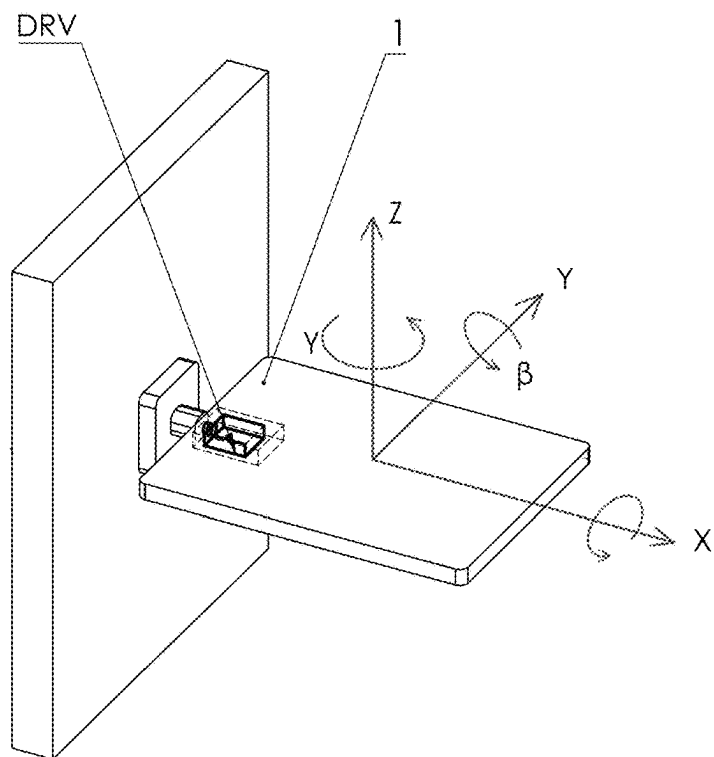


FIG. 6

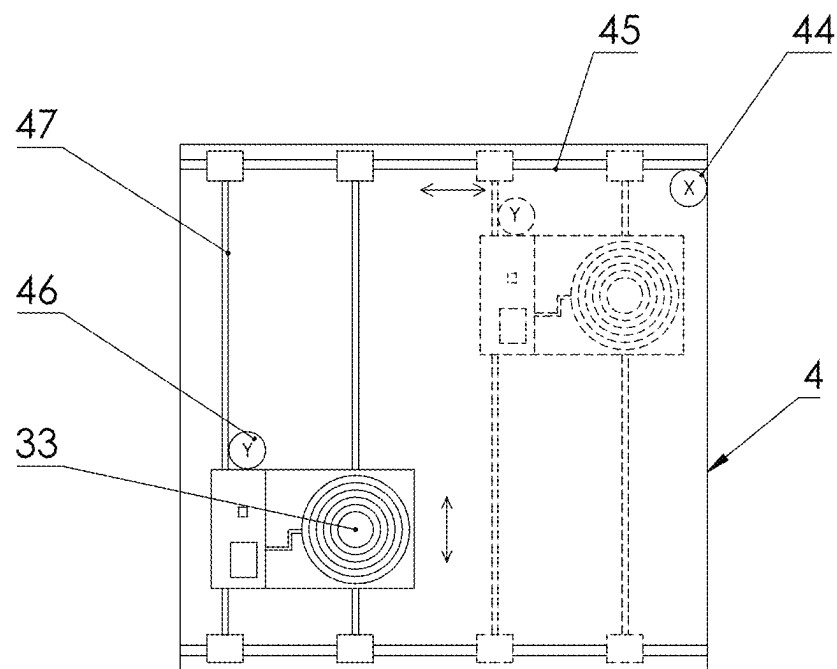


FIG. 7

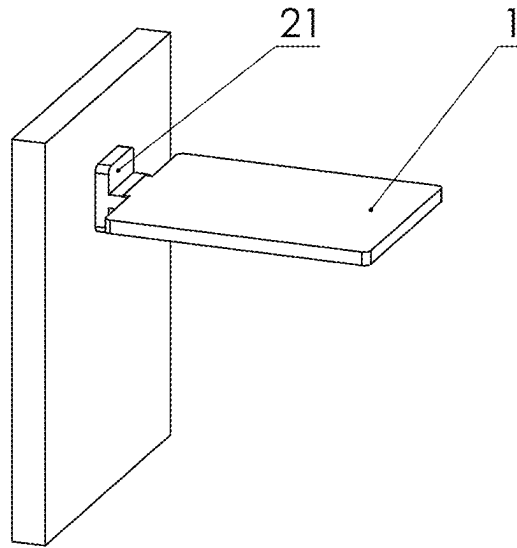


FIG. 8

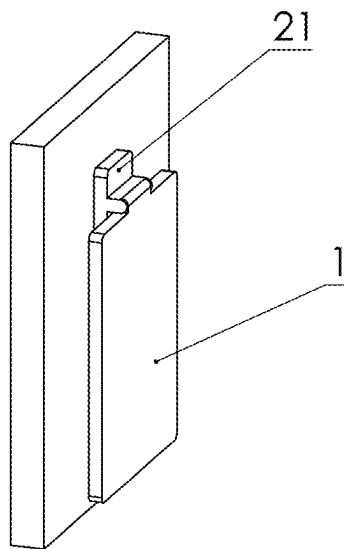


FIG. 9

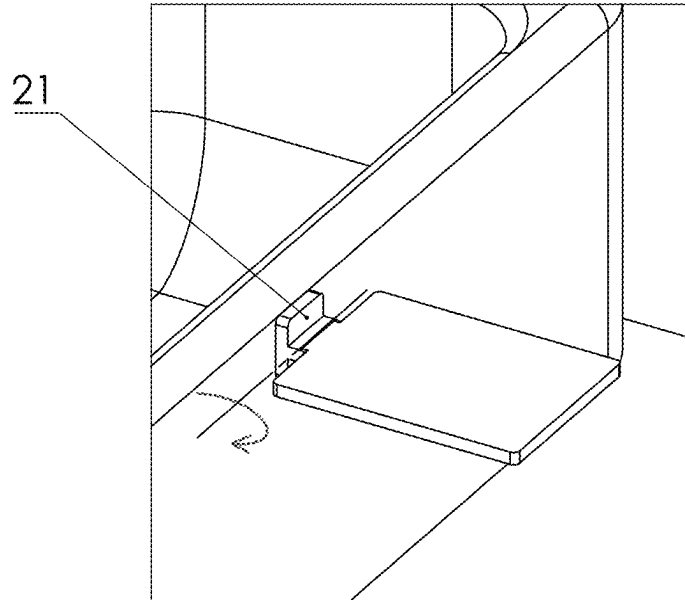


FIG. 10

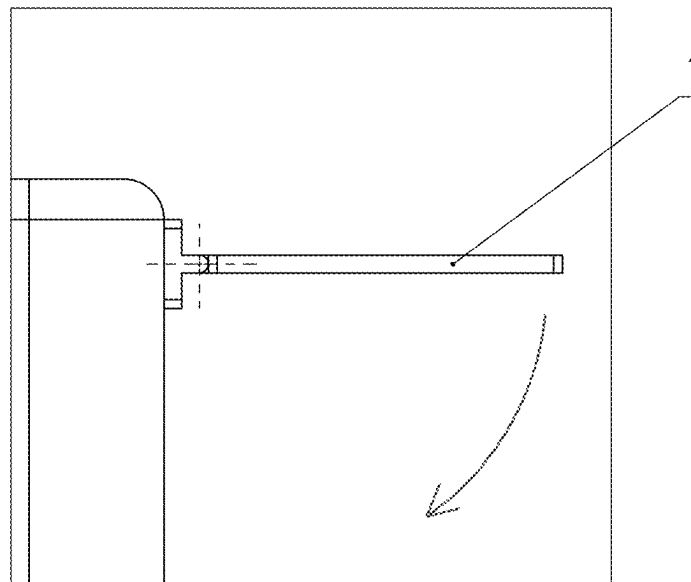


FIG. 11

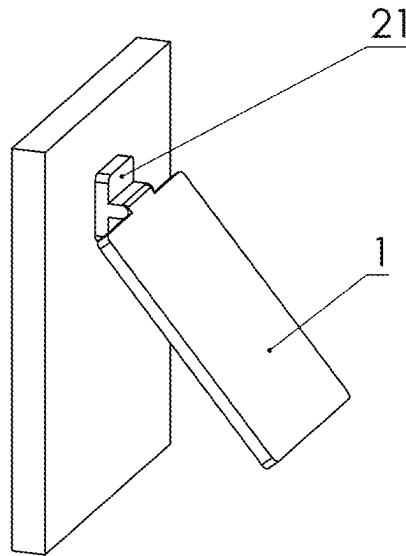


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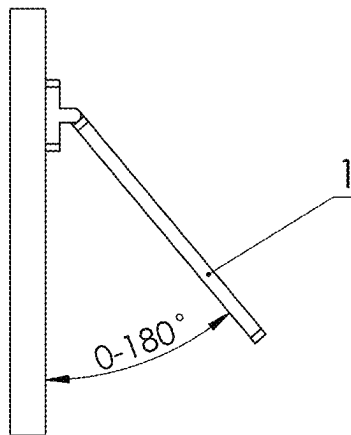


FIG. 13

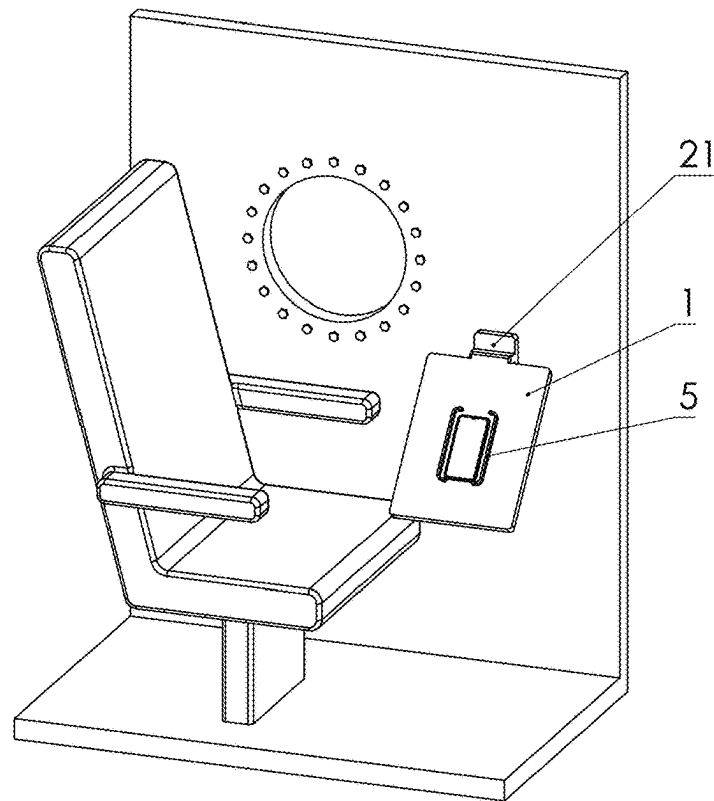


FIG. 14

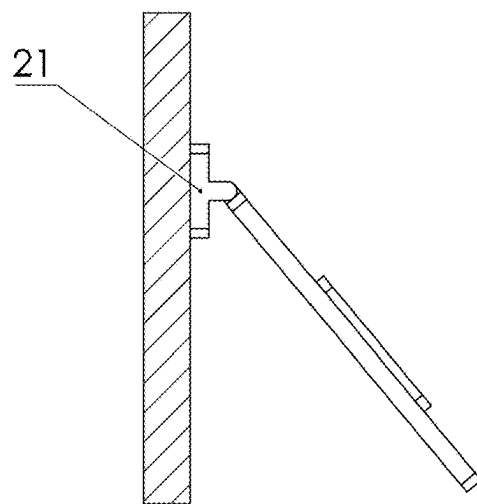


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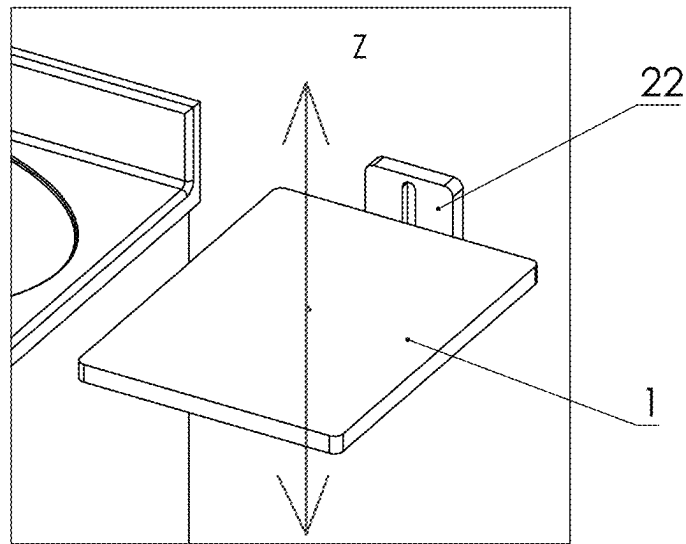


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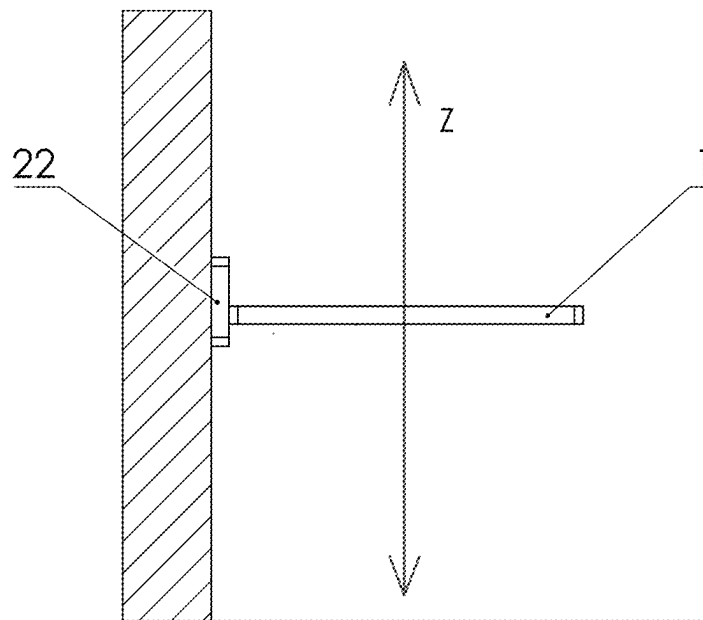


FIG. 17

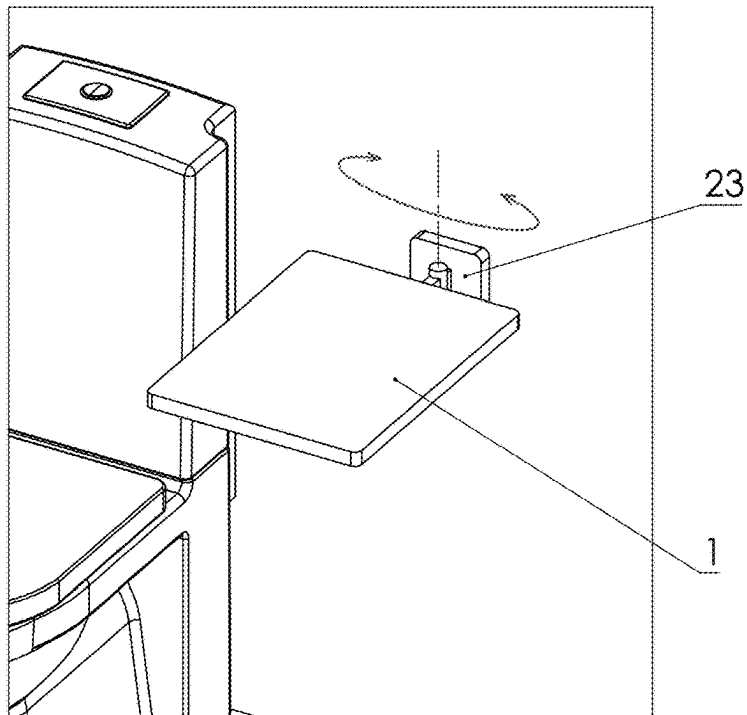


FIG. 18

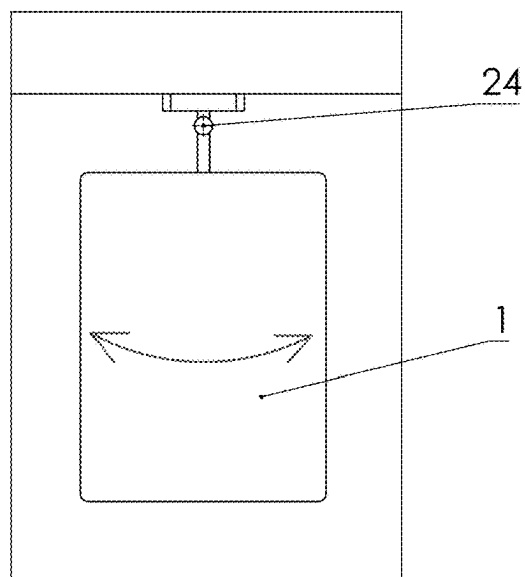


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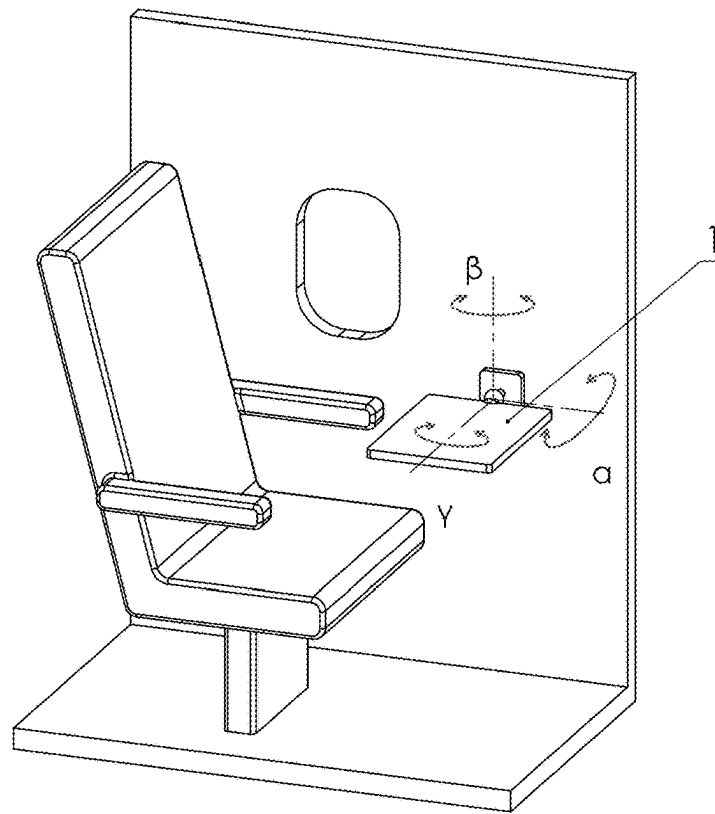


FIG. 20

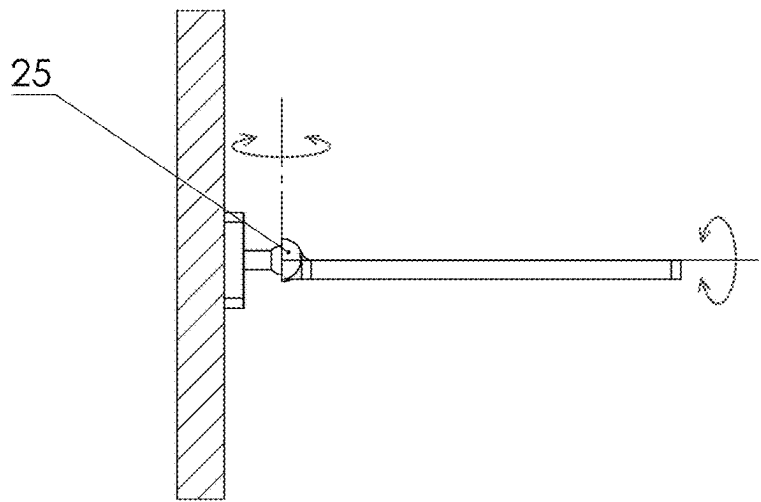


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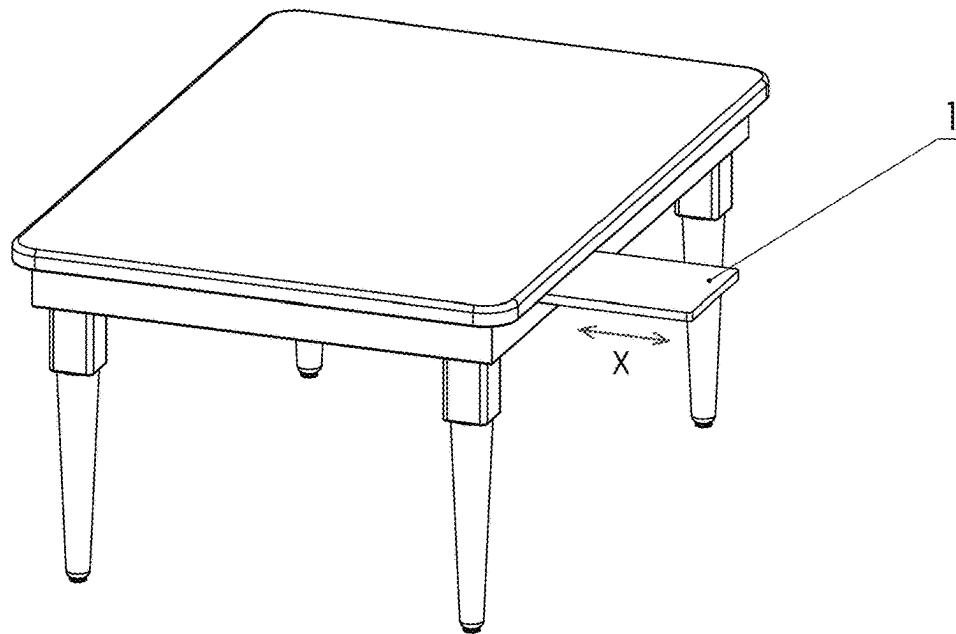


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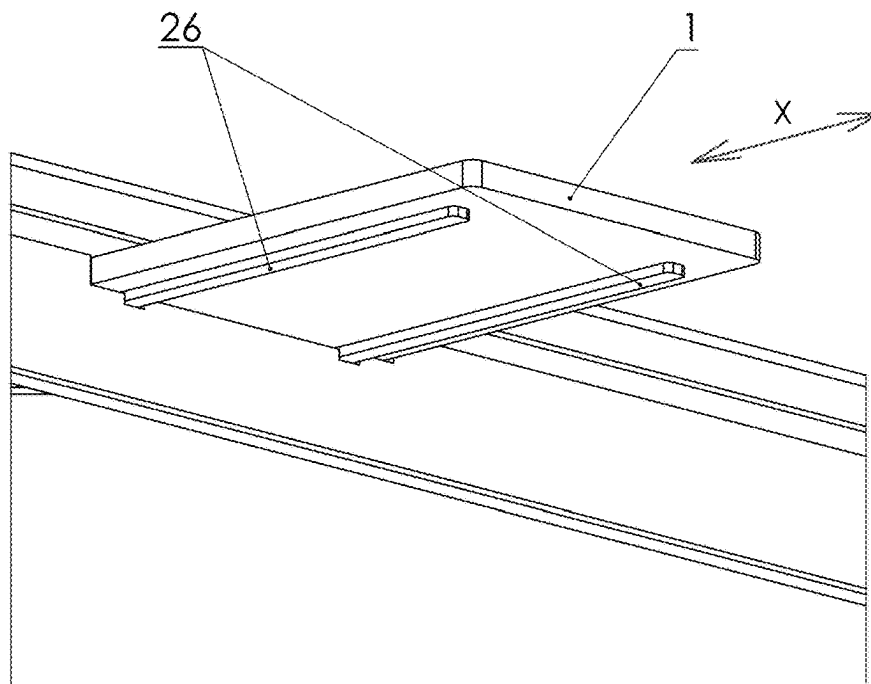


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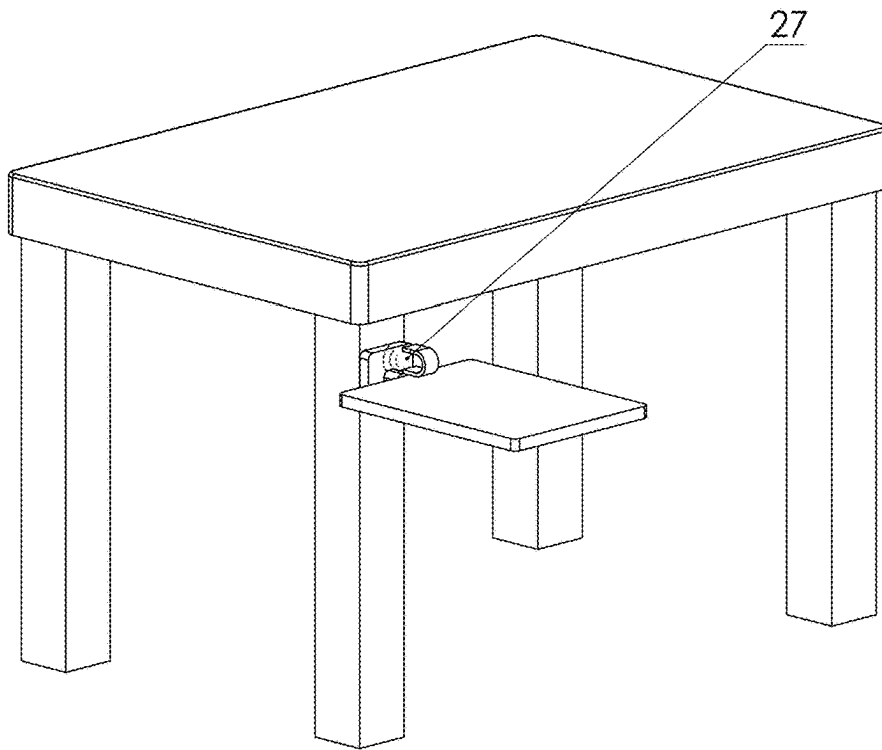


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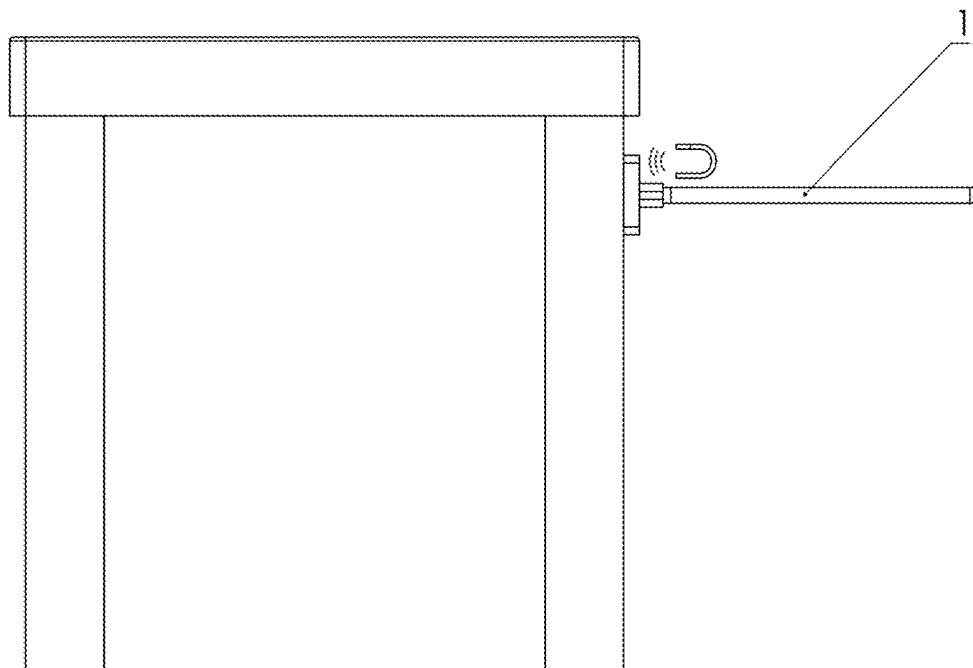


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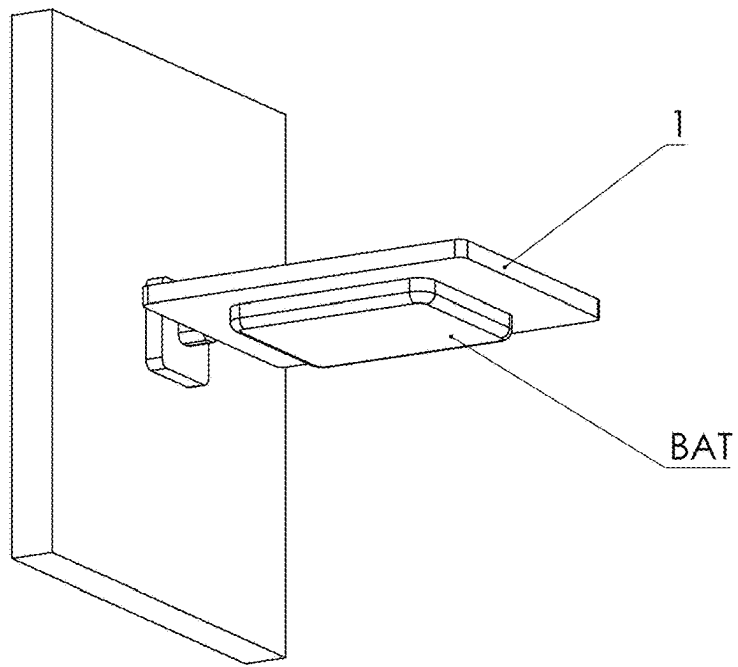


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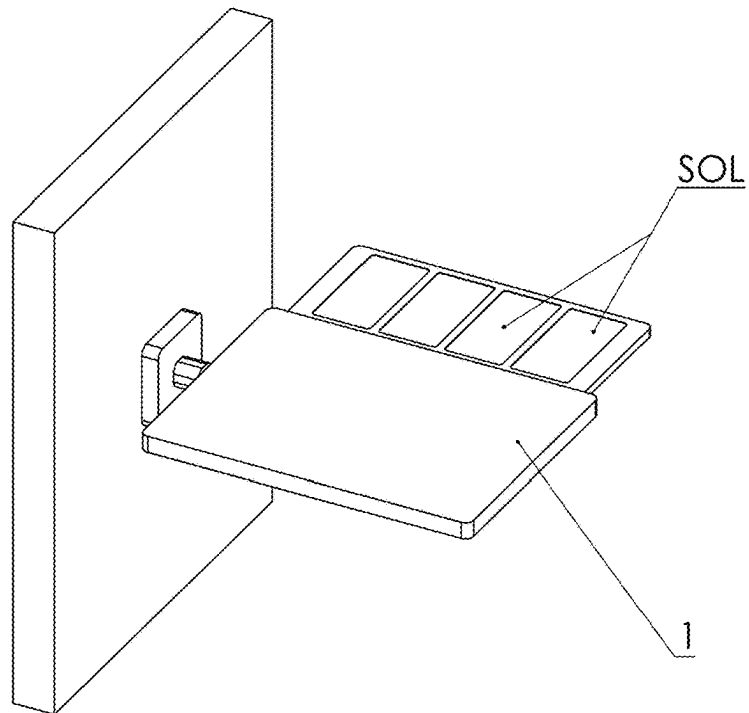


FIG. 27

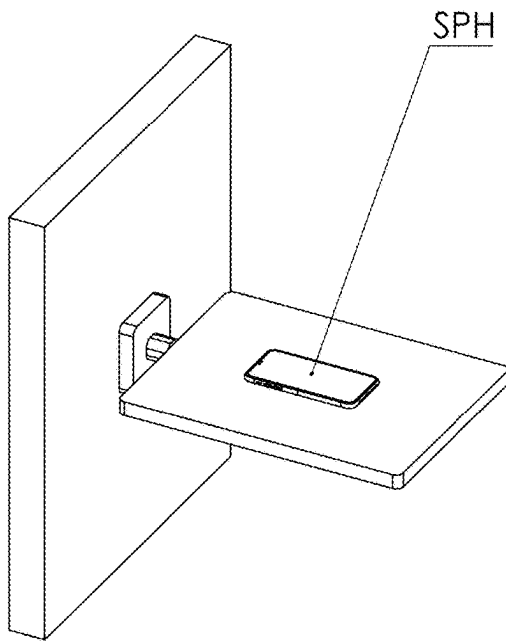


FIG. 28

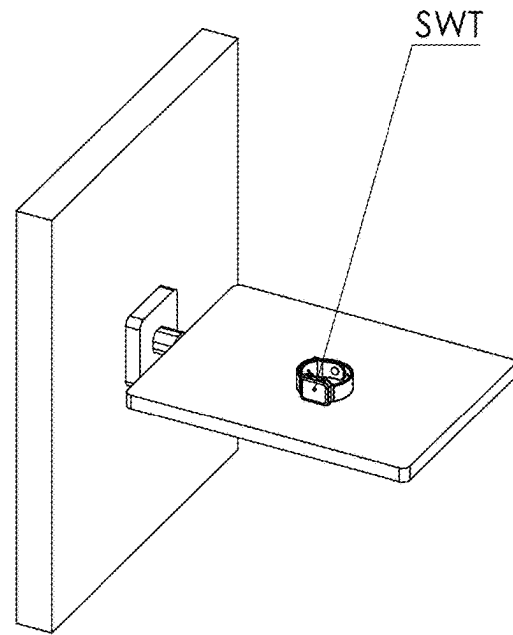


FIG. 29

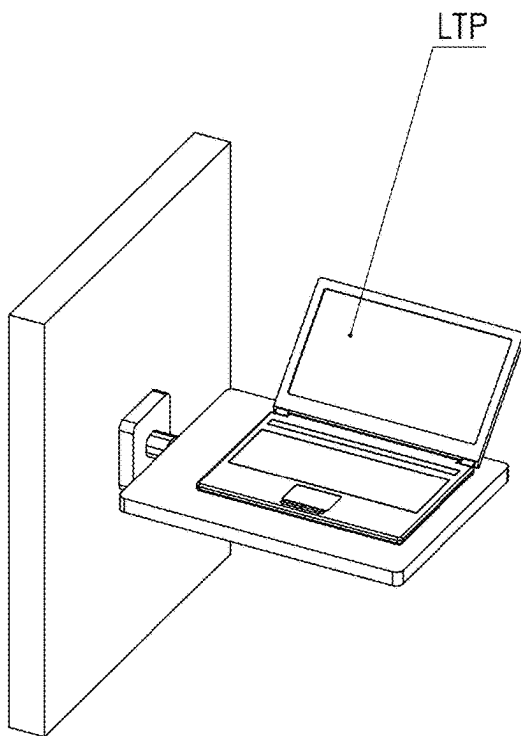


FIG. 30

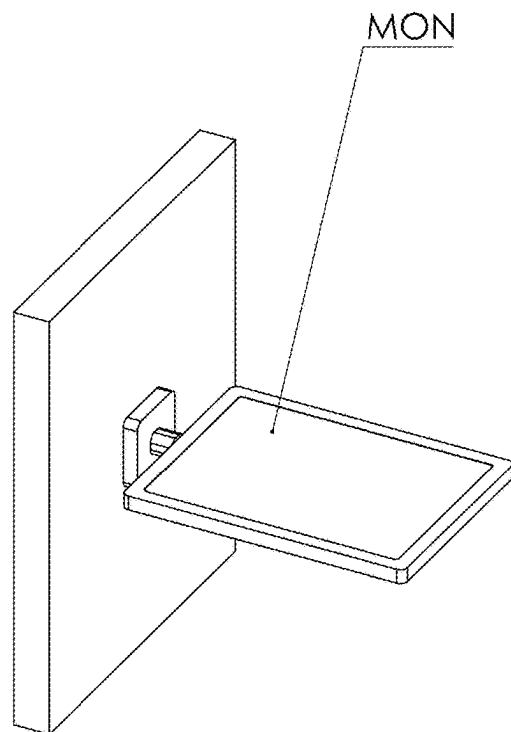


FIG. 31

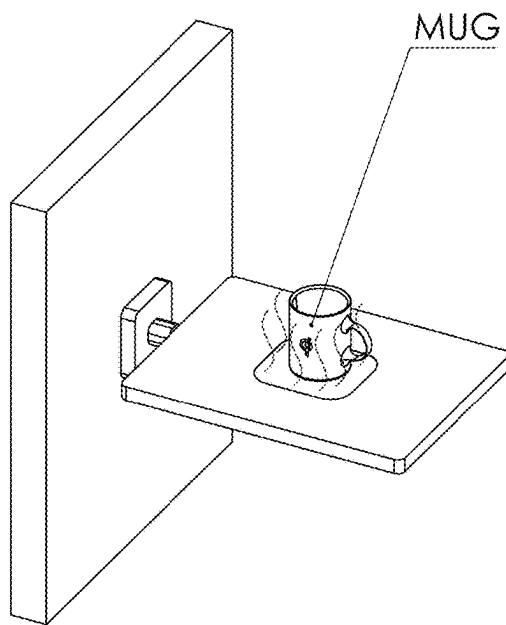


FIG. 32

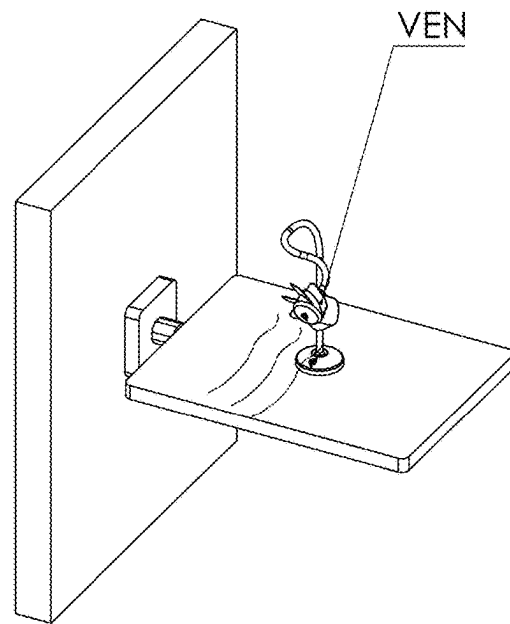


FIG. 33

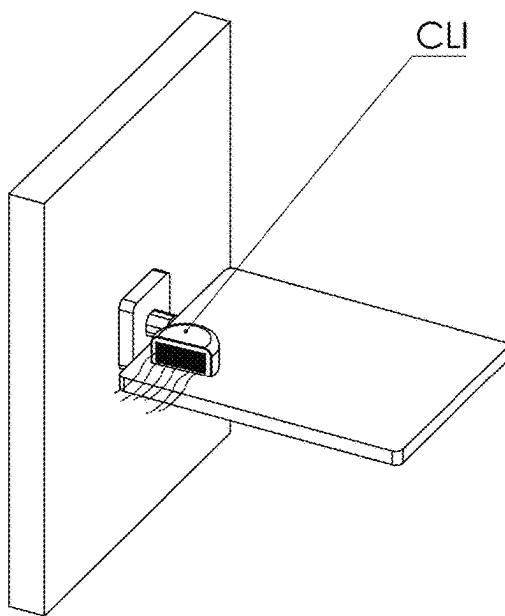


FIG. 34

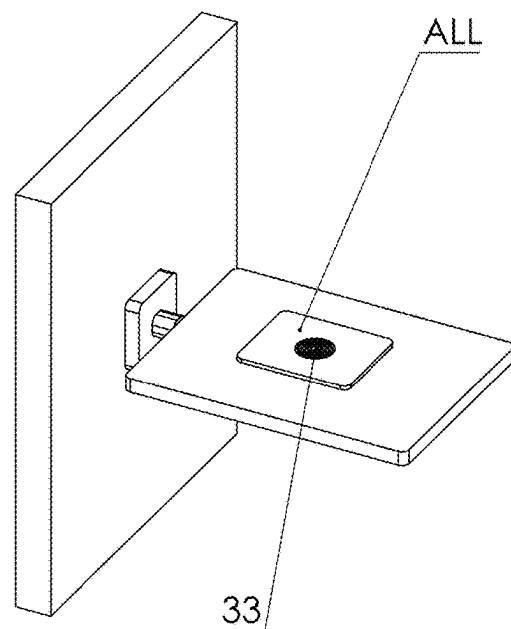


FIG. 35

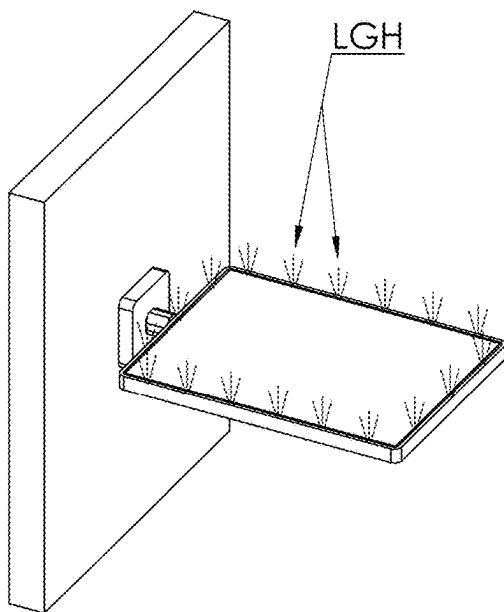


FIG. 36

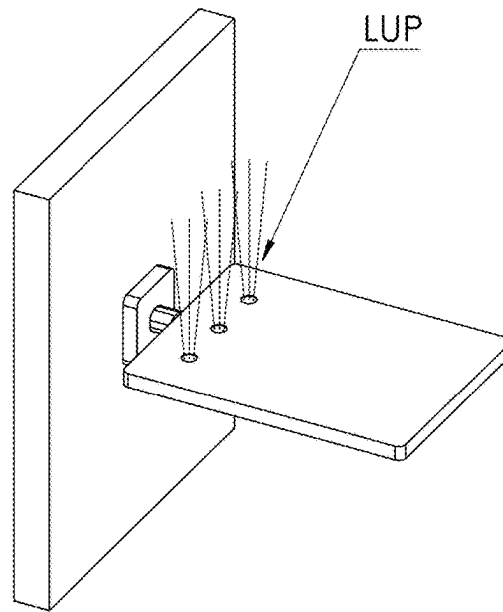


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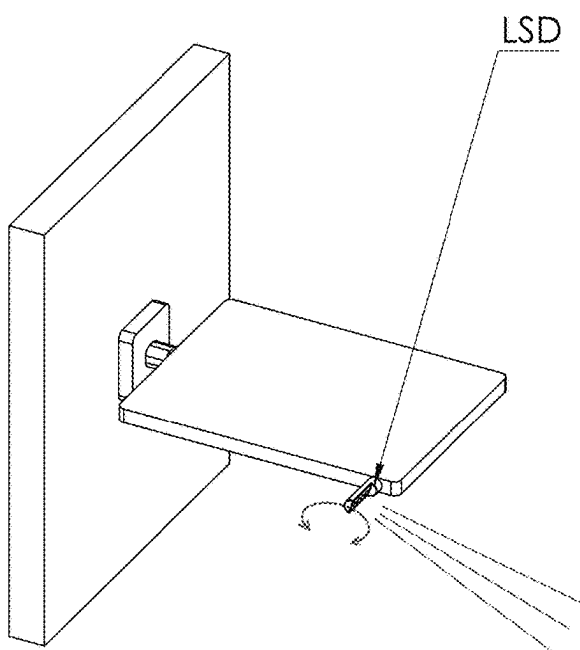


FIG. 38

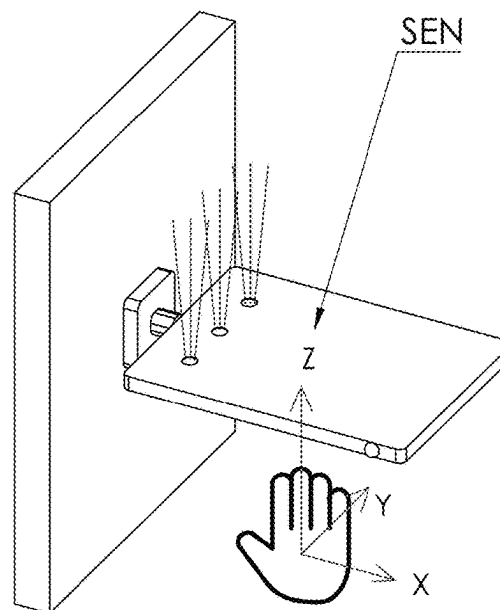


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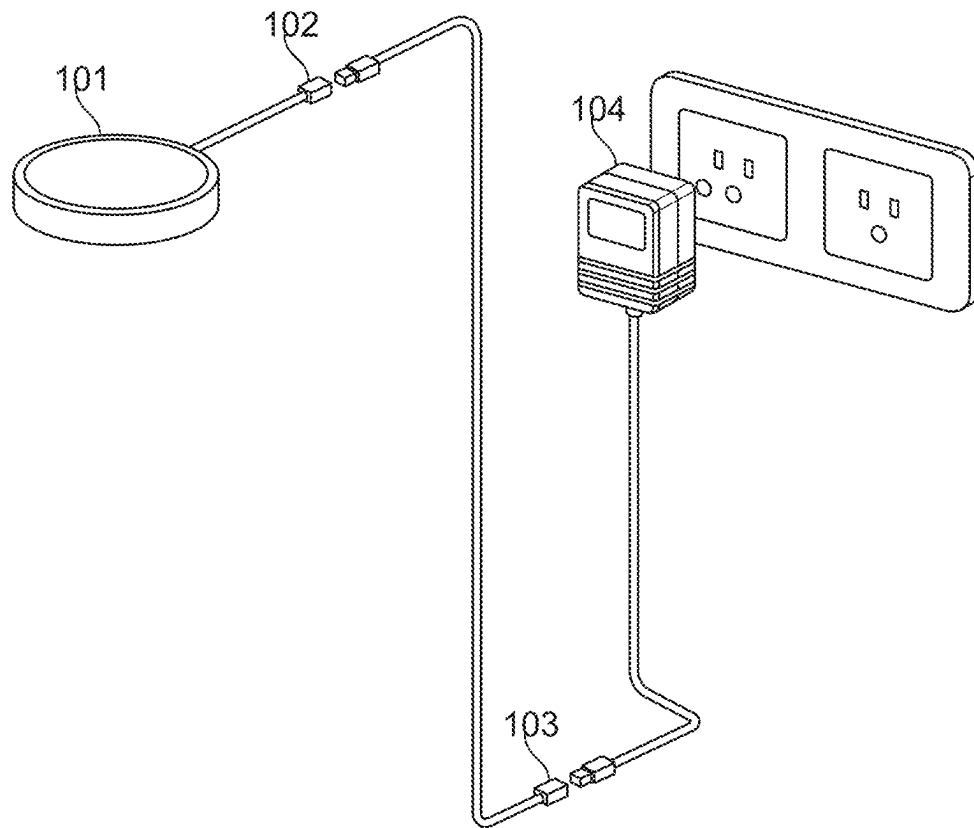


FIG. 40

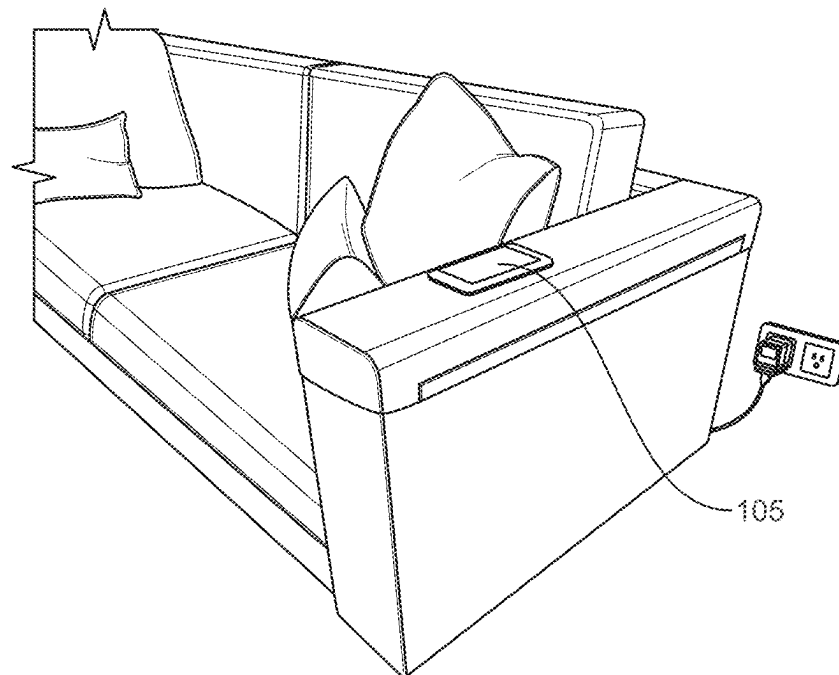


FIG. 41

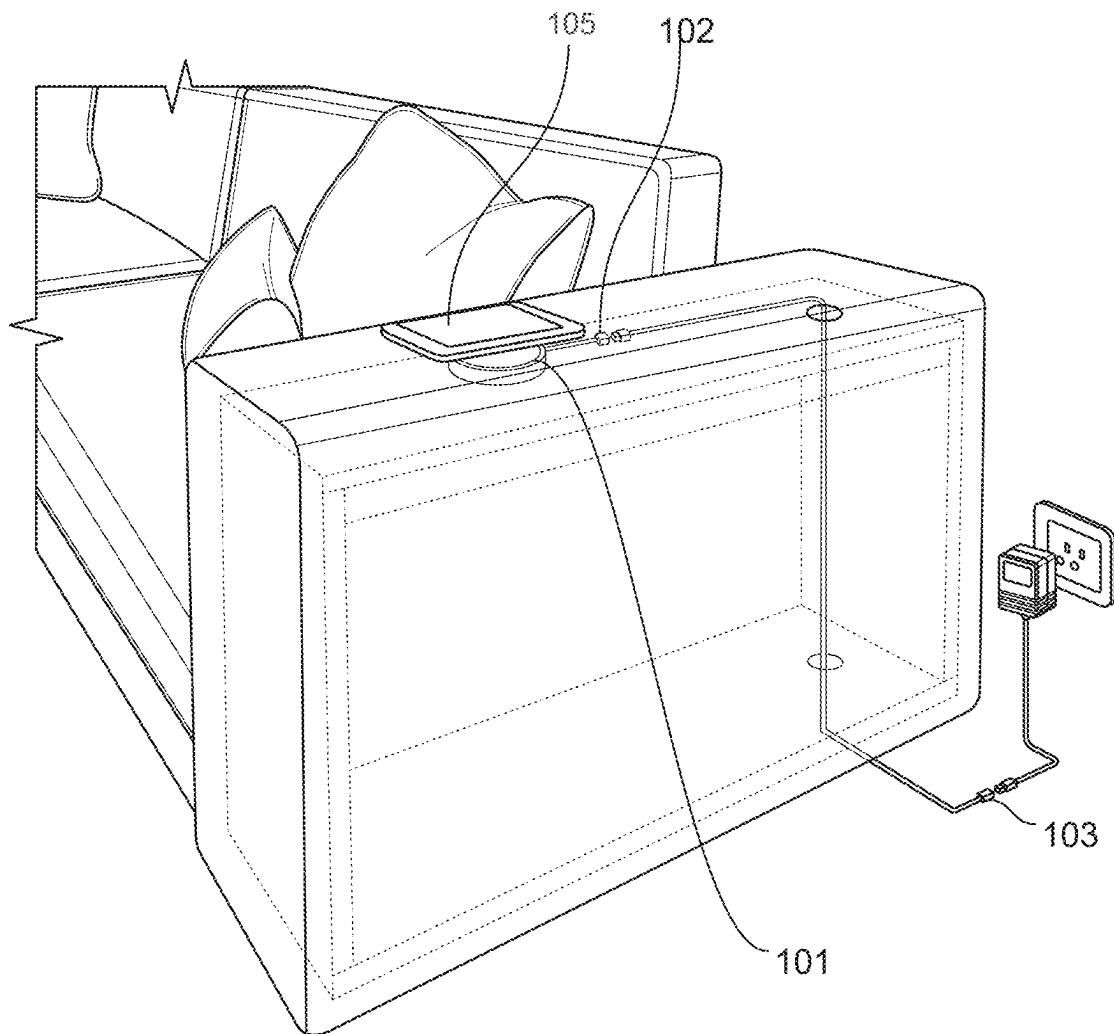


FIG. 42

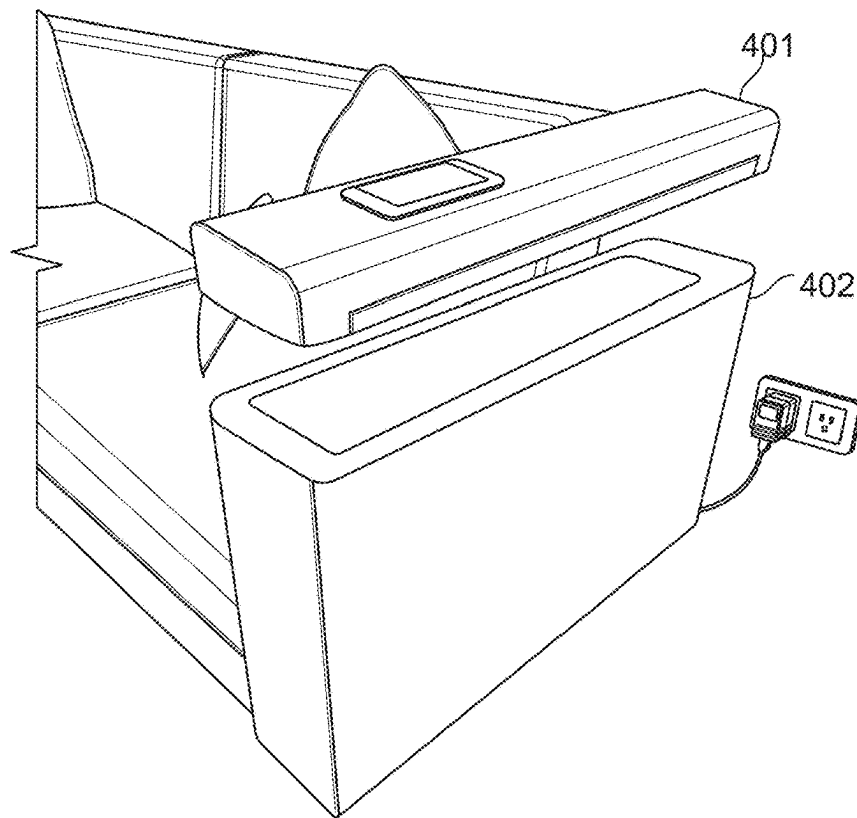


FIG. 43

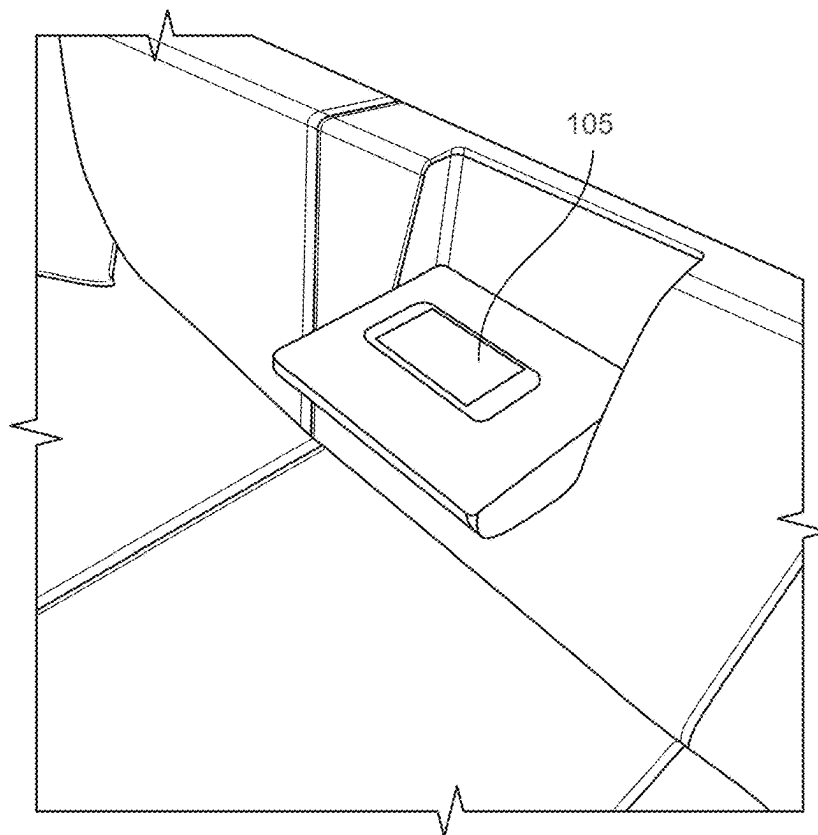


FIG. 44

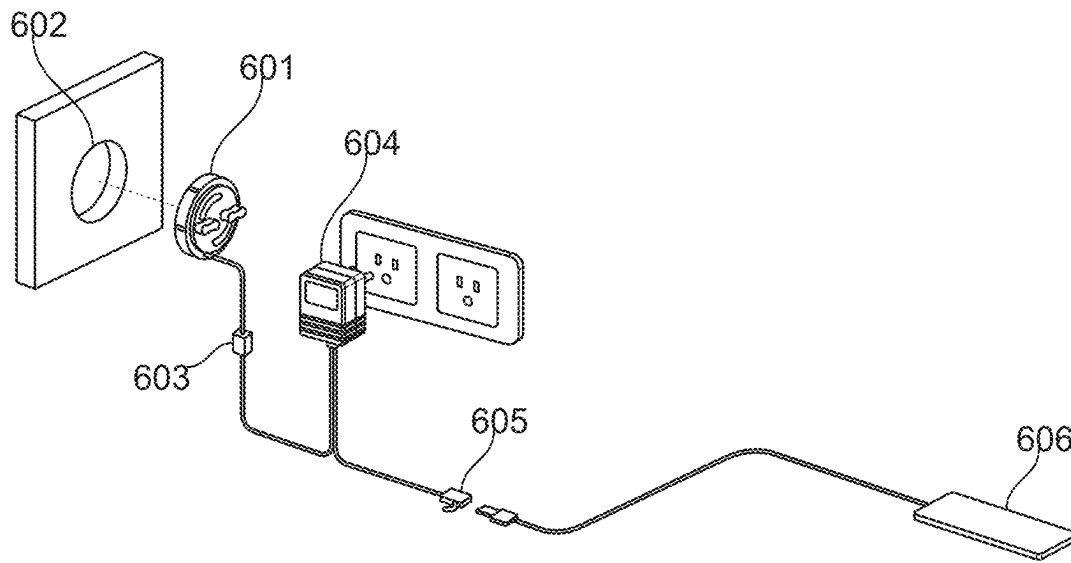


FIG. 45

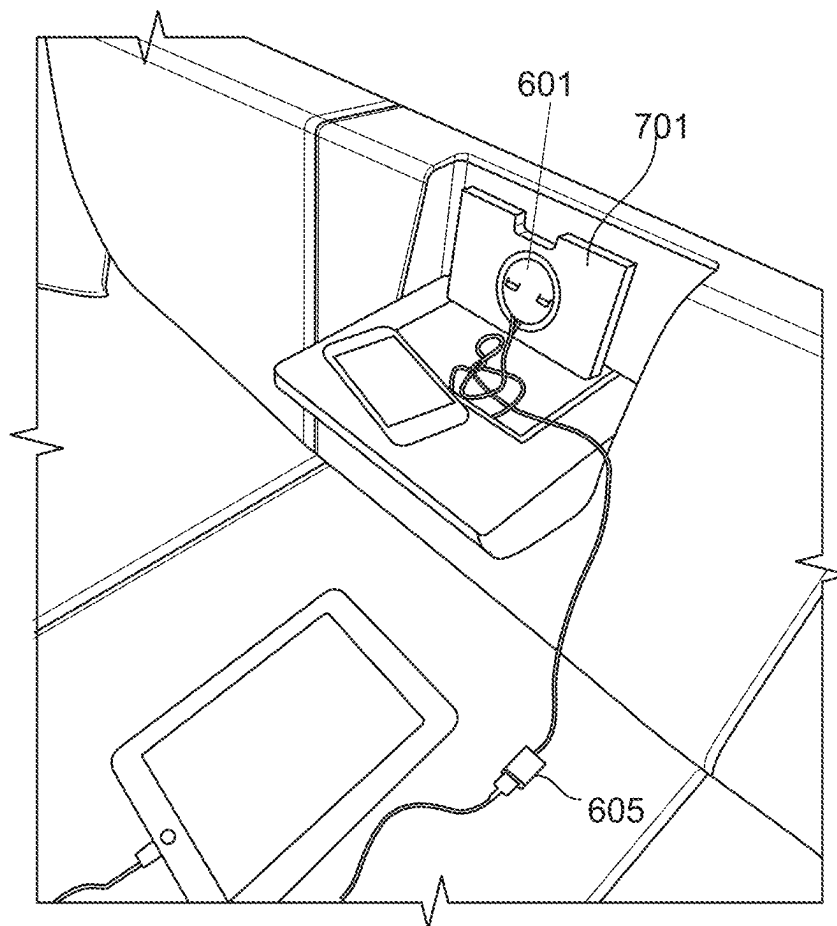


FIG. 46

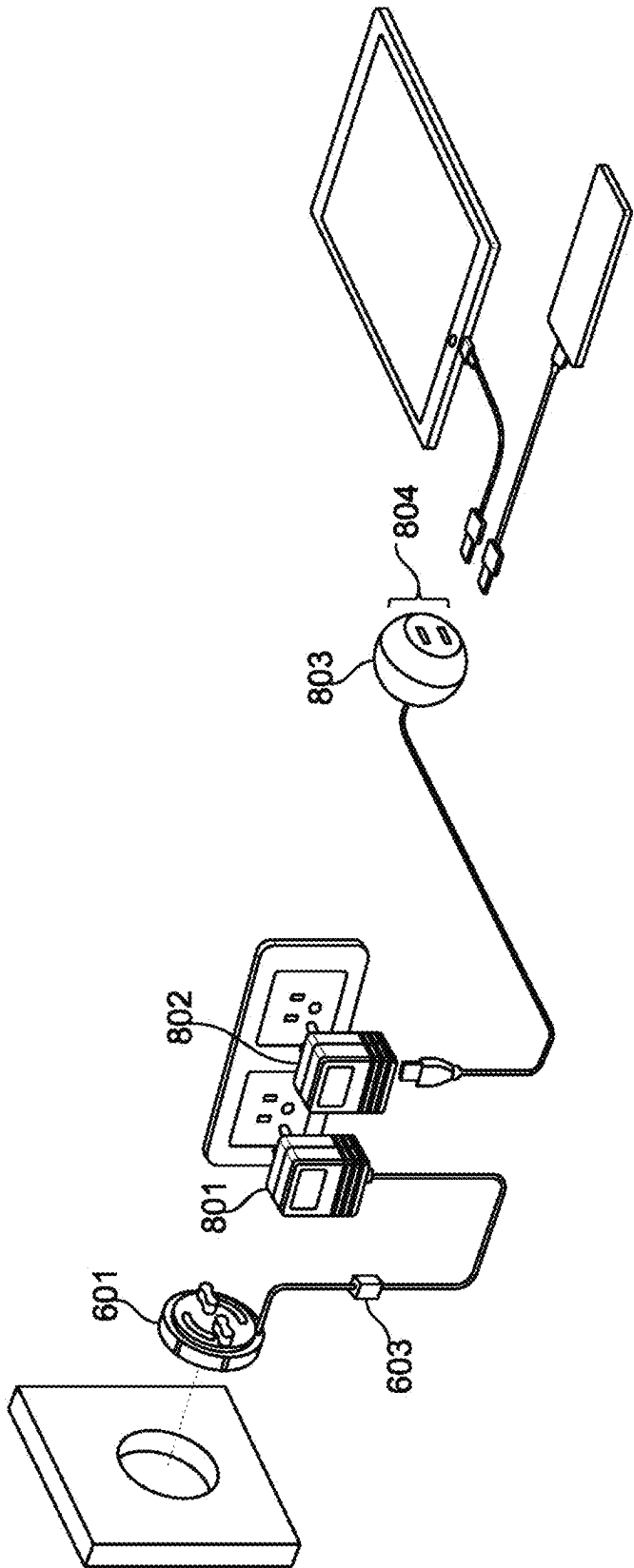


FIG. 47

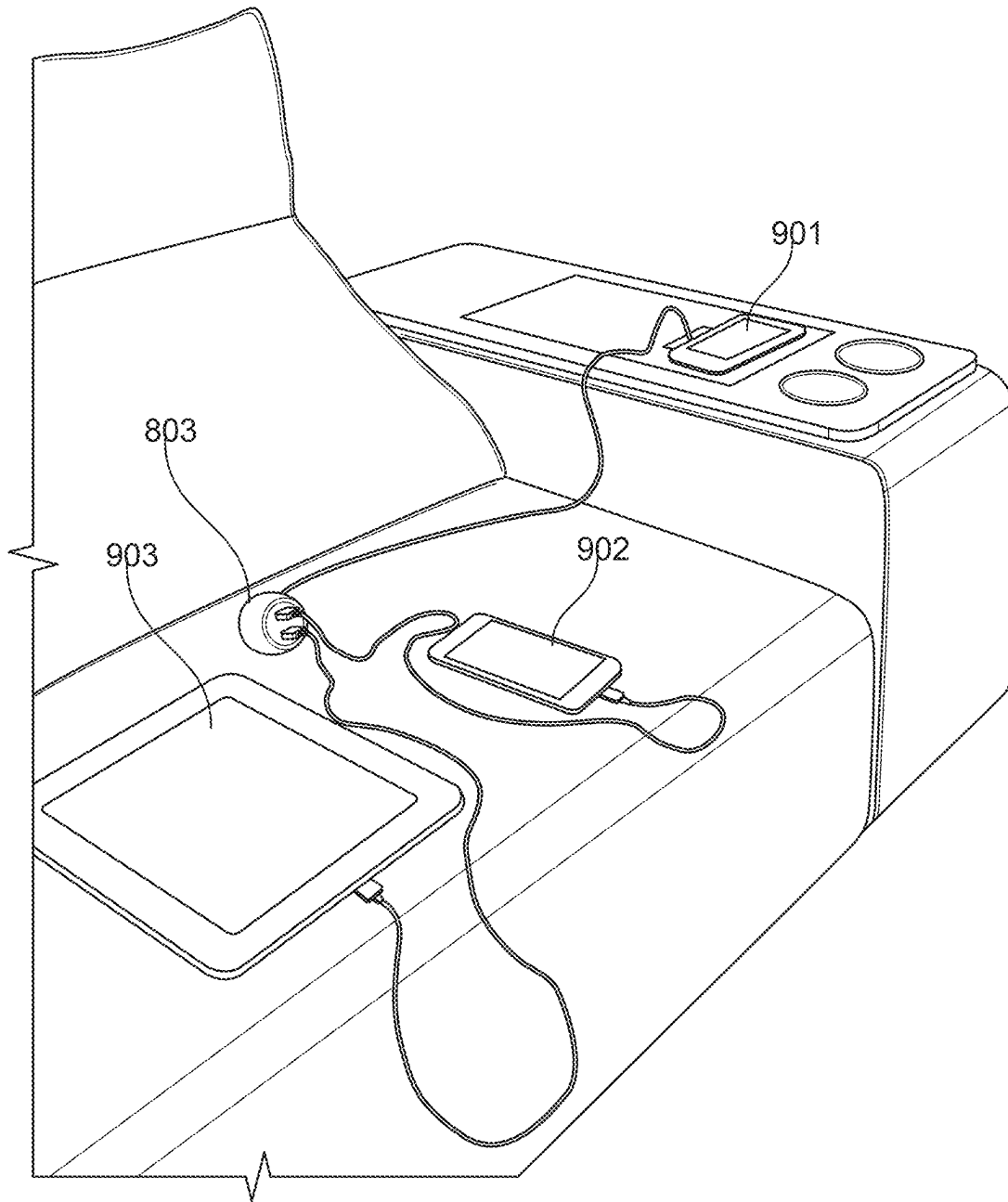


FIG. 48

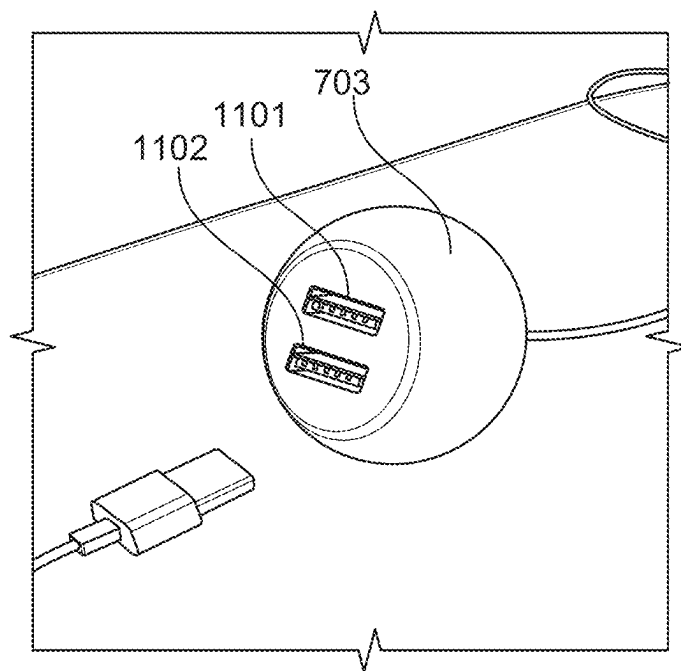
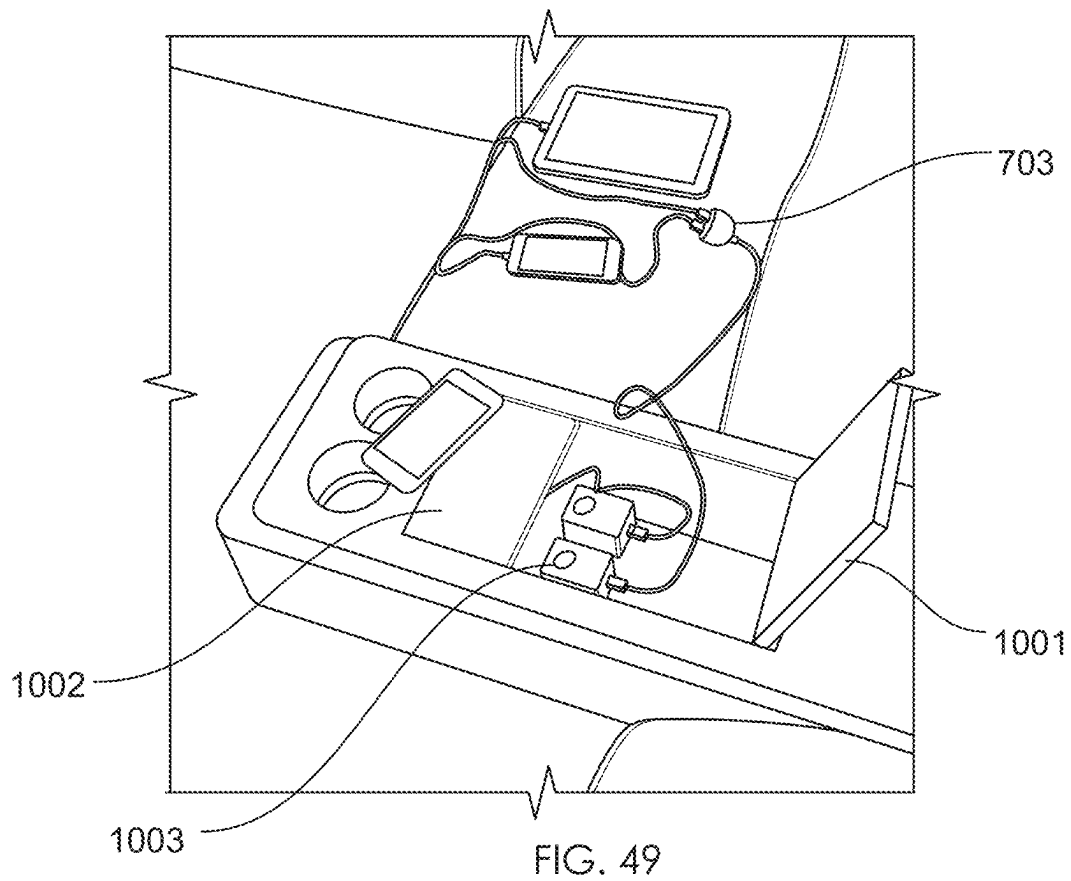


FIG. 50

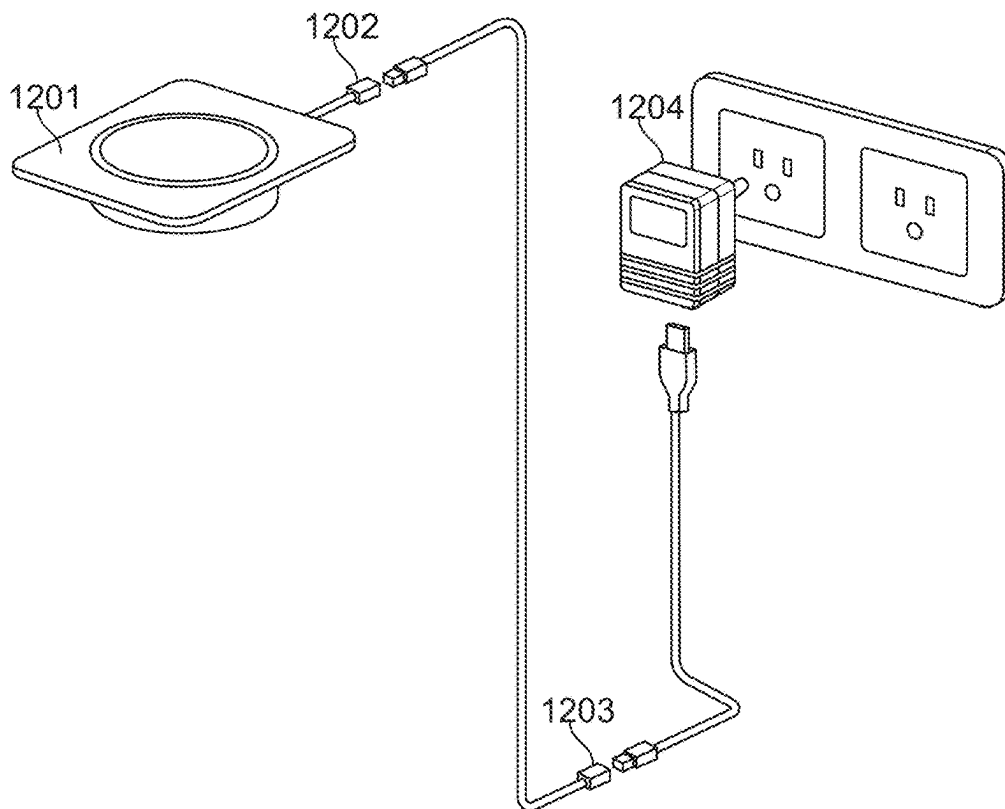


FIG. 51

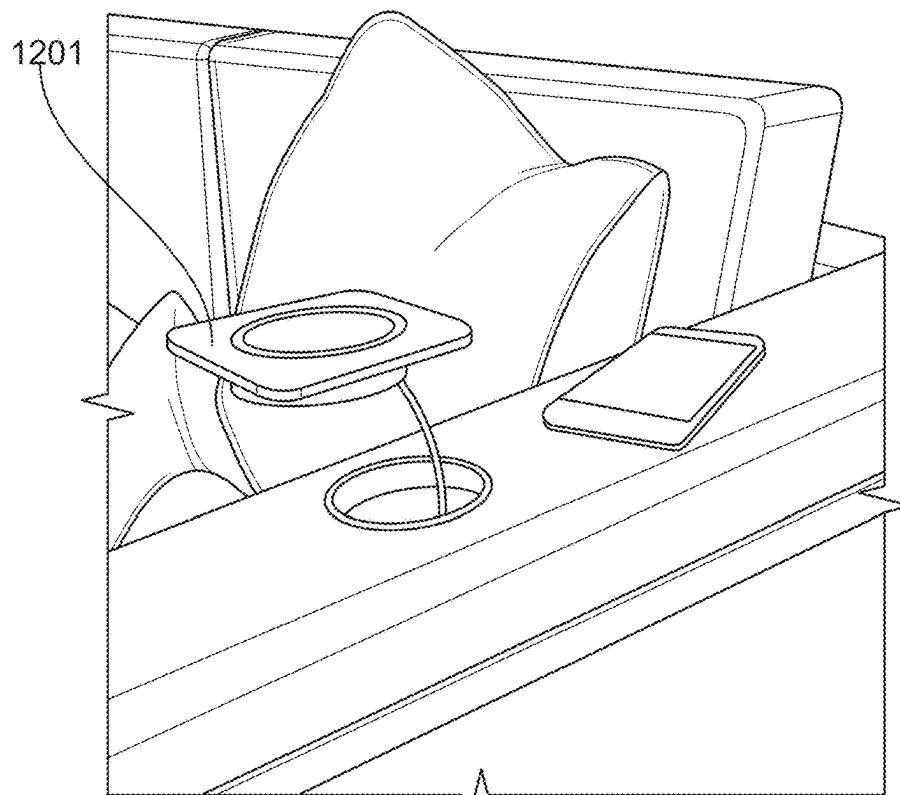


FIG. 52

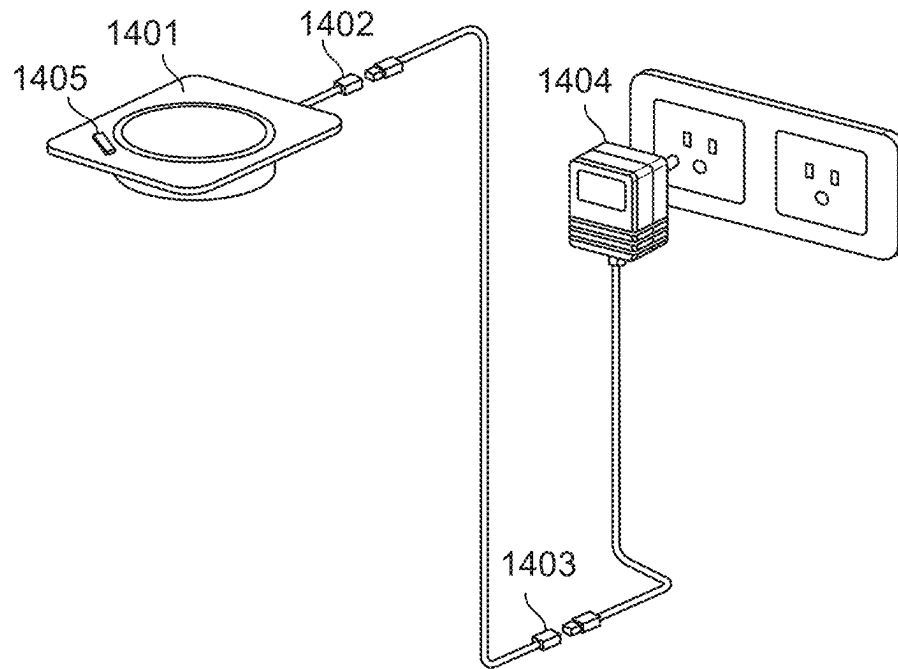


FIG. 53

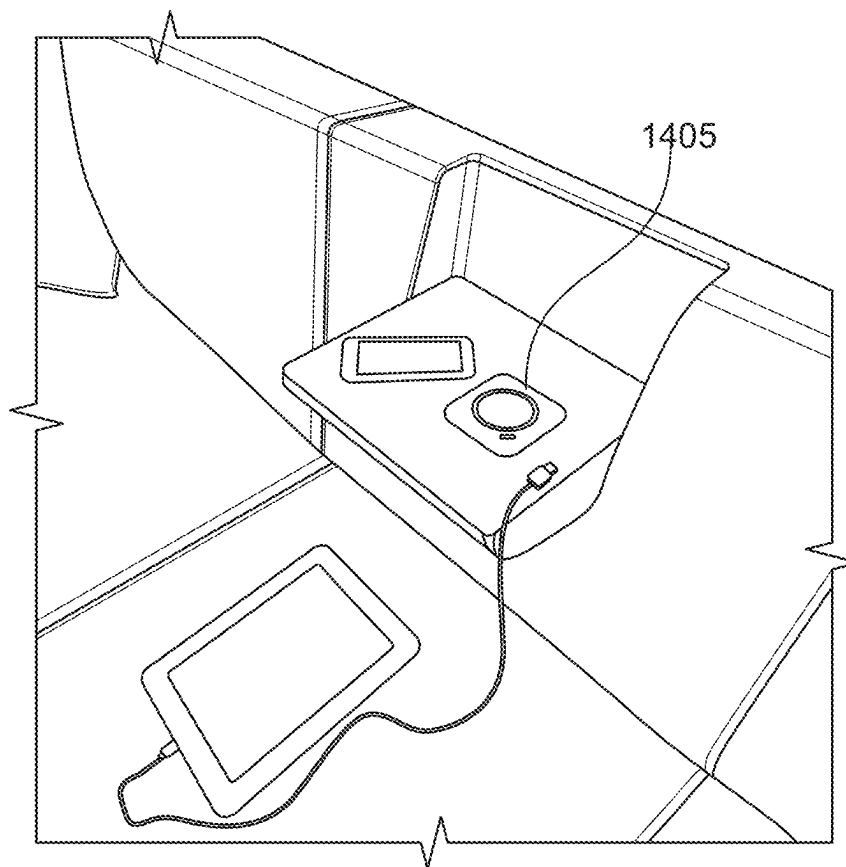


FIG. 54

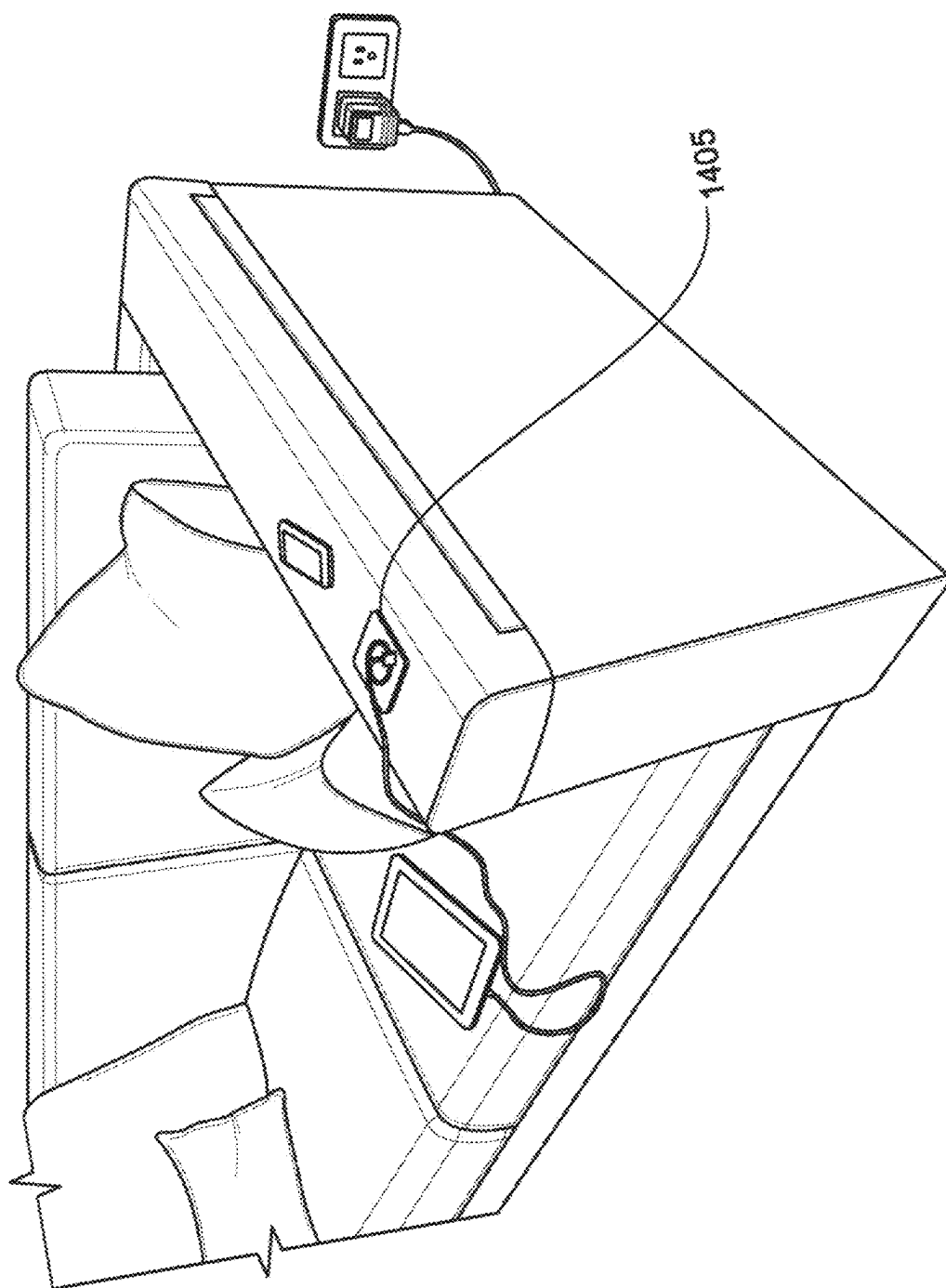


FIG. 55

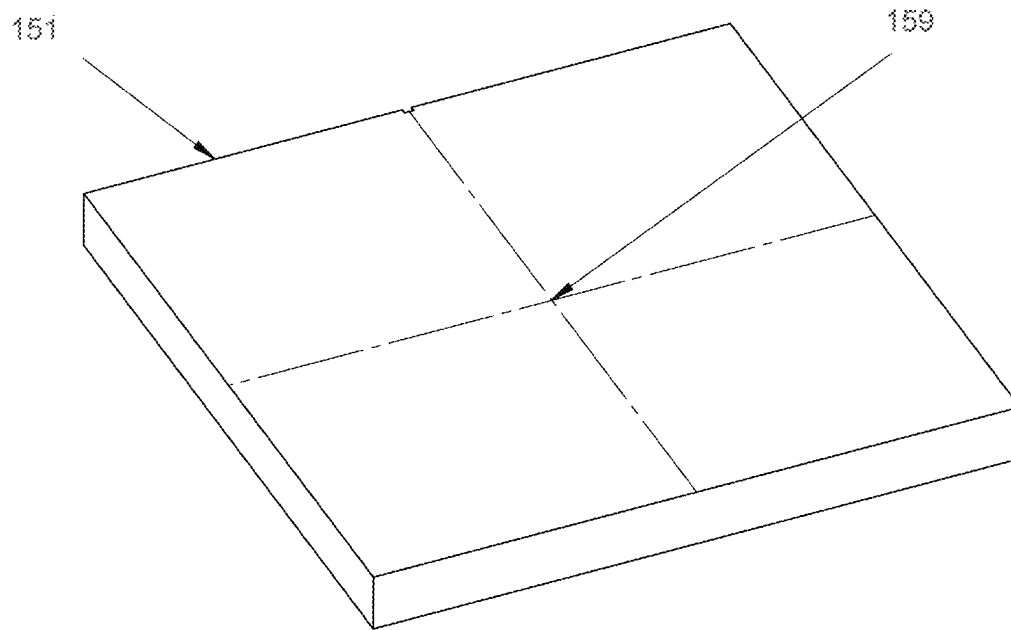


FIG. 56

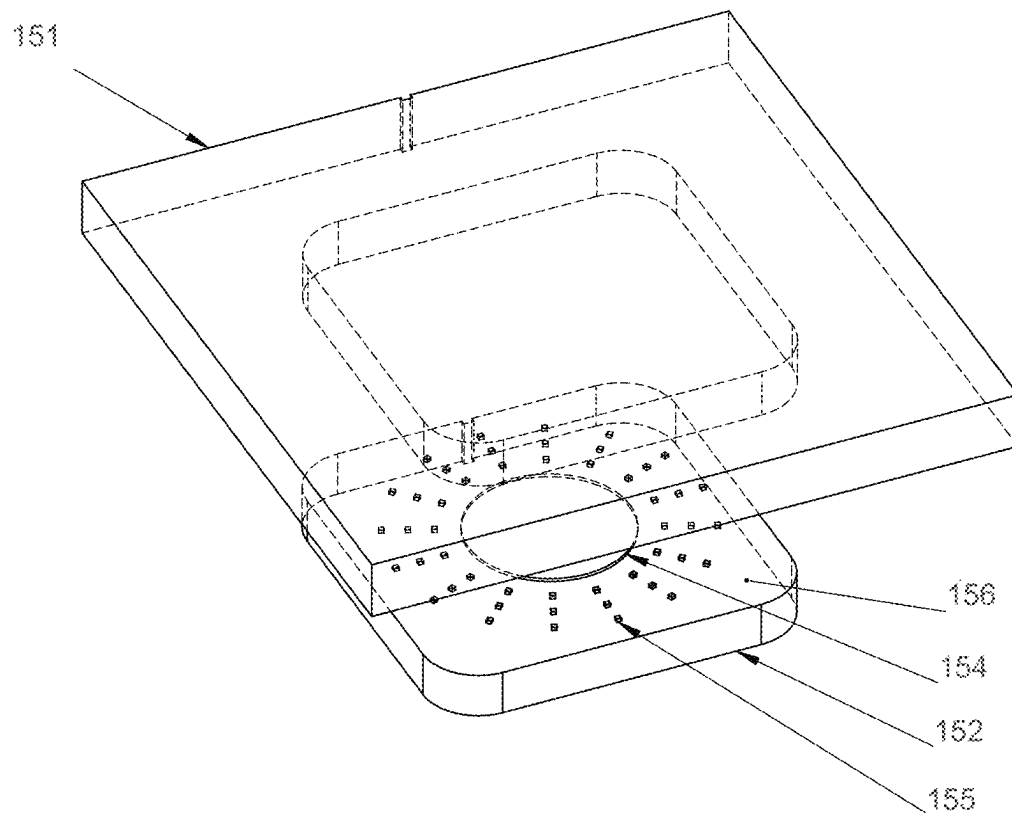


FIG. 57

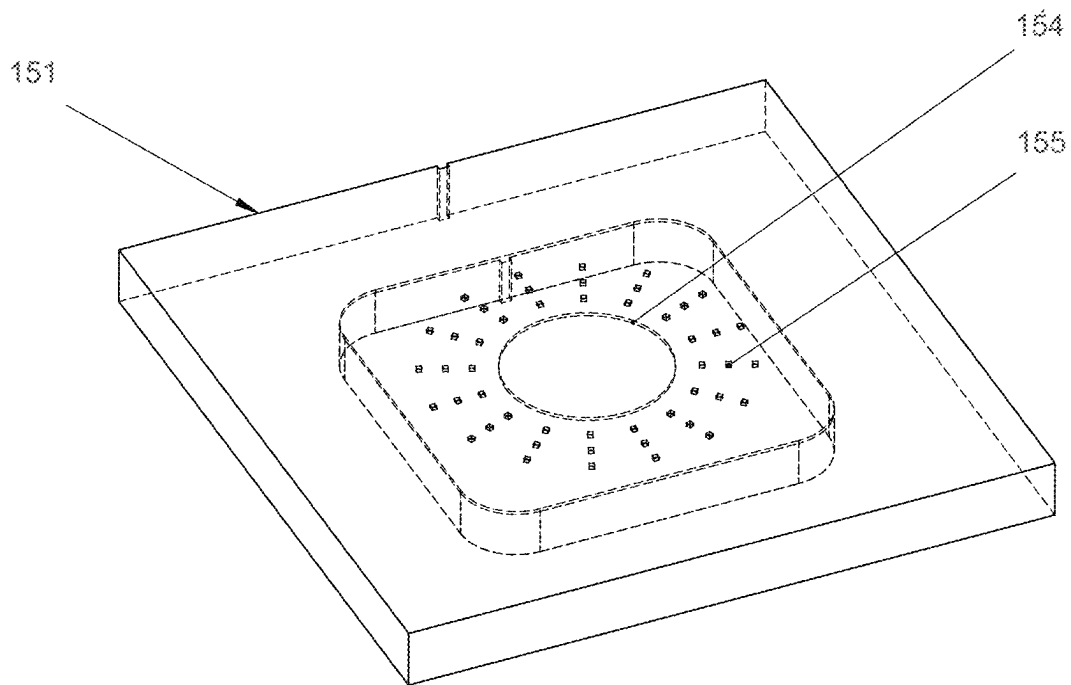


FIG. 58

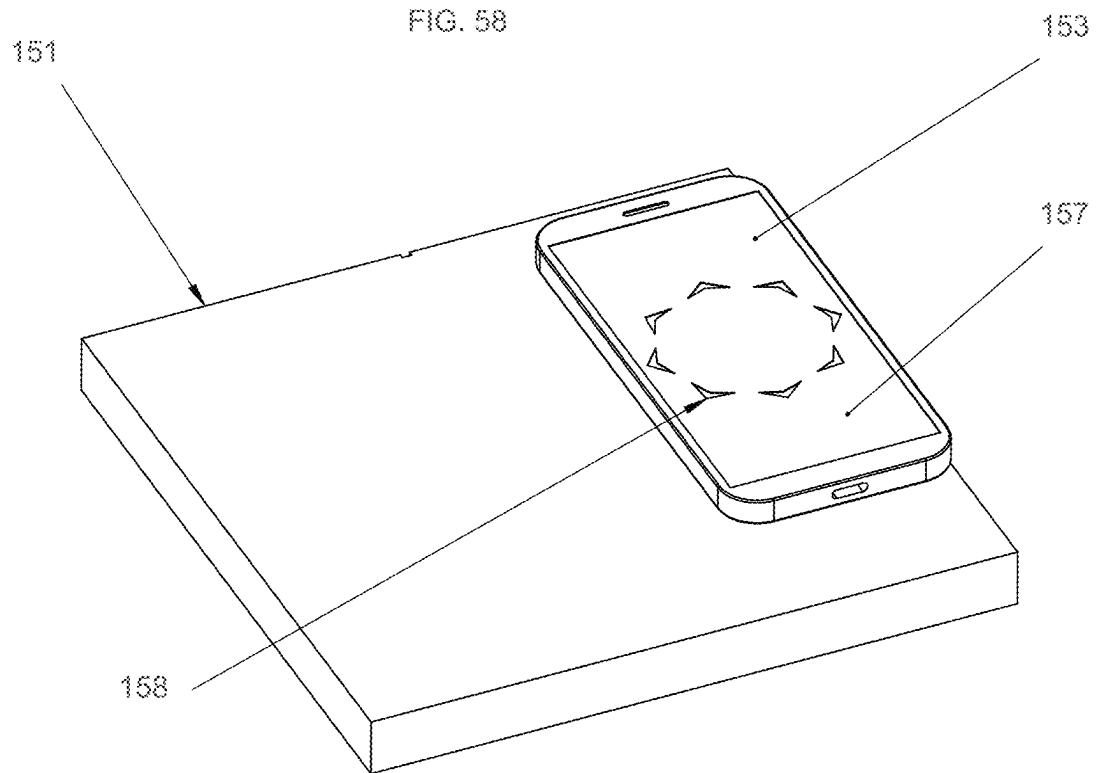
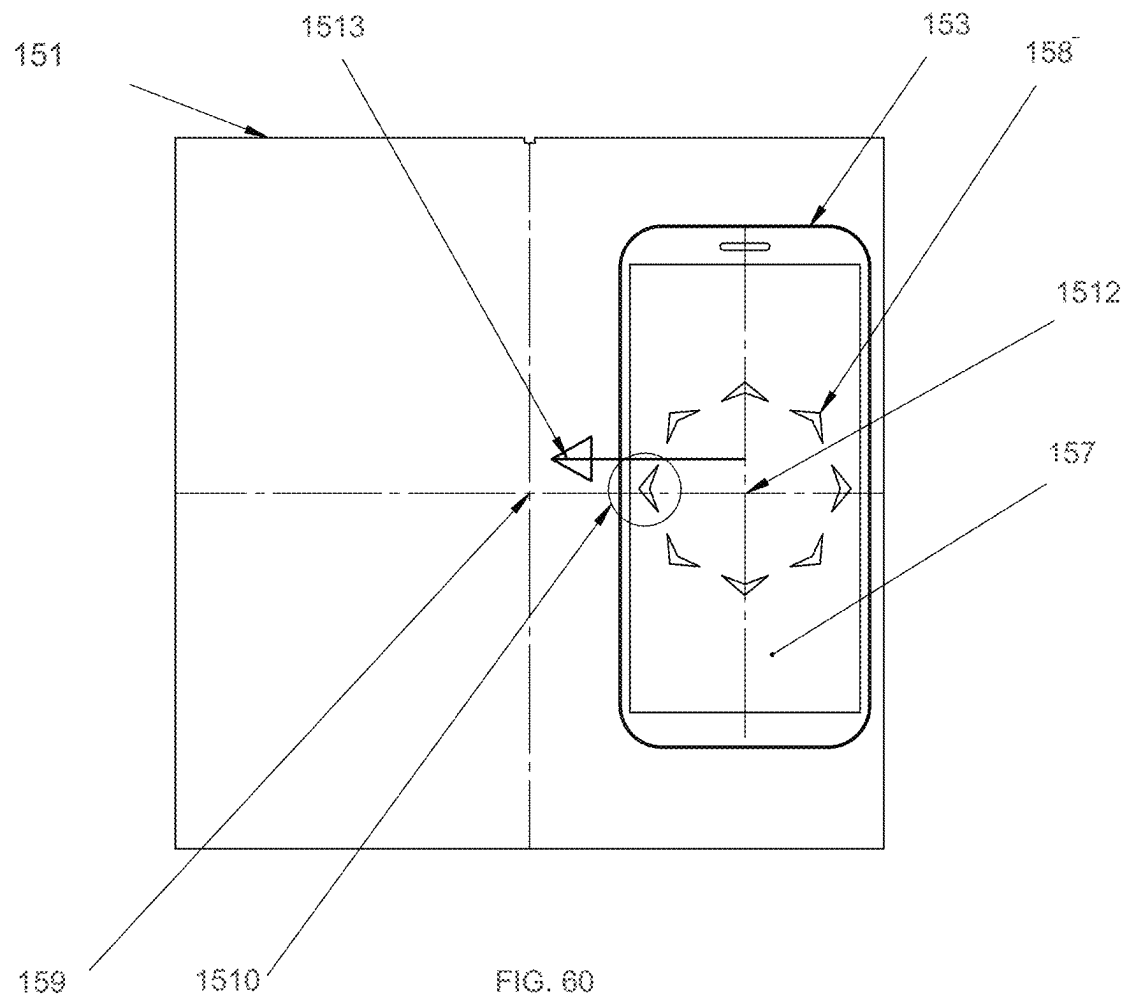


FIG. 59



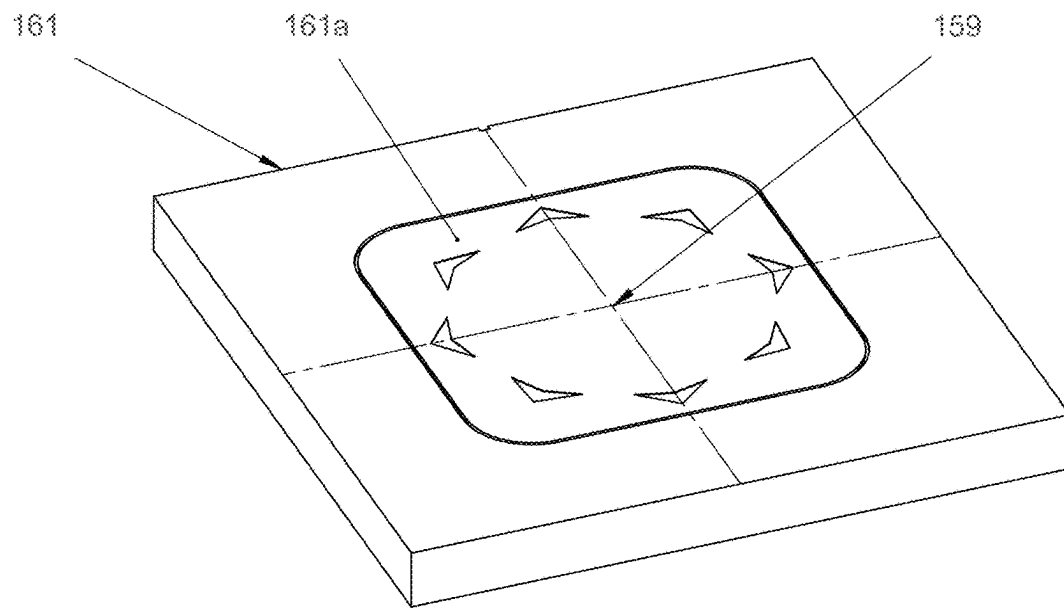


FIG. 61

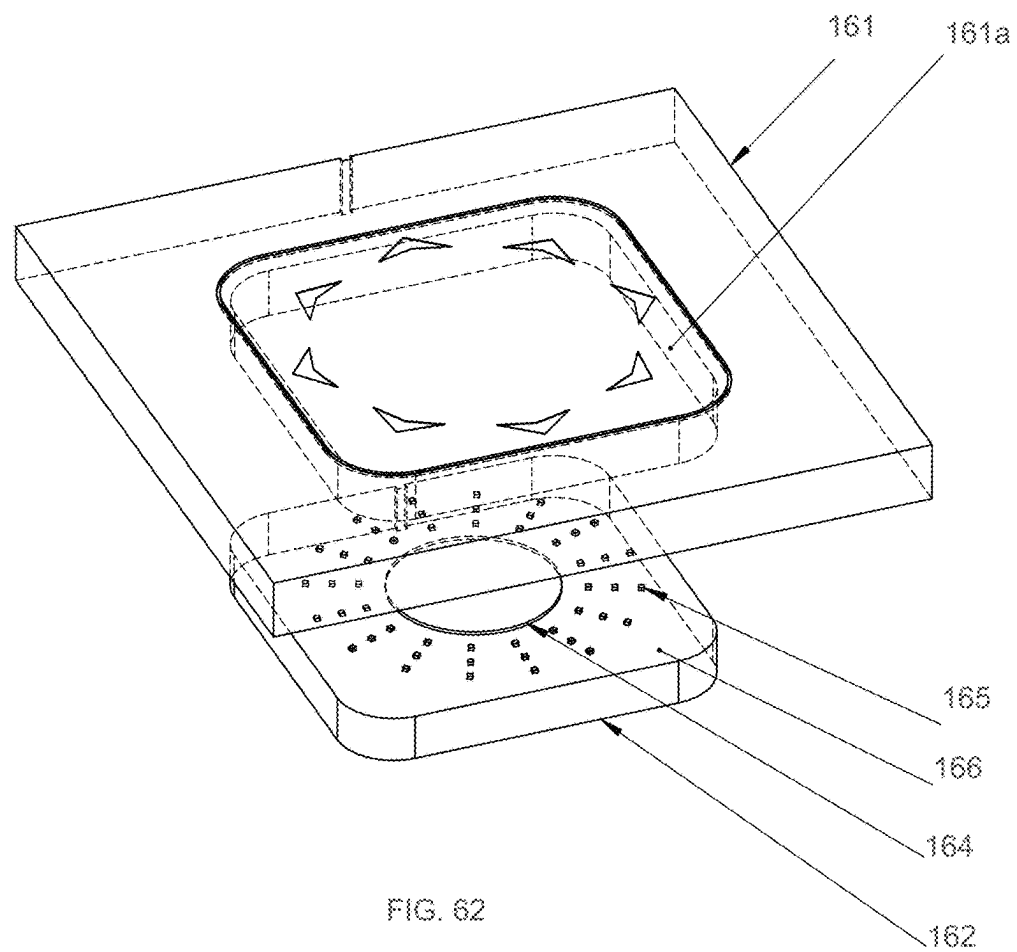


FIG. 62

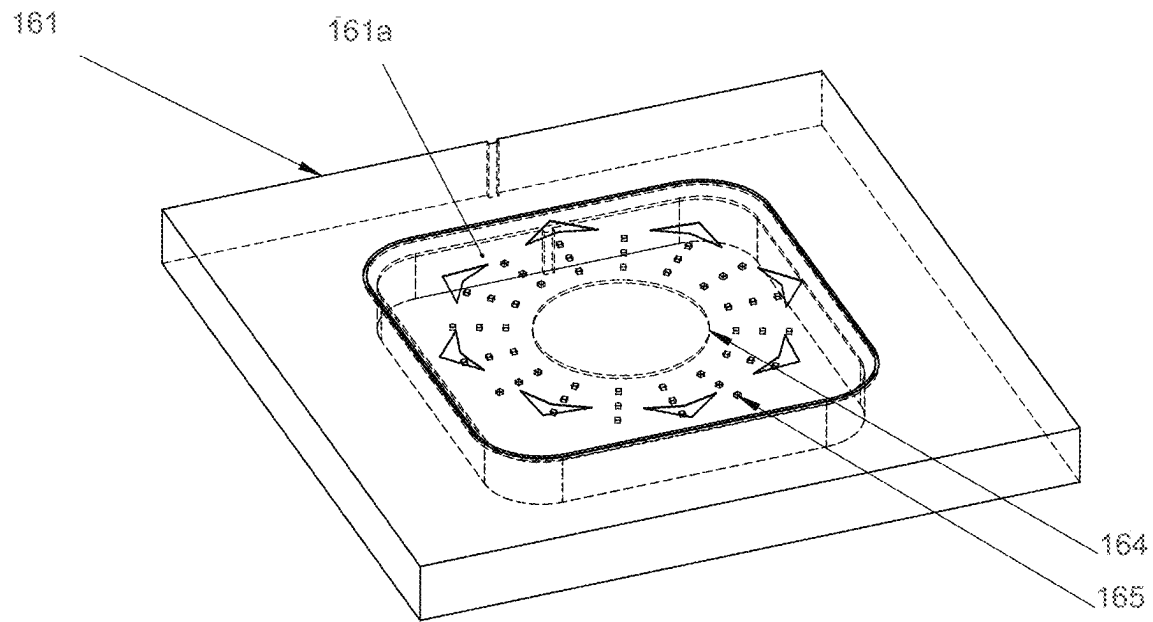


FIG. 63

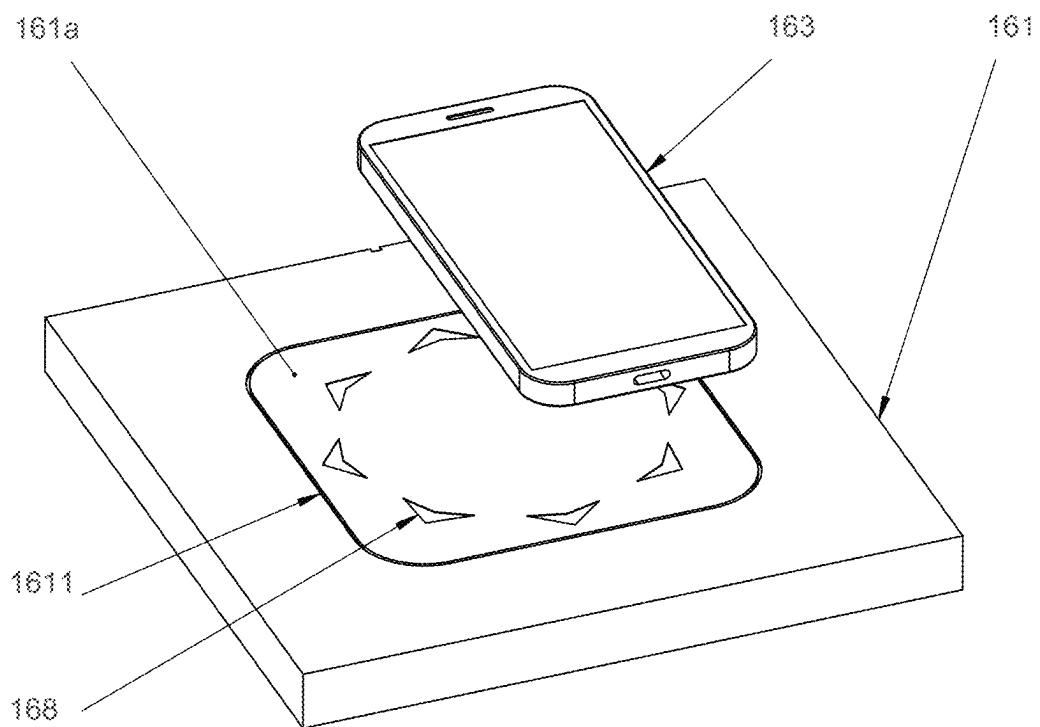


FIG. 64

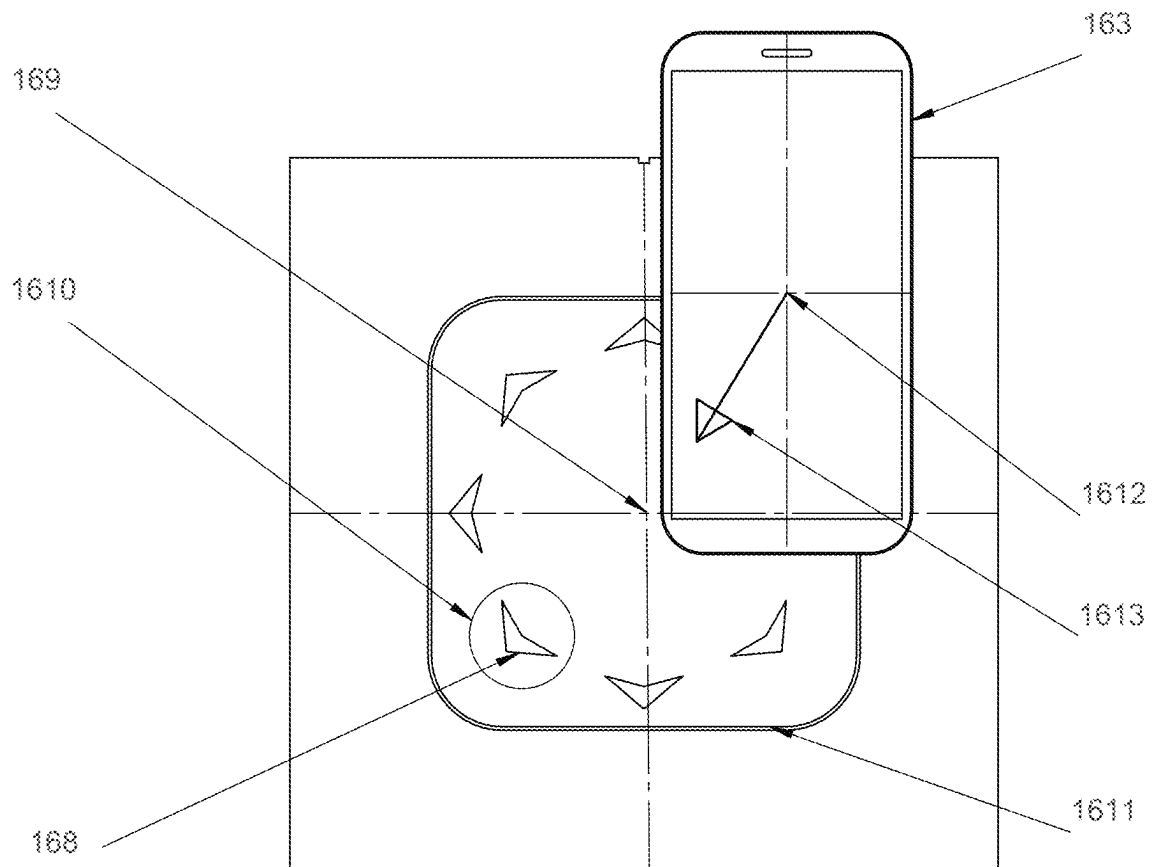


FIG. 65

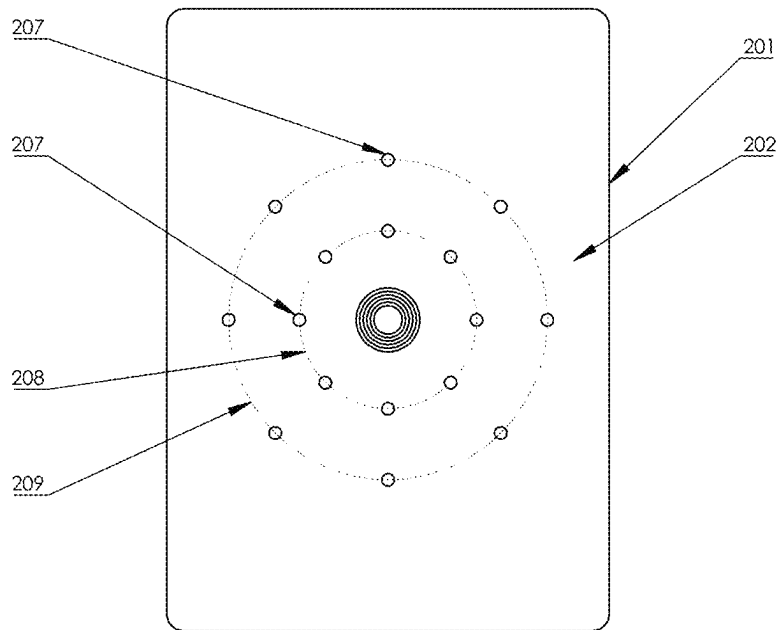


FIG. 66

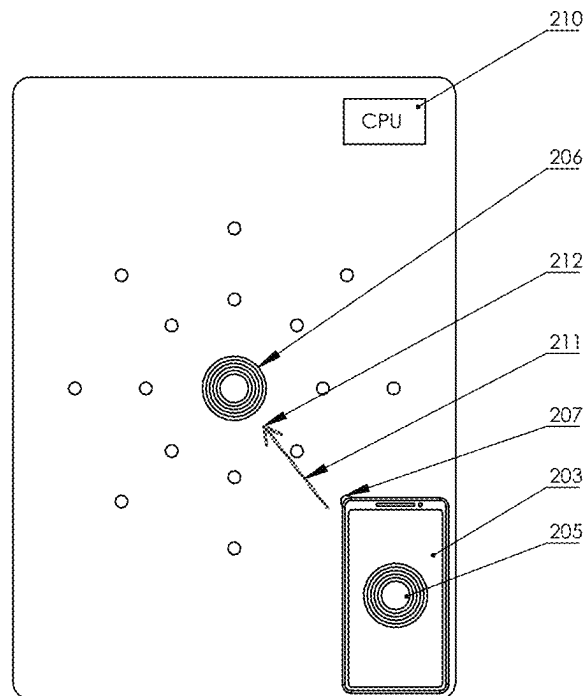


FIG. 67

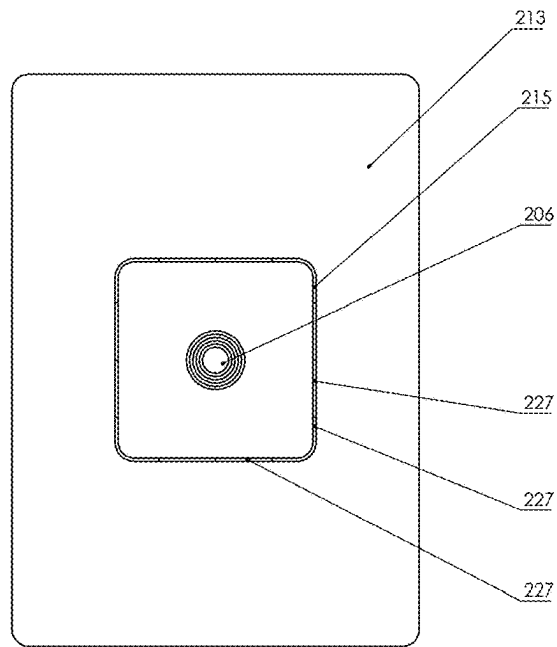


FIG. 68

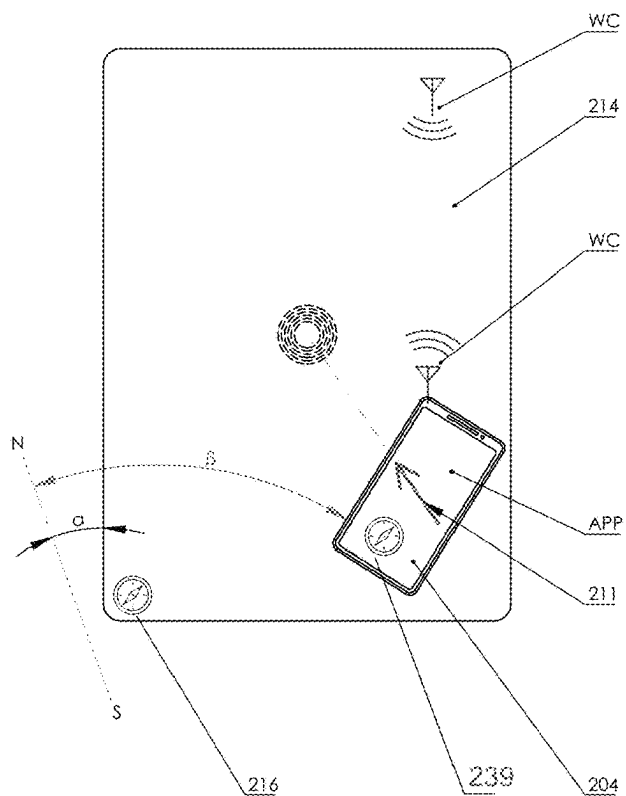


FIG. 69

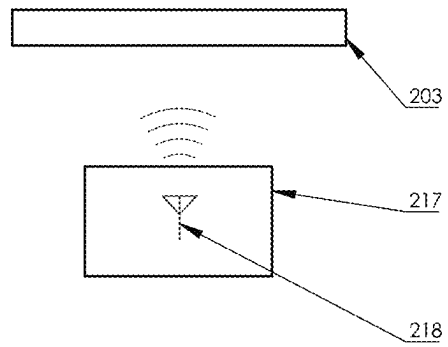


FIG. 70

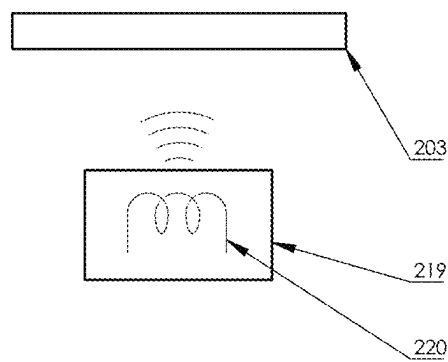


FIG. 71

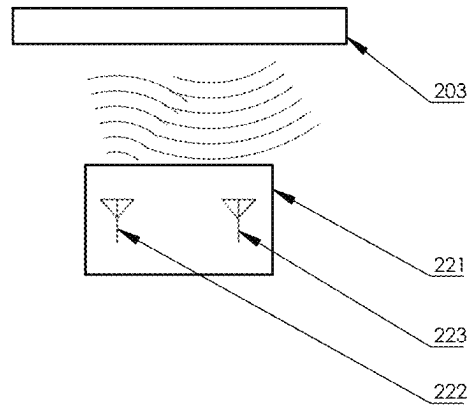


FIG. 72

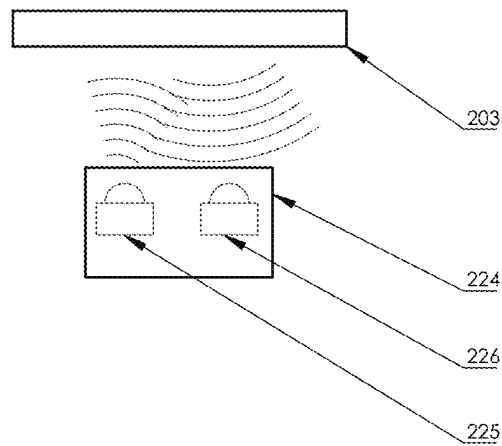


FIG. 73

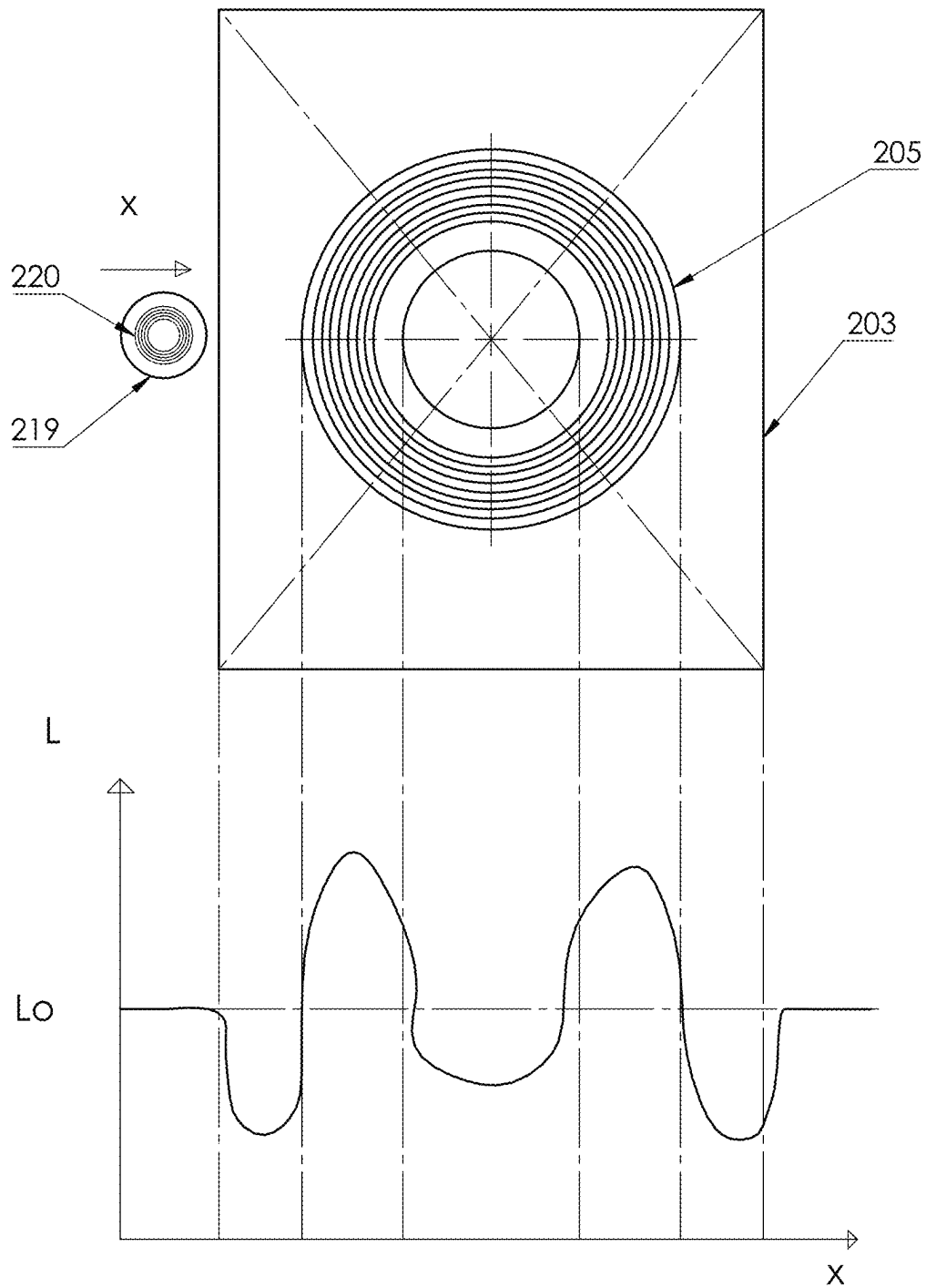


FIG. 74

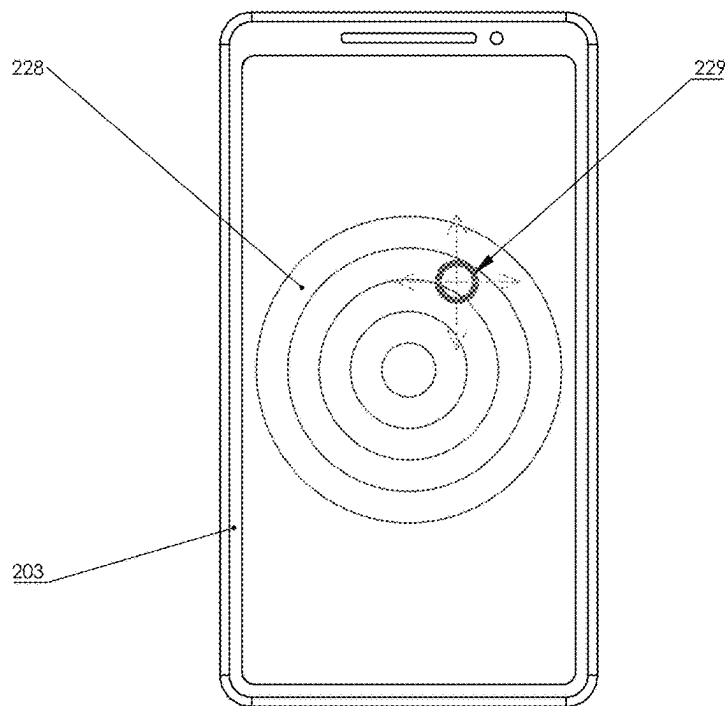


FIG. 75

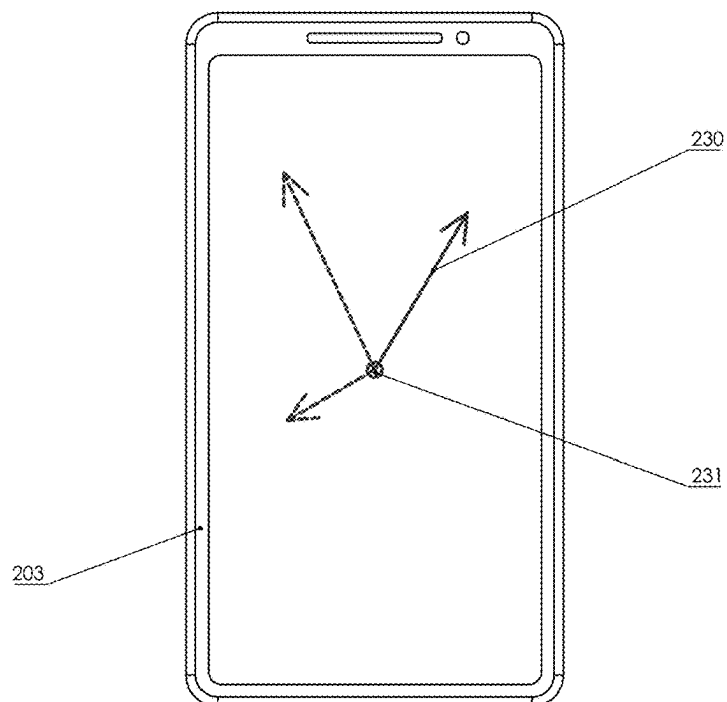


FIG. 76

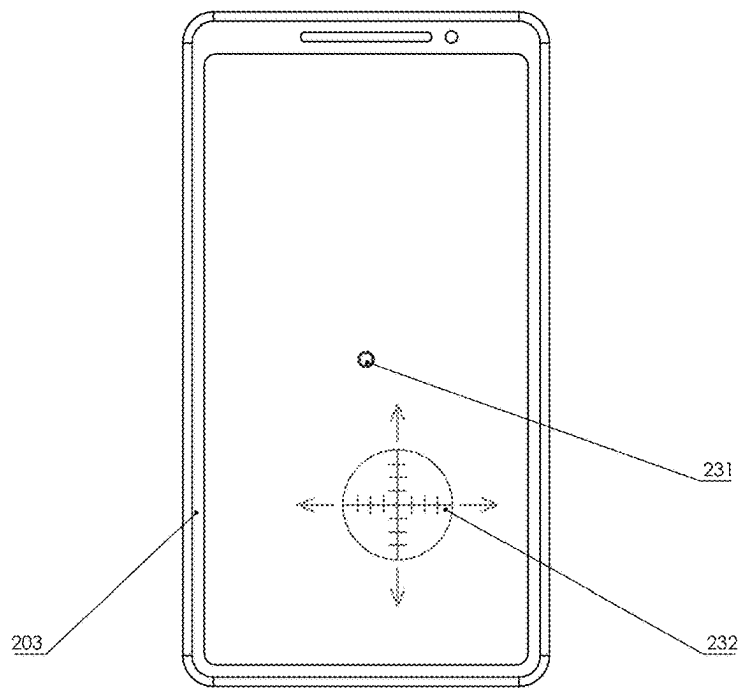


FIG. 77

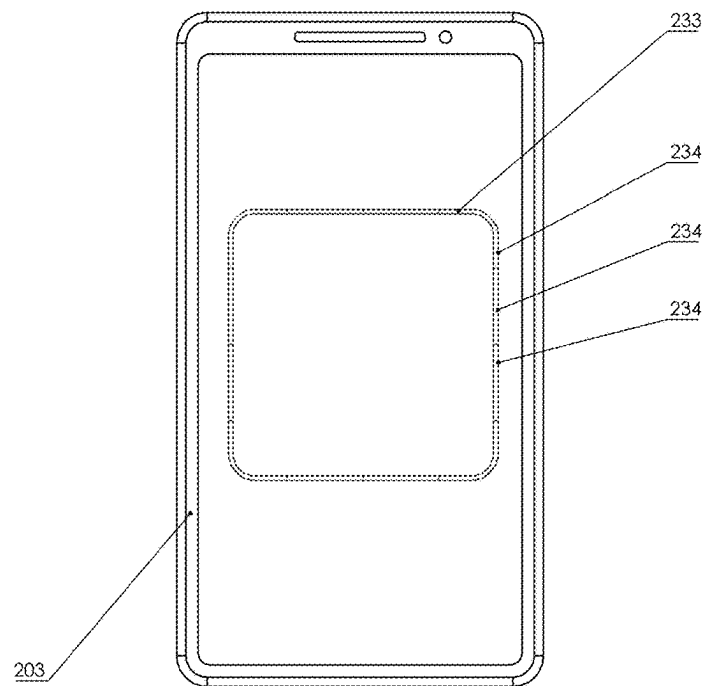


FIG. 78

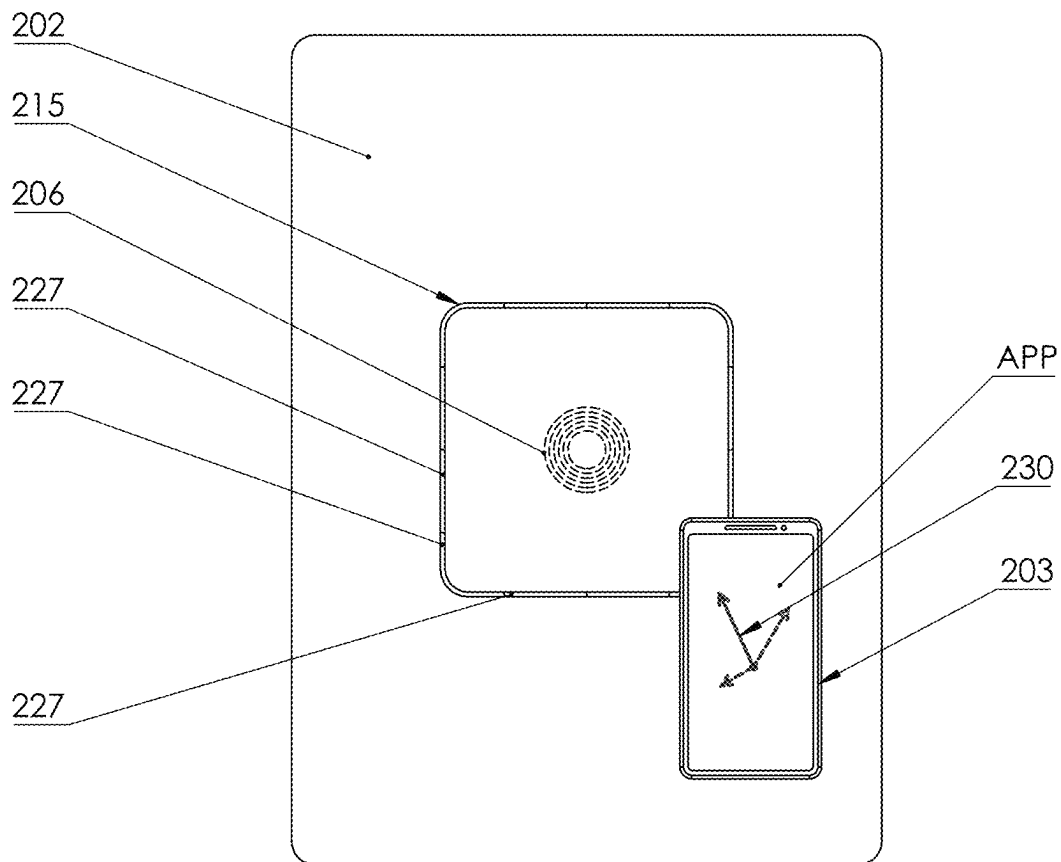


FIG. 79

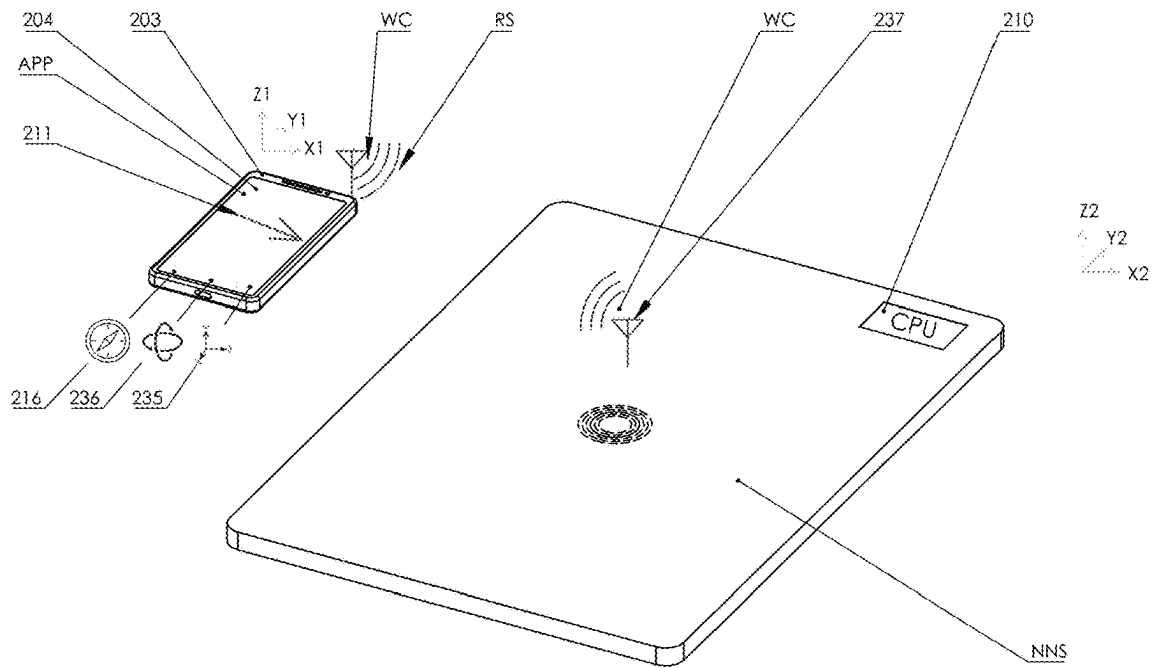


FIG. 80

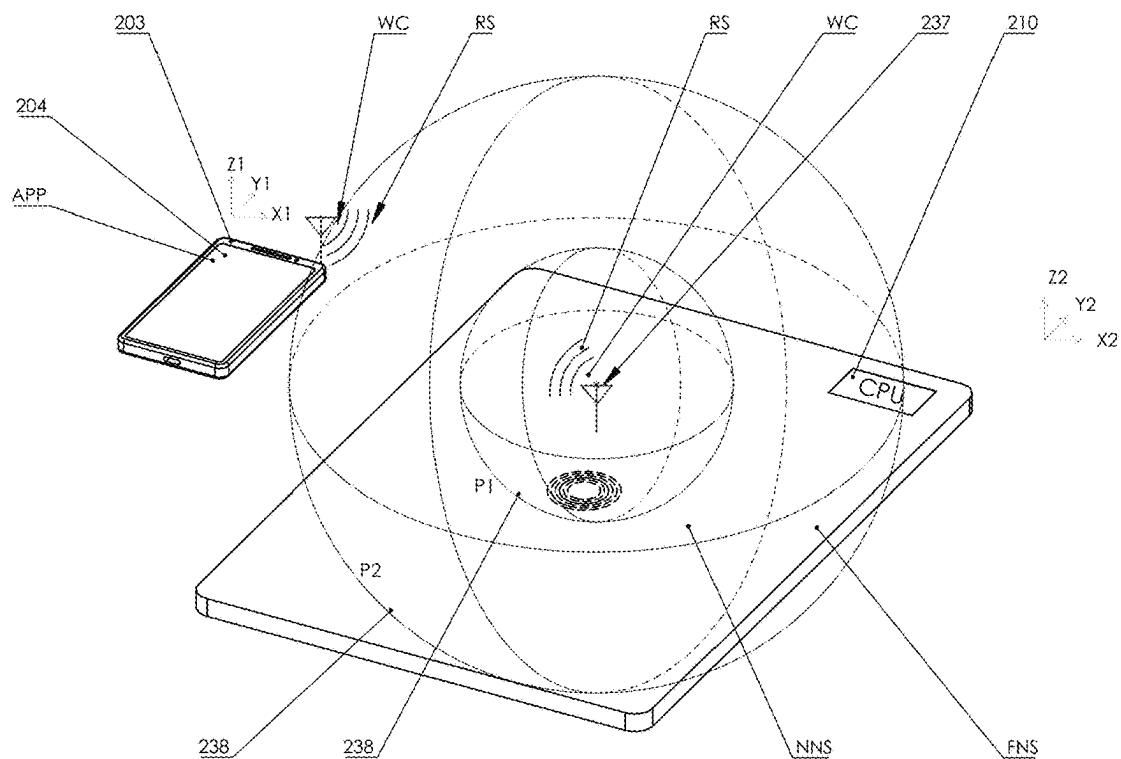


FIG. 81

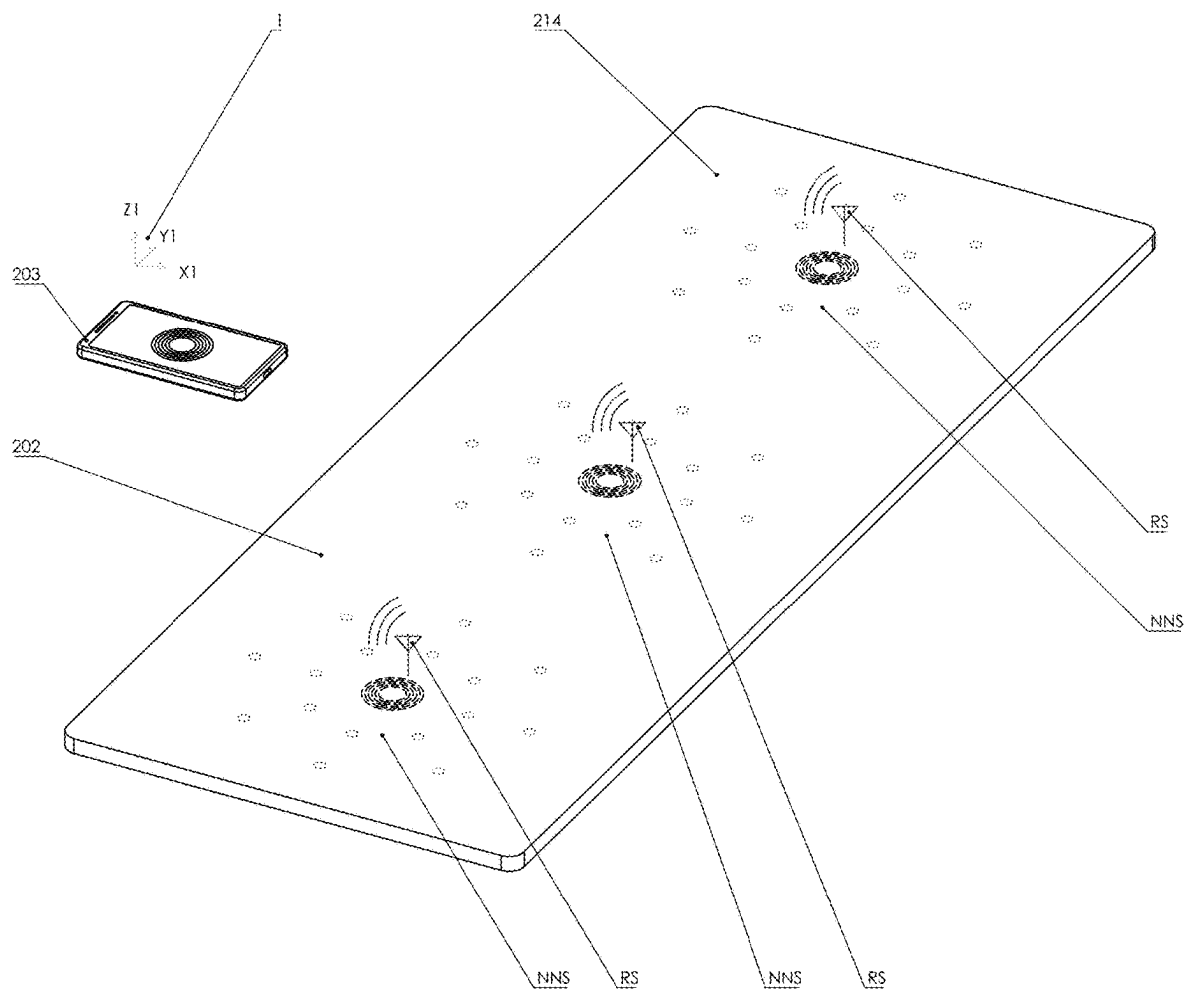


FIG. 82

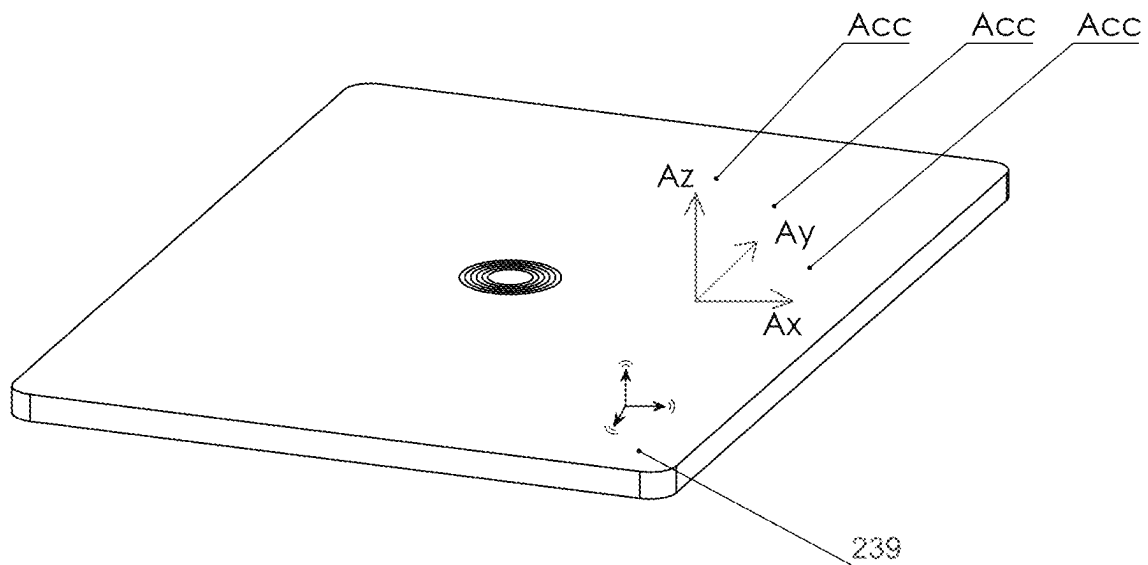


FIG. 83

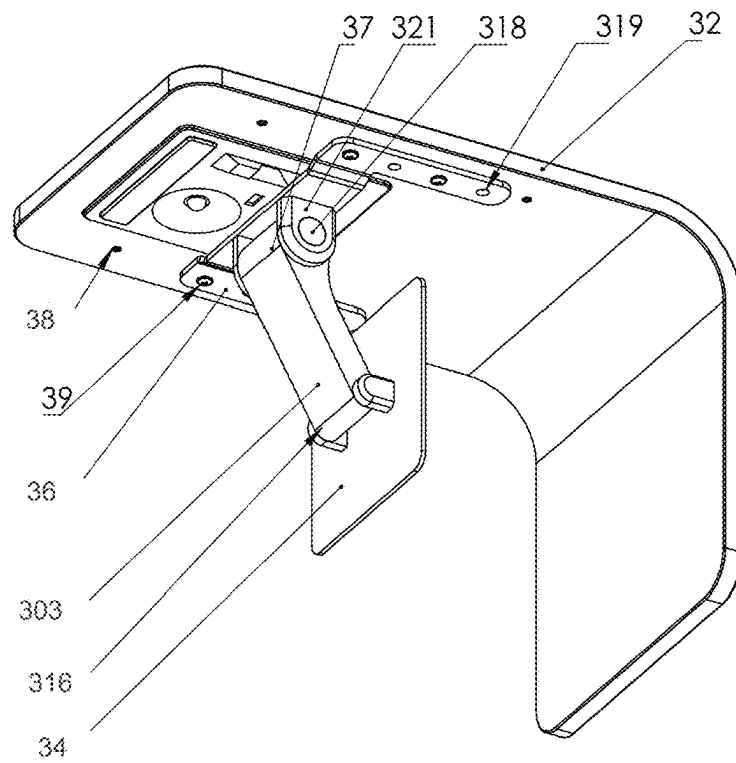


FIG. 84

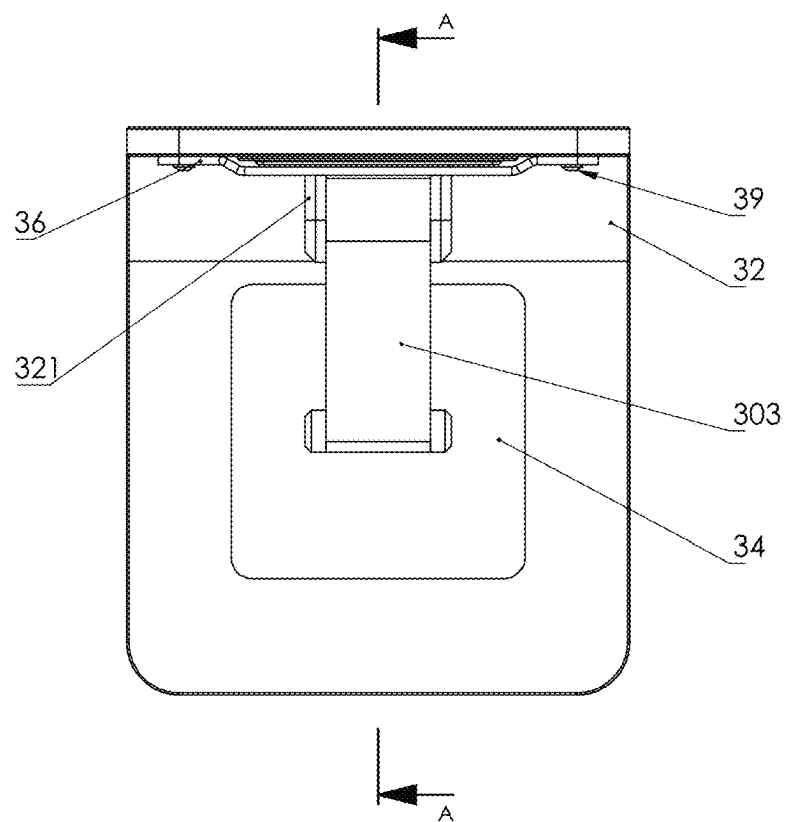
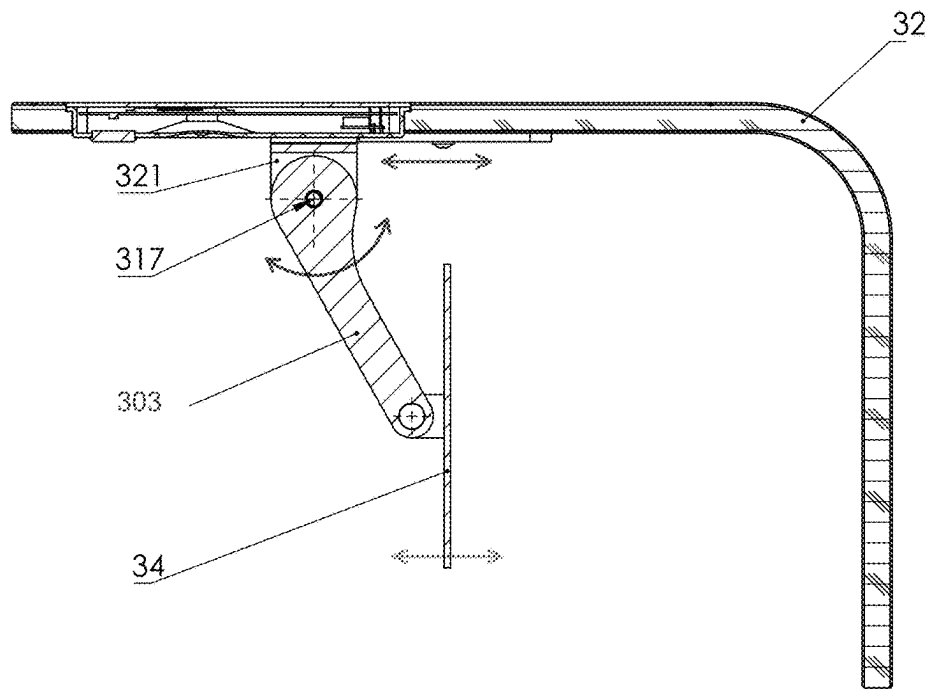


FIG. 85



SECTION A-A

FIG. 86

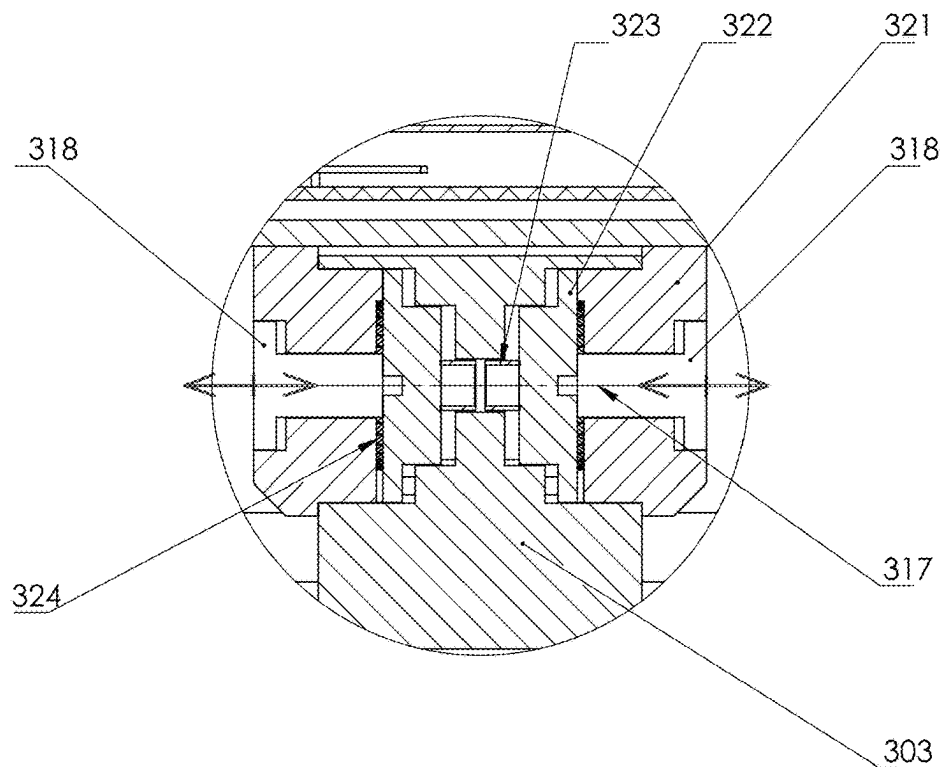


FIG. 87

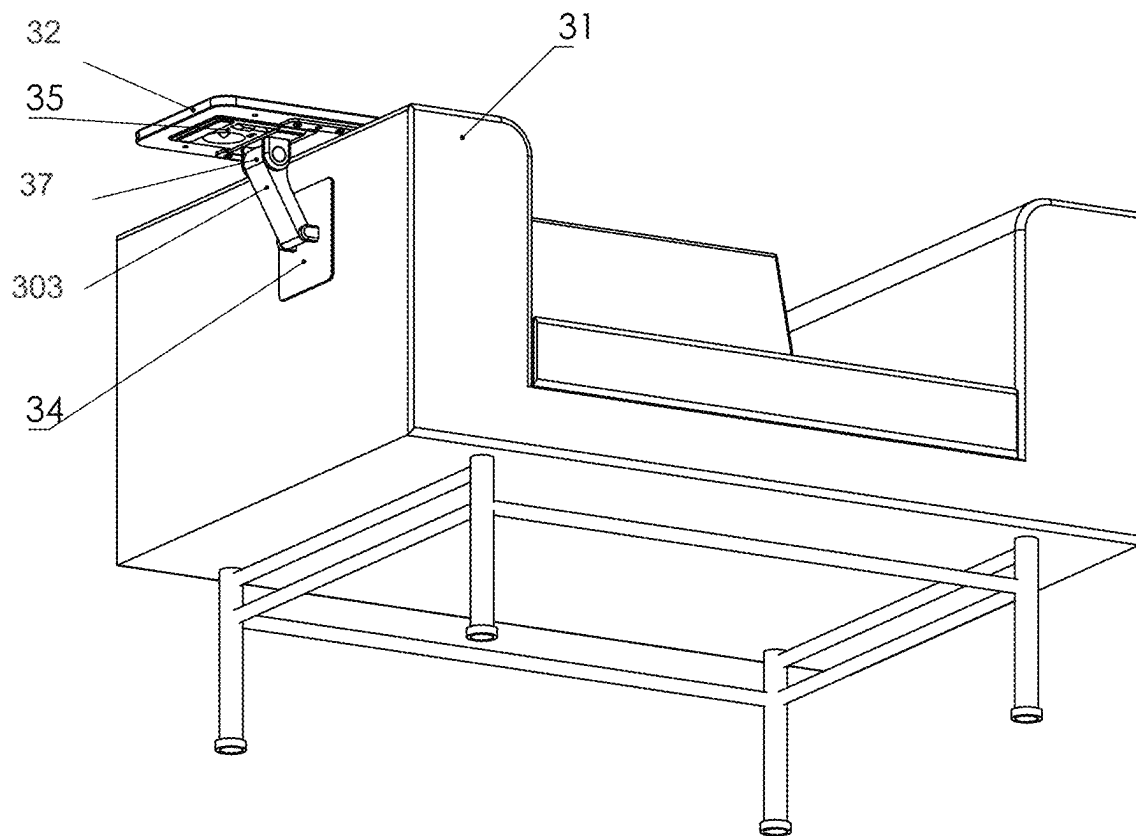


FIG. 88

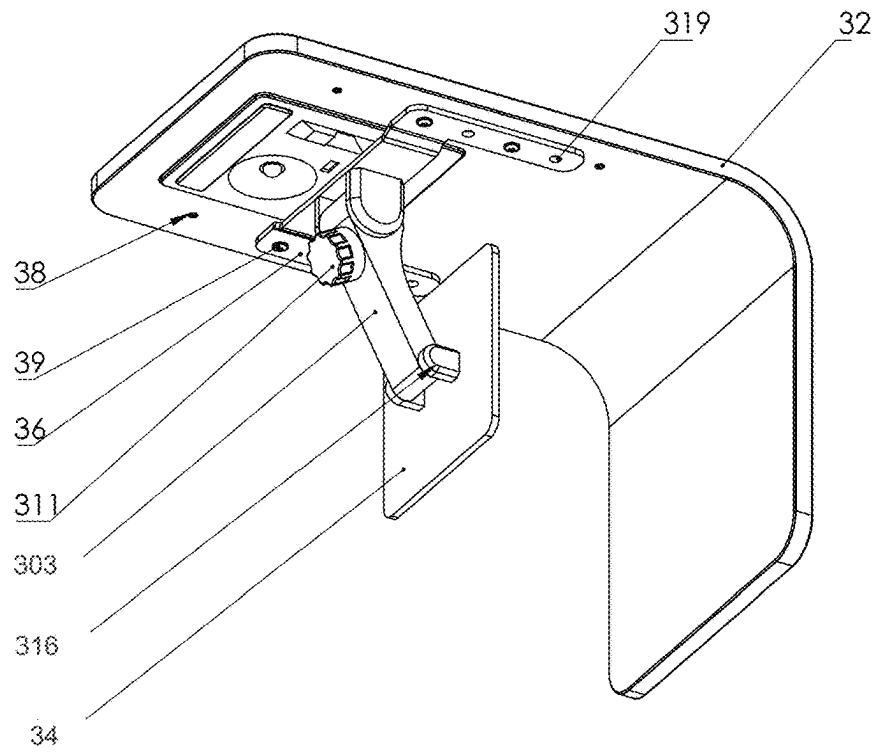


FIG. 89

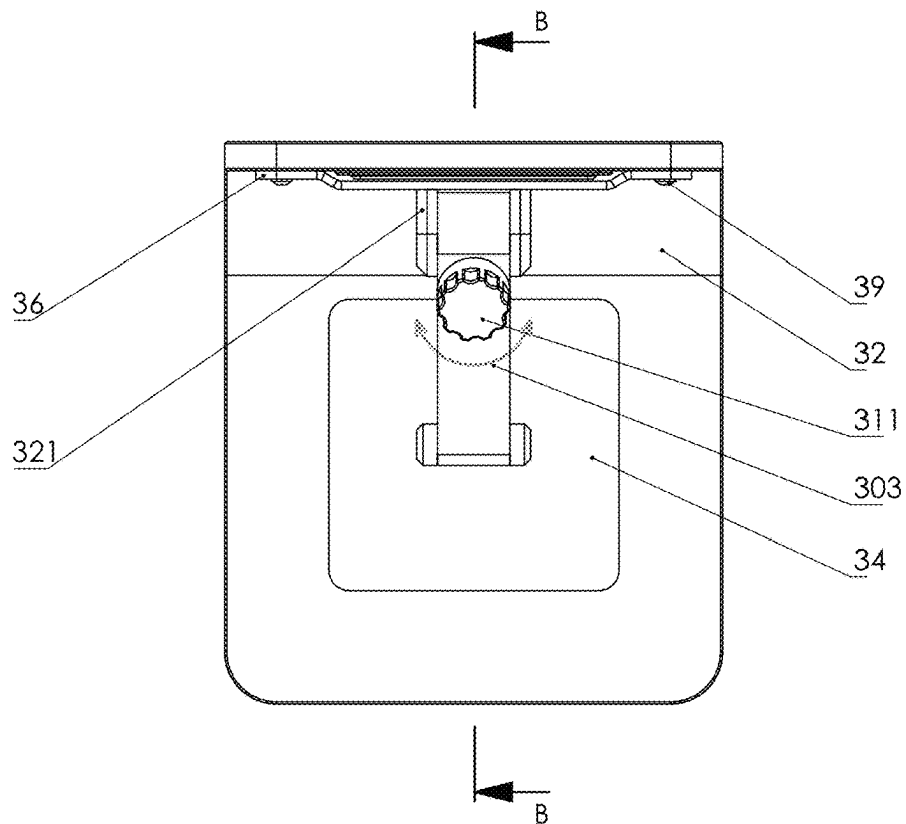
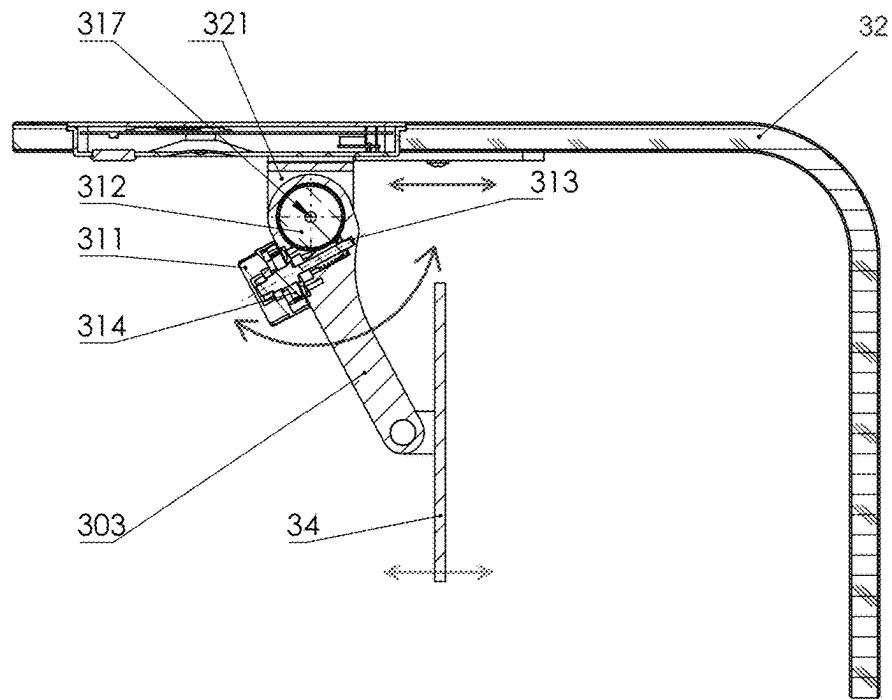


FIG. 90



SECTION B-B

FIG. 91

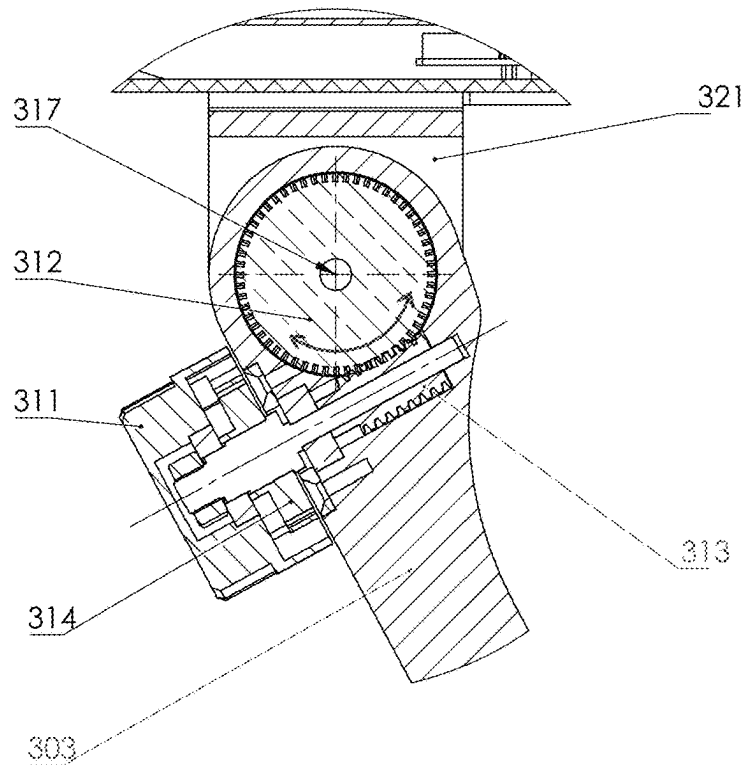


FIG. 92

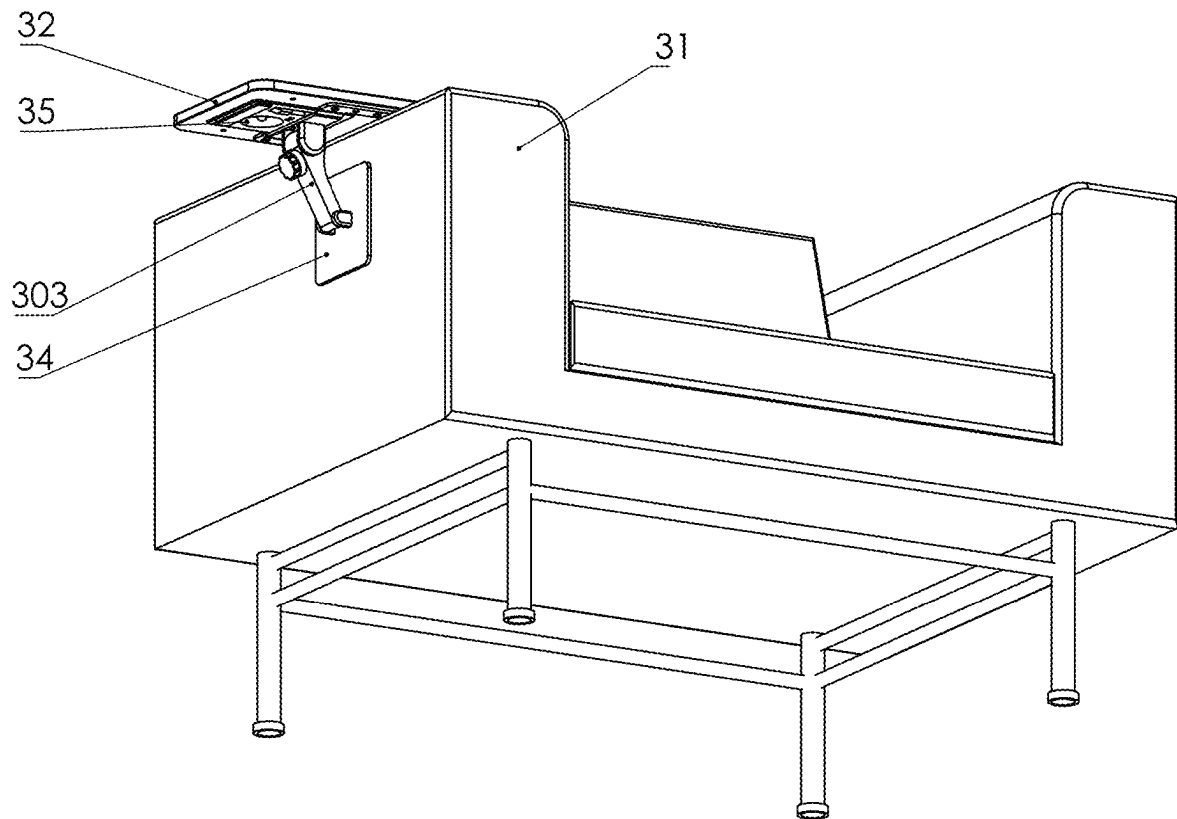


FIG. 93

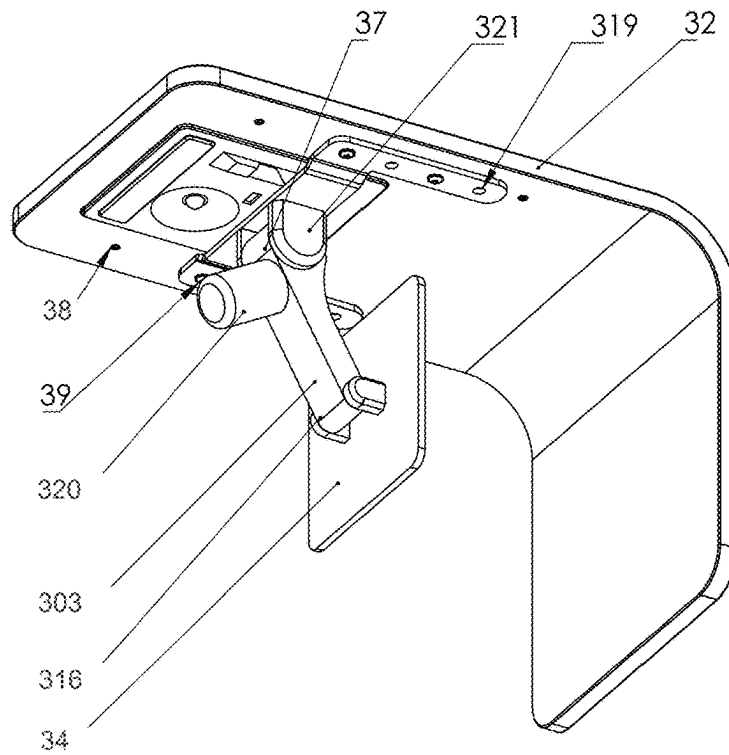


FIG. 94

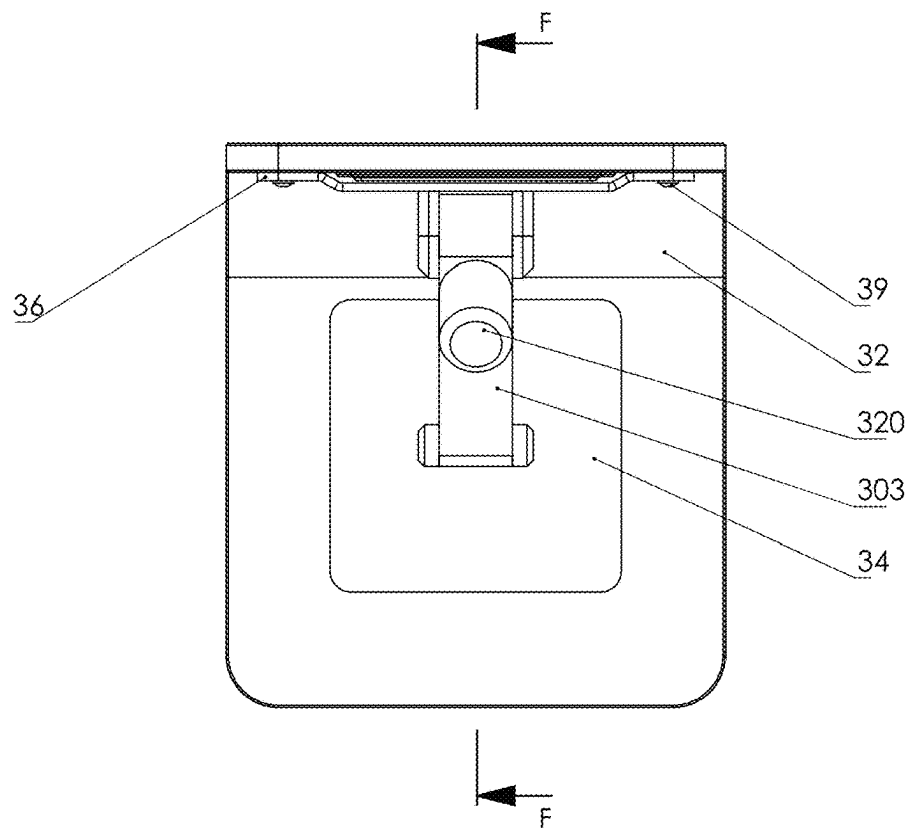


FIG. 95

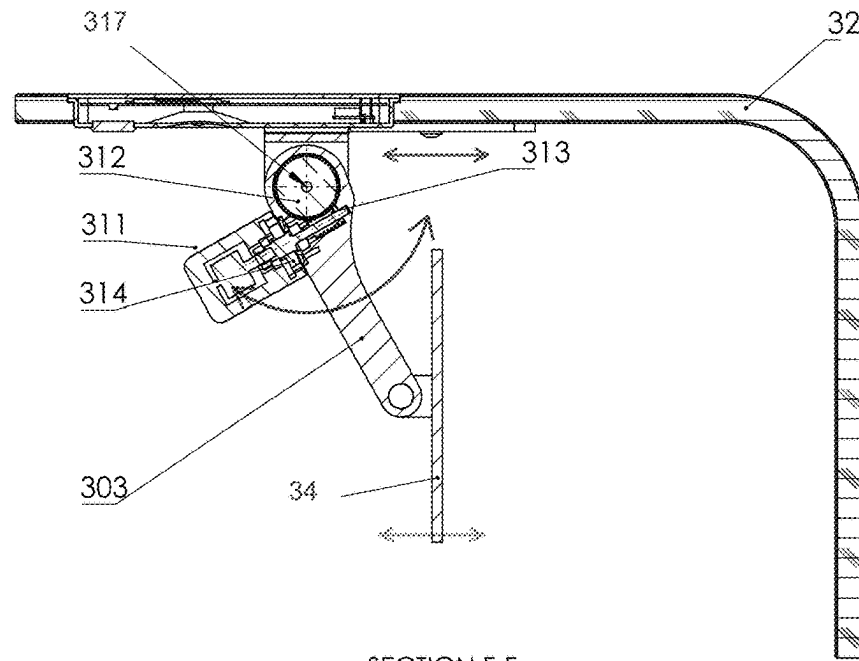


FIG. 96

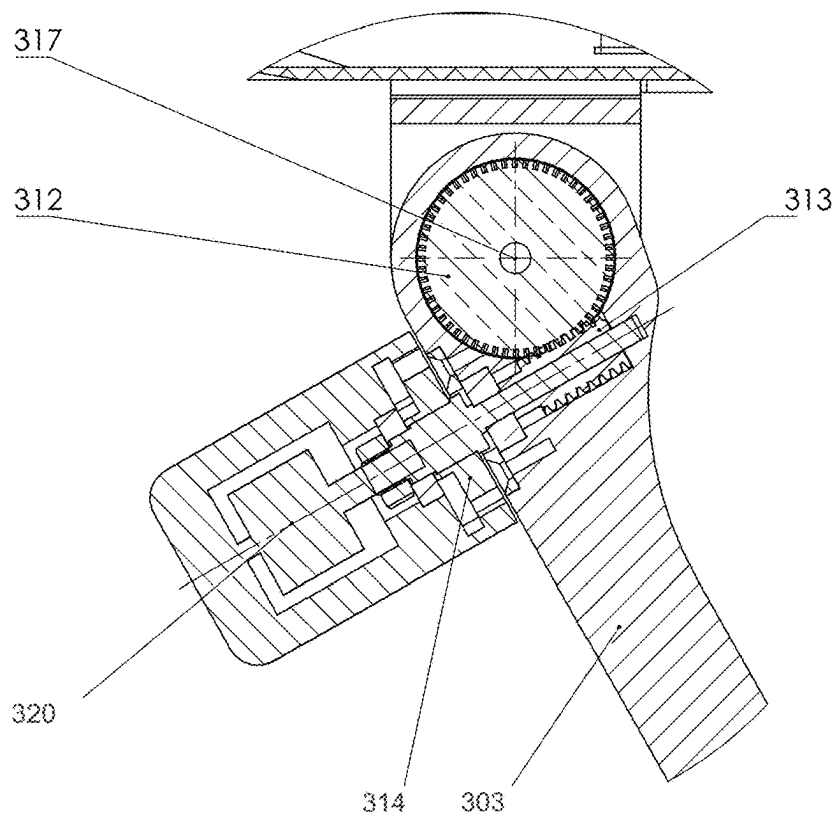


FIG. 97

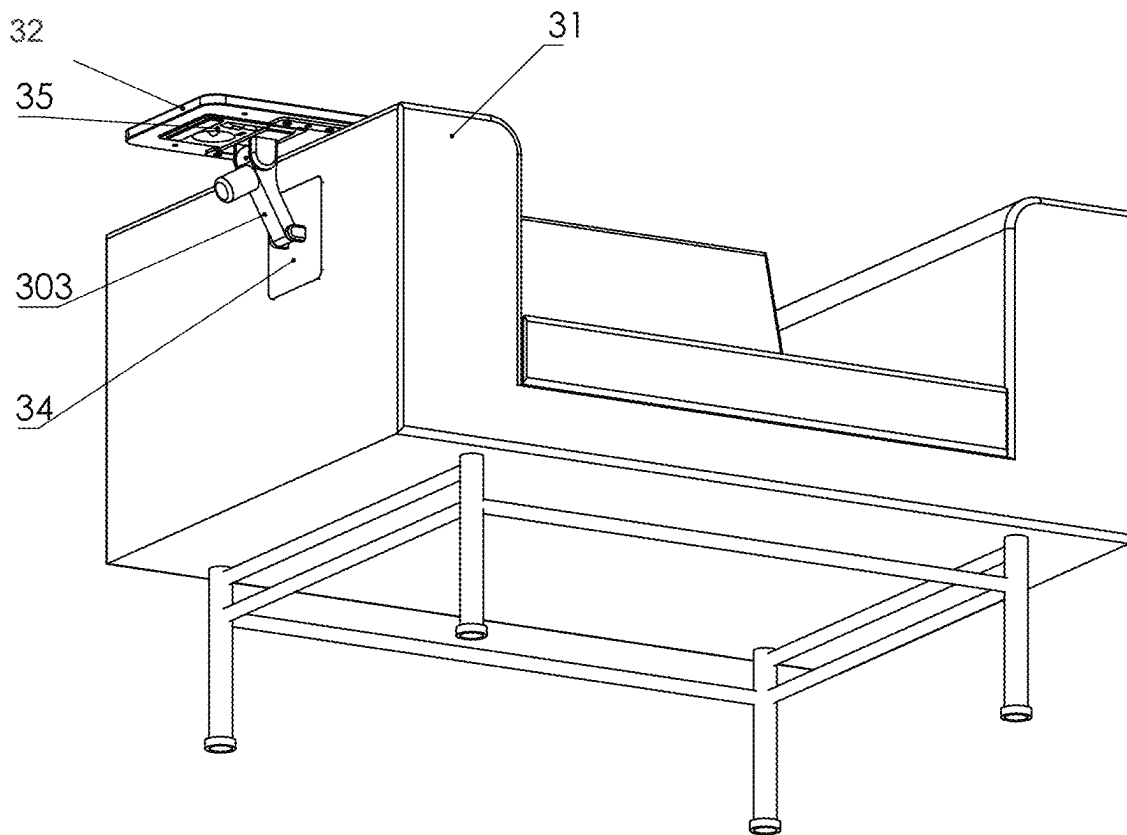


FIG. 98

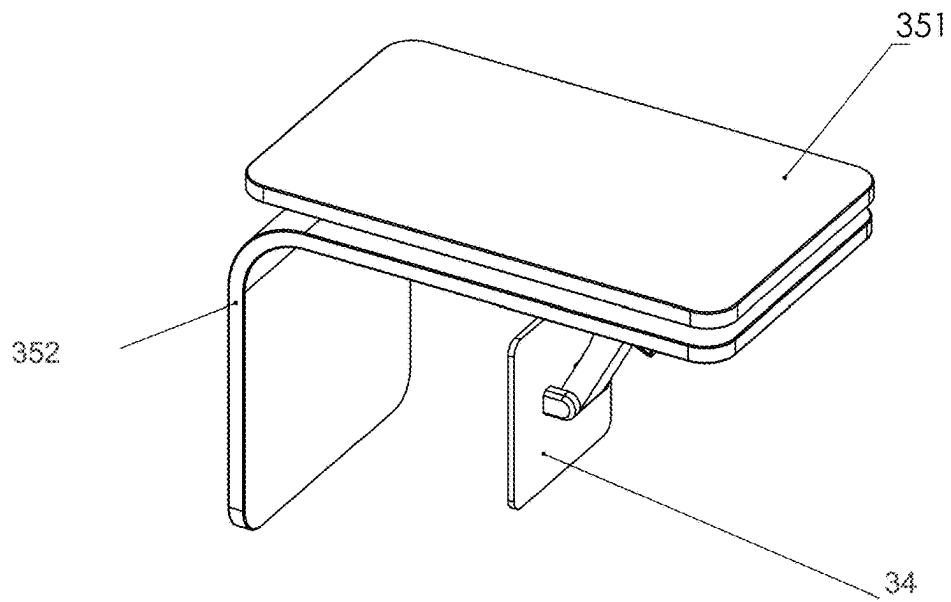


FIG. 99

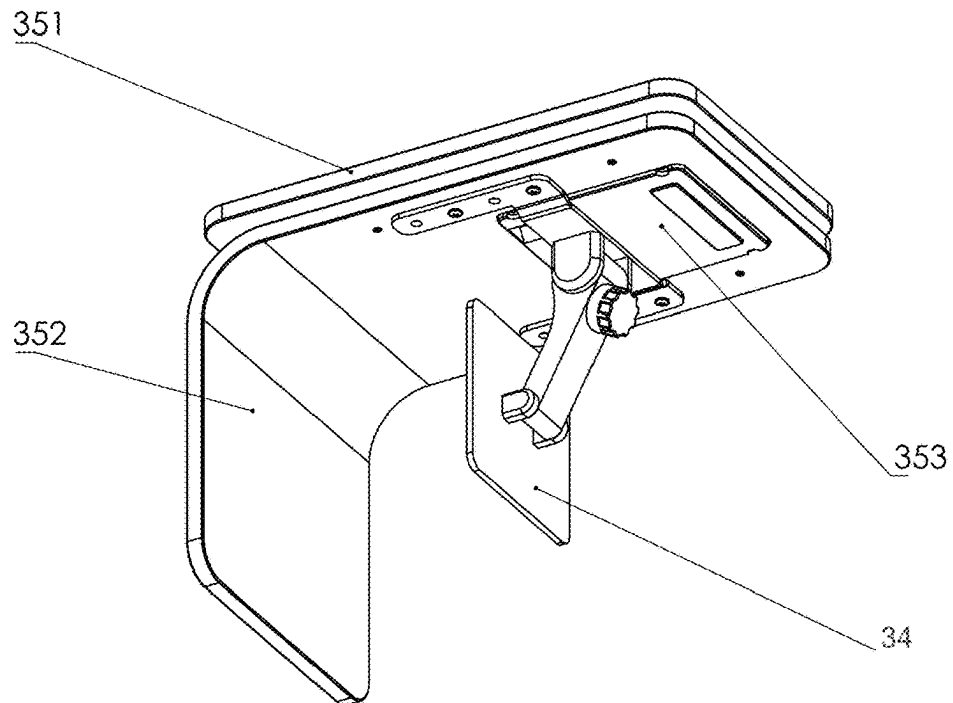


FIG. 100

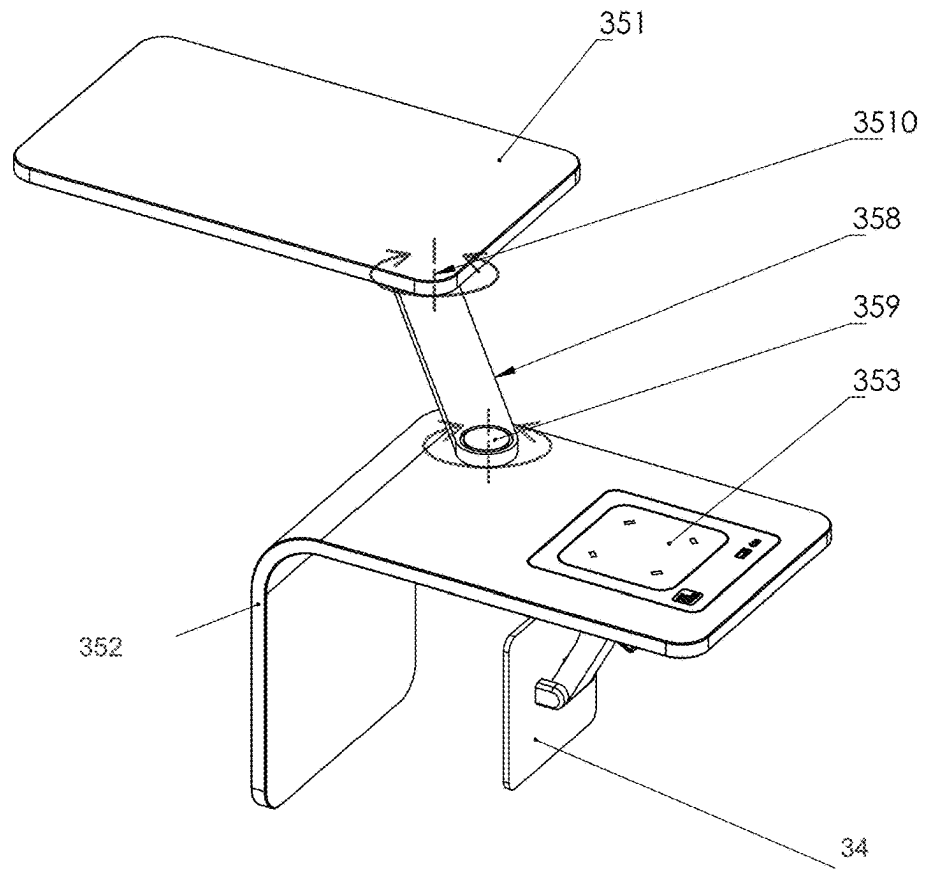


FIG. 101

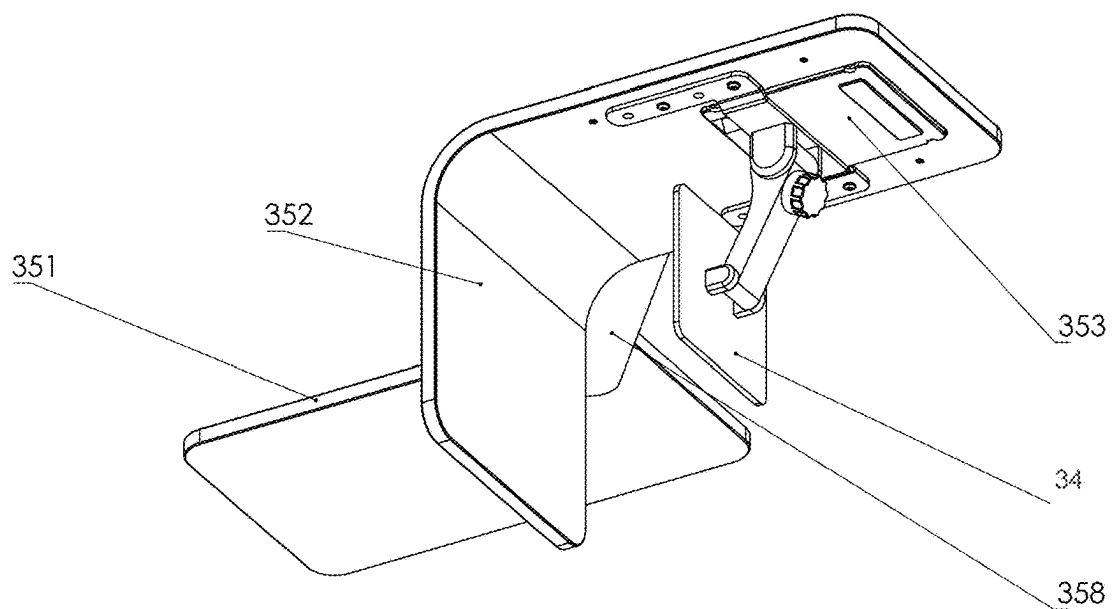


FIG. 102

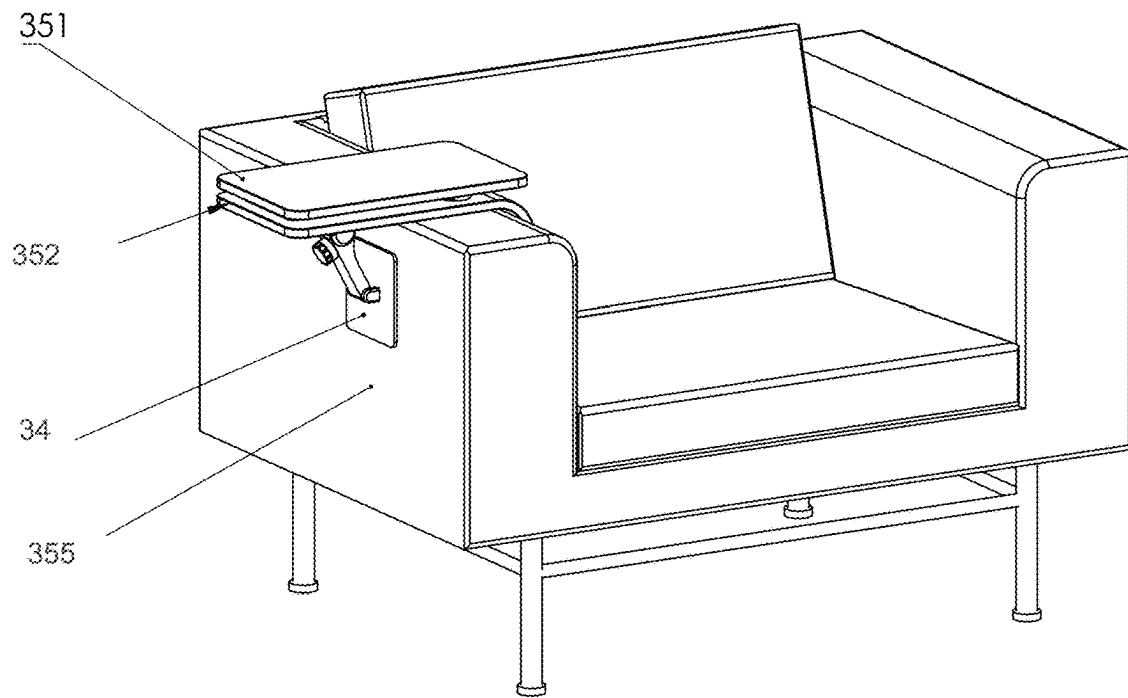


FIG. 103

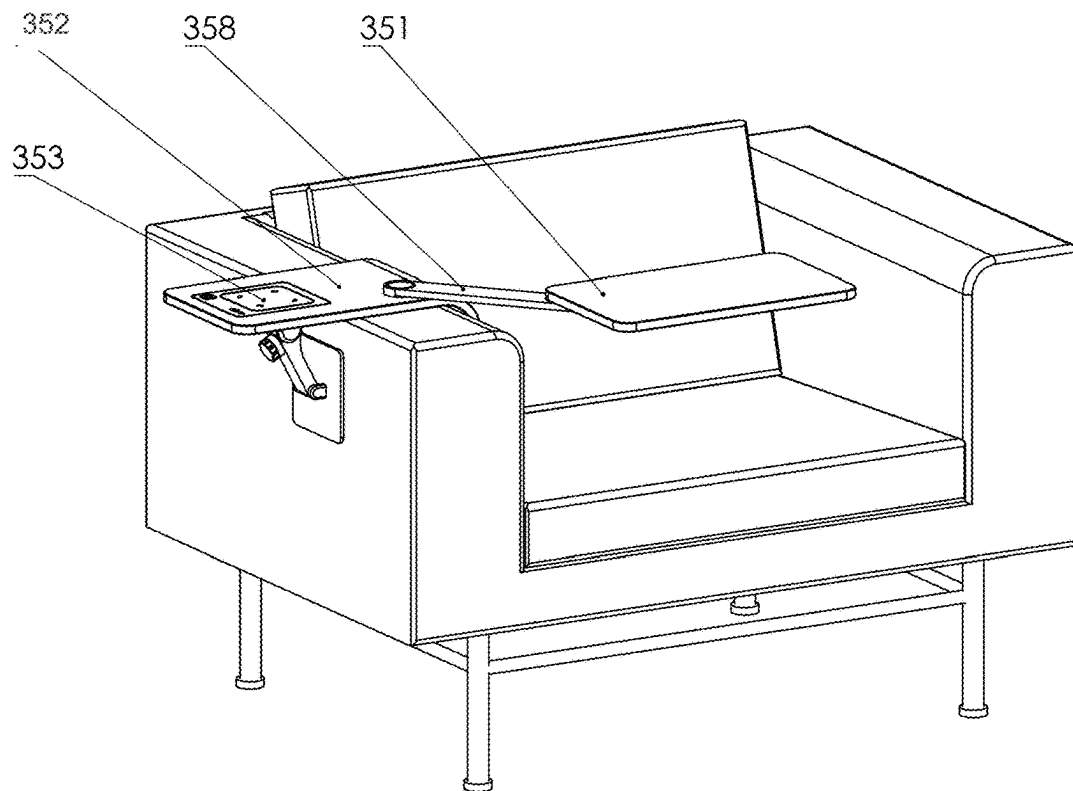
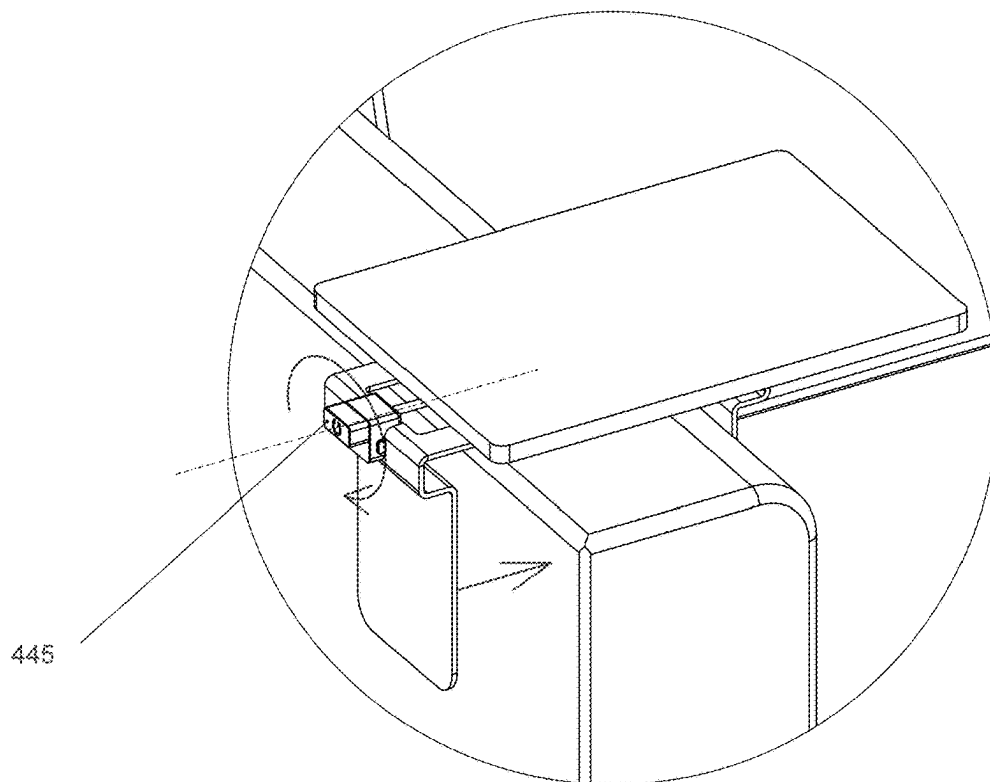
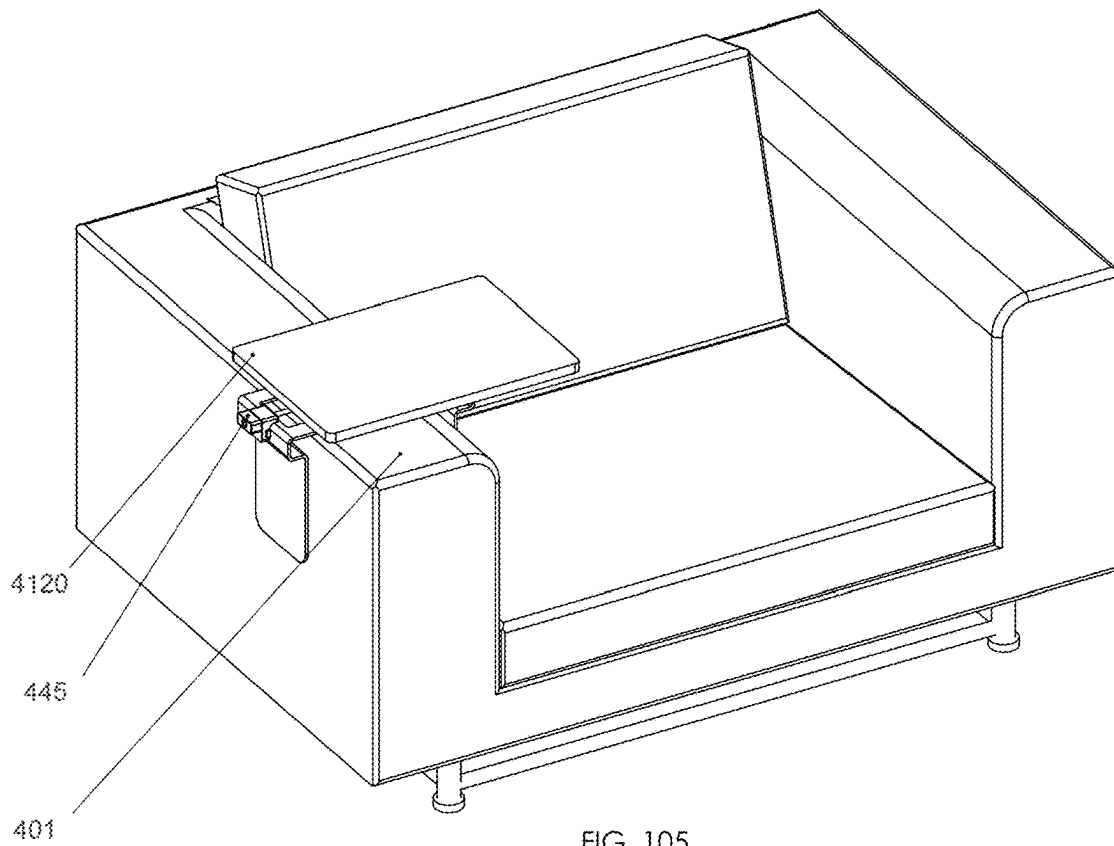


FIG. 104



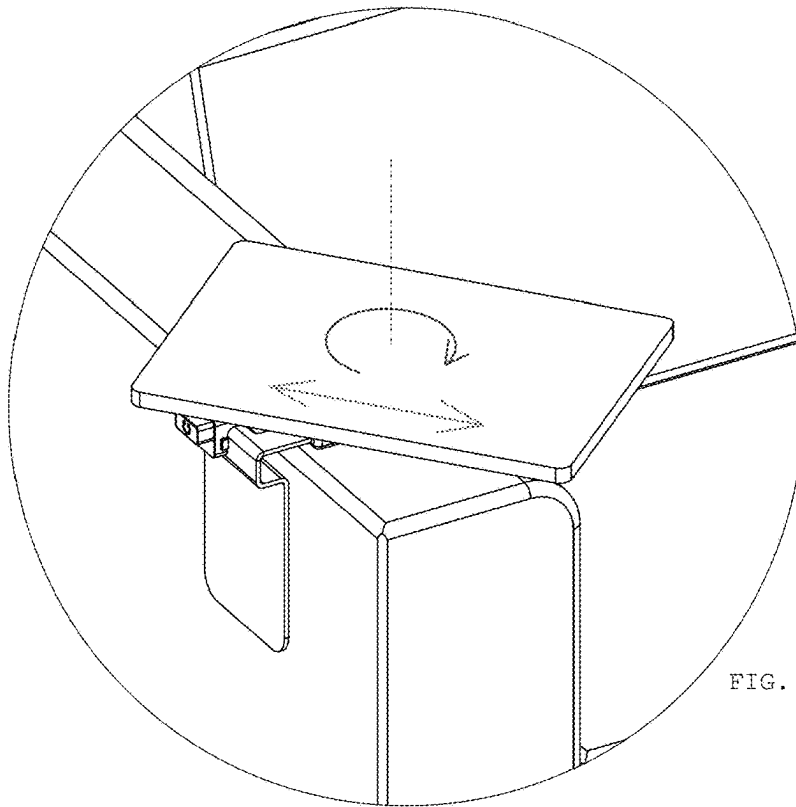


FIG. 107

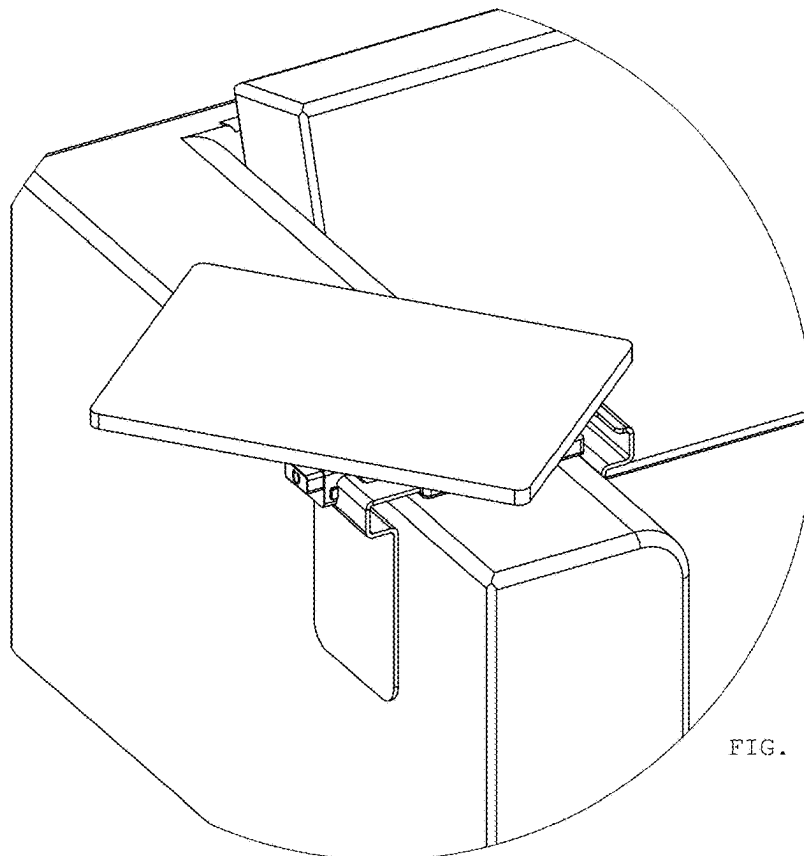


FIG. 108

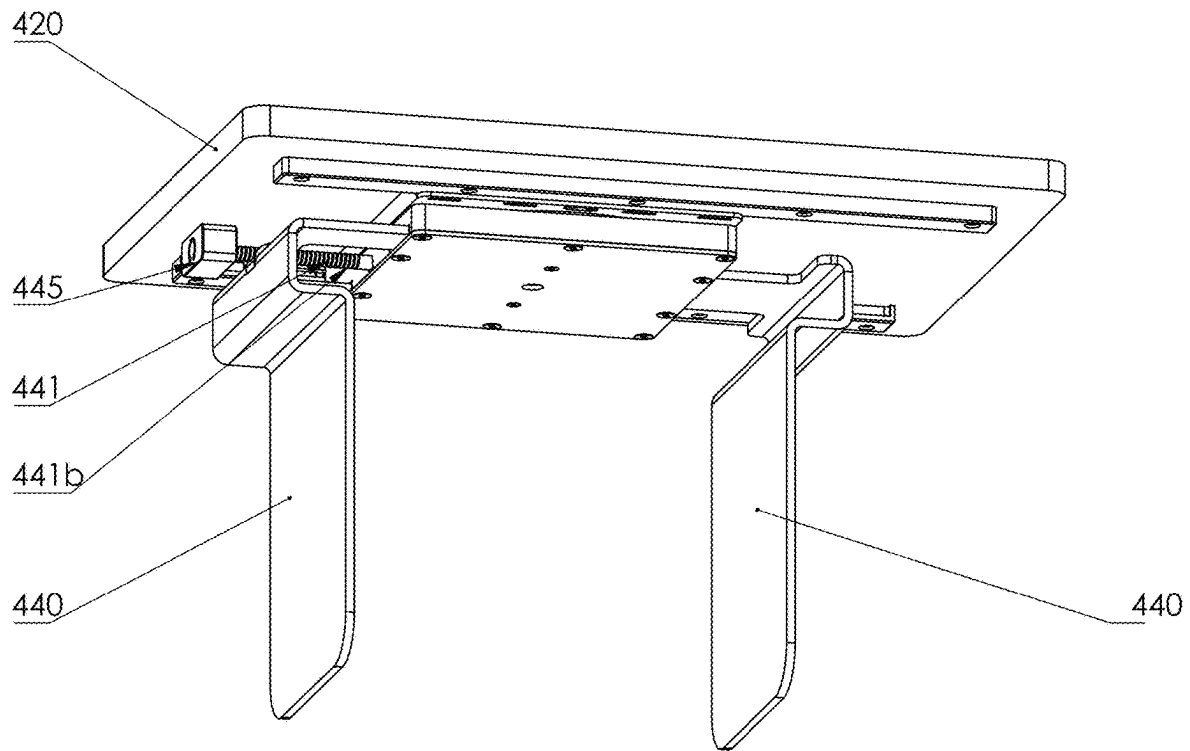


FIG. 109

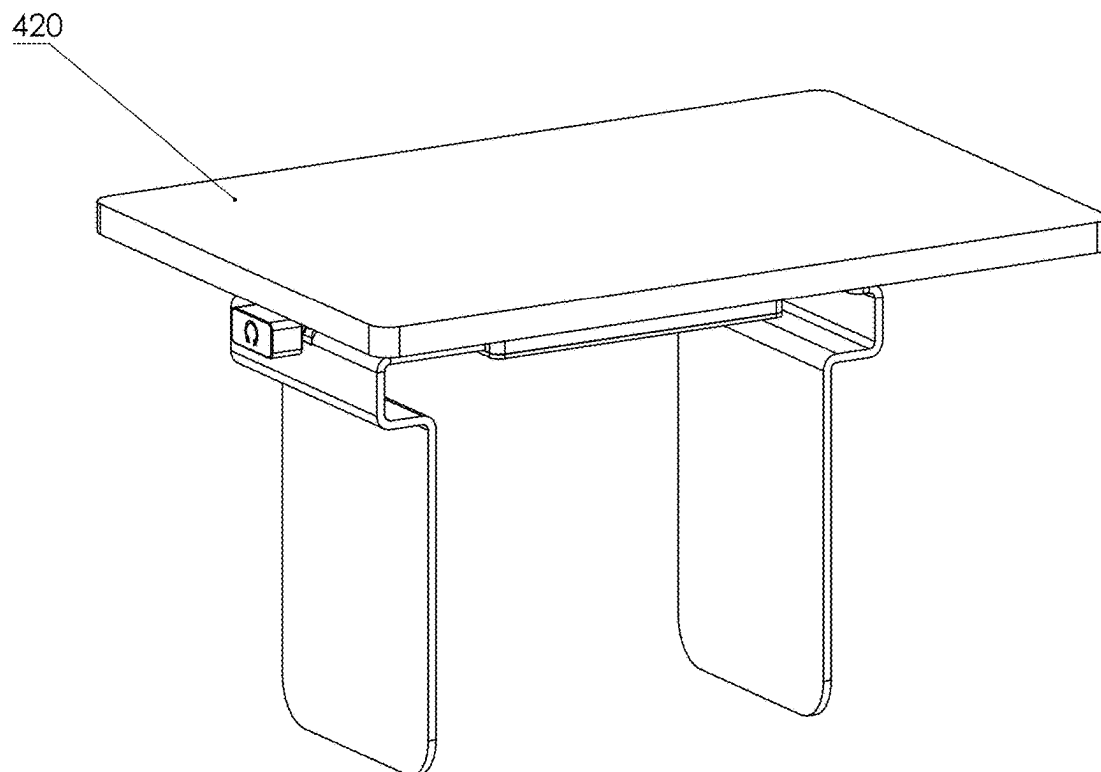


FIG. 110

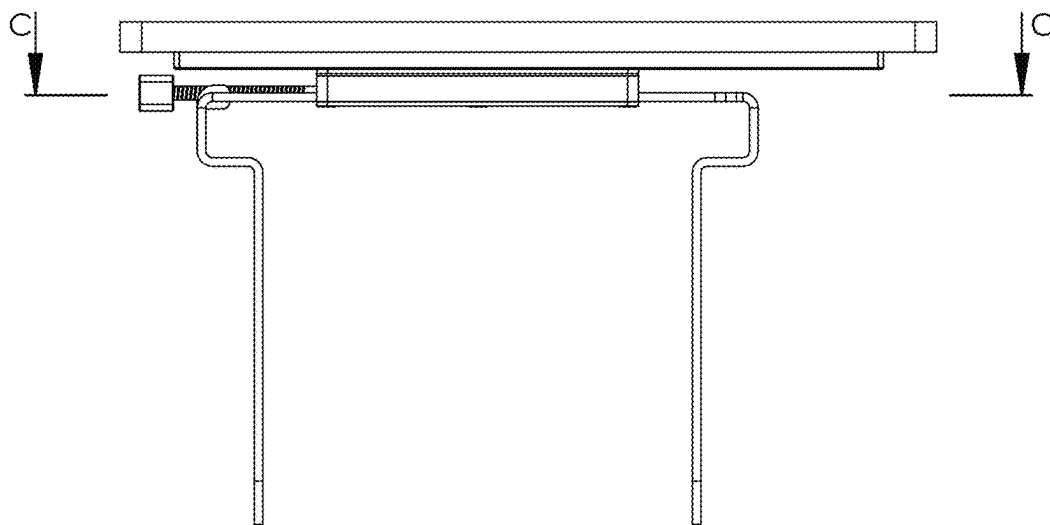
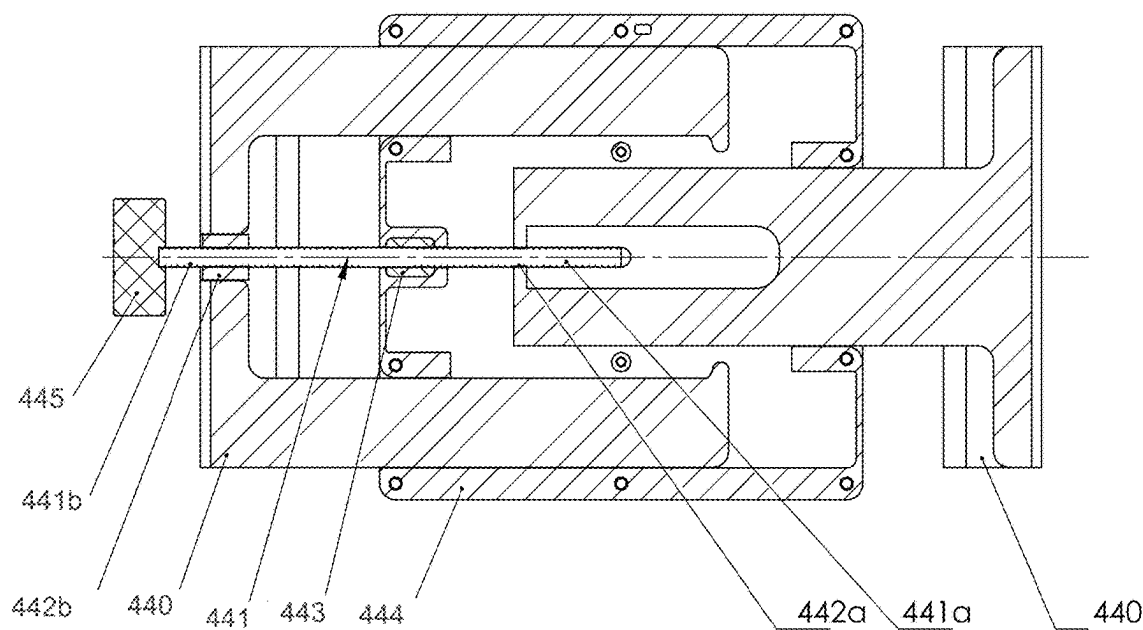


FIG. 111



SECTION C-C

FIG. 112

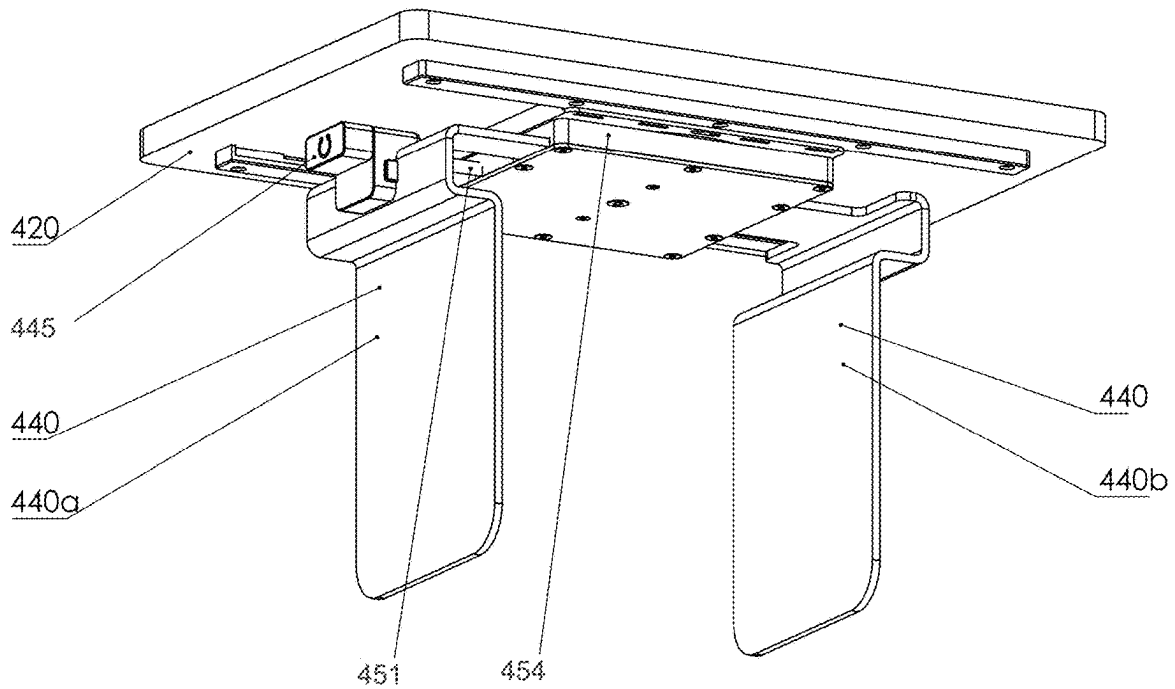


FIG. 113

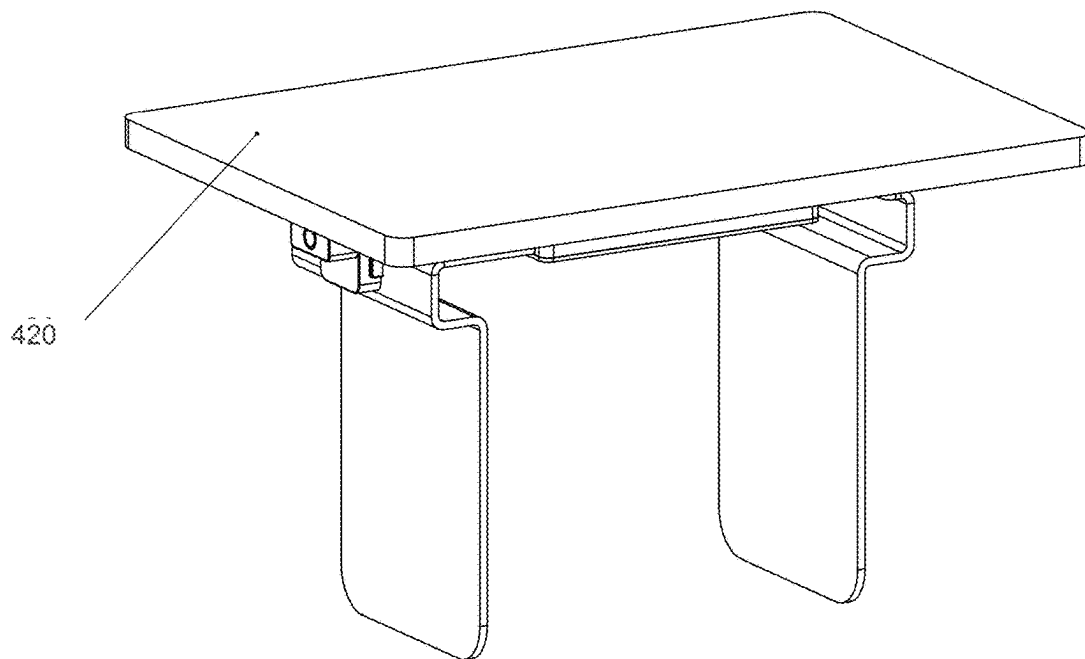


FIG. 114

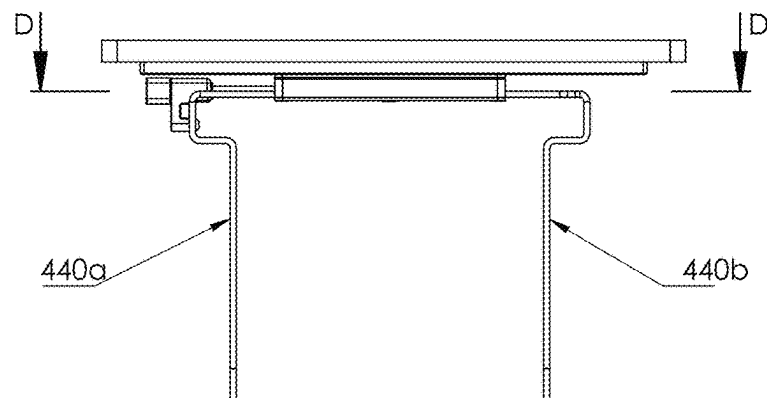
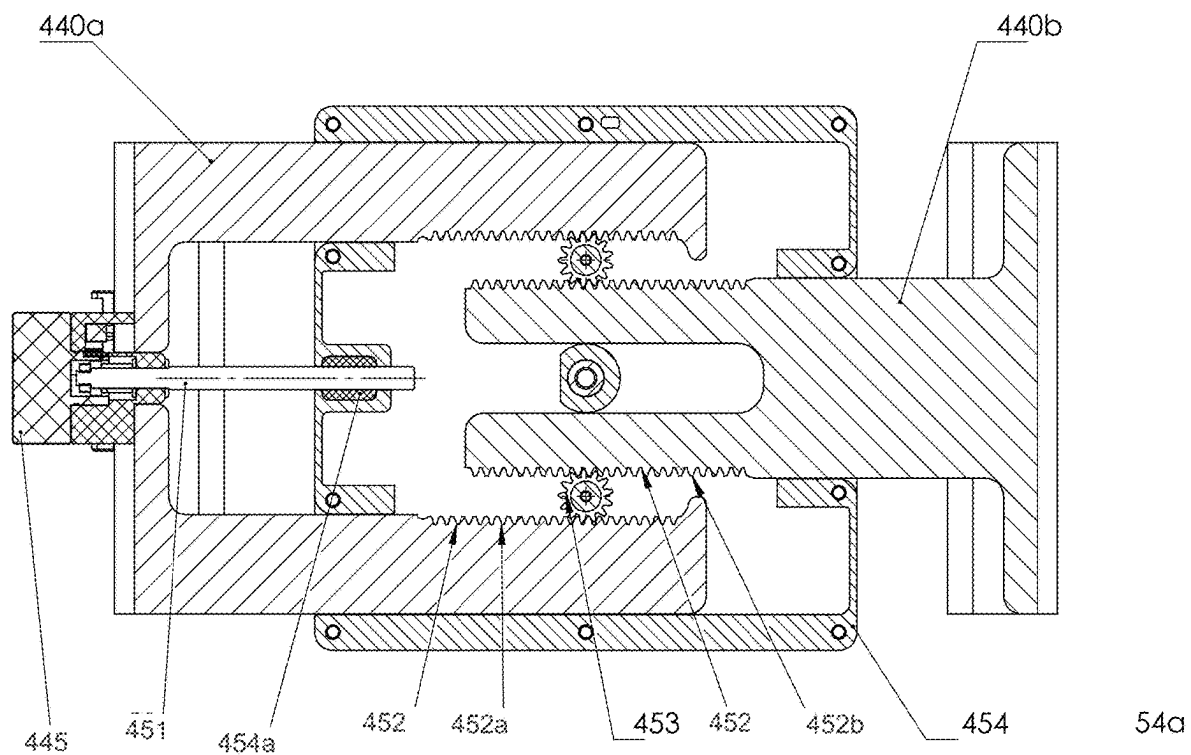


FIG. 115



SECTION D-D

FIG. 116

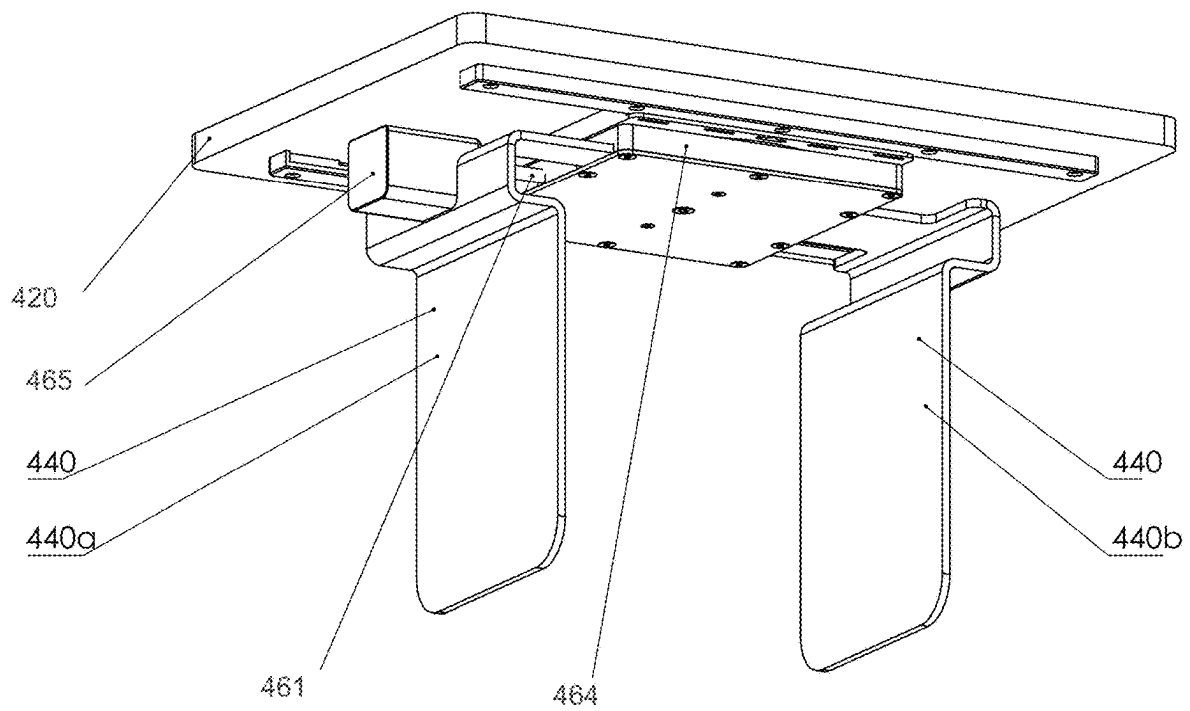


FIG. 117

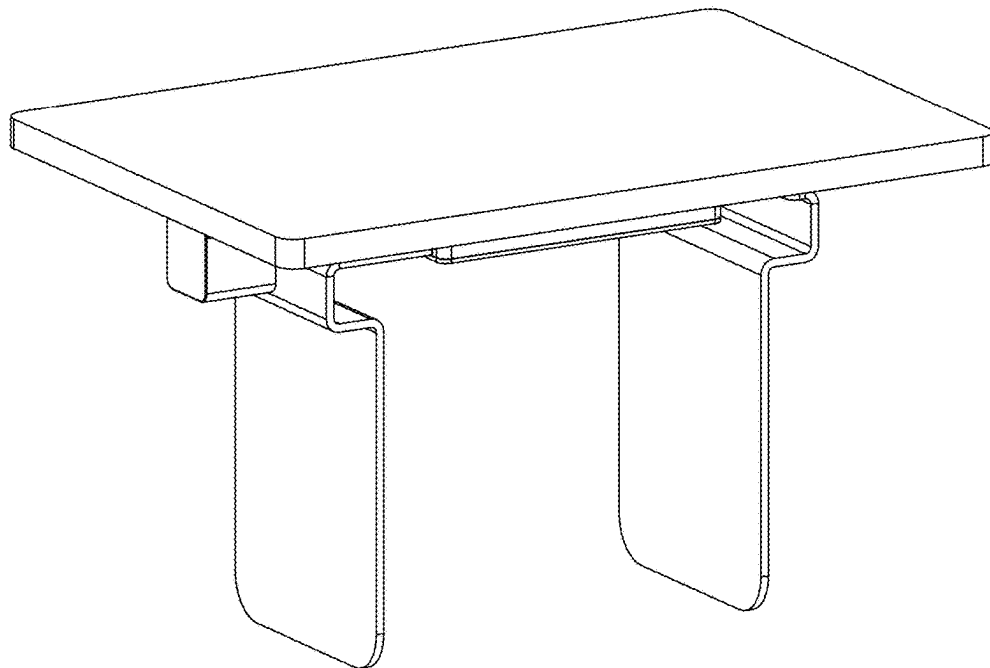


FIG. 118

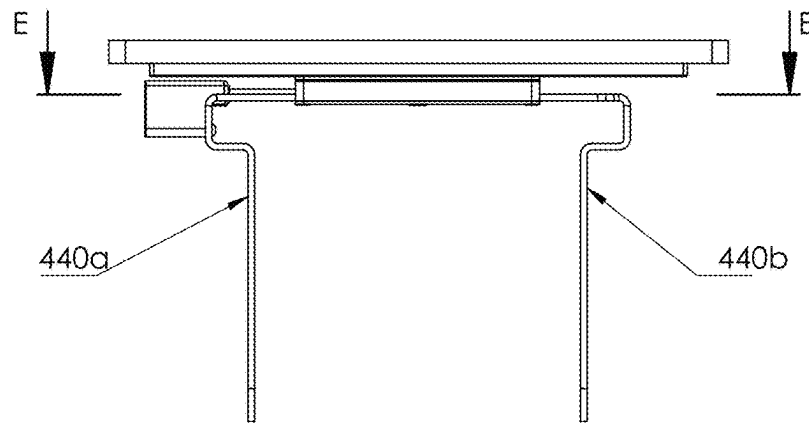
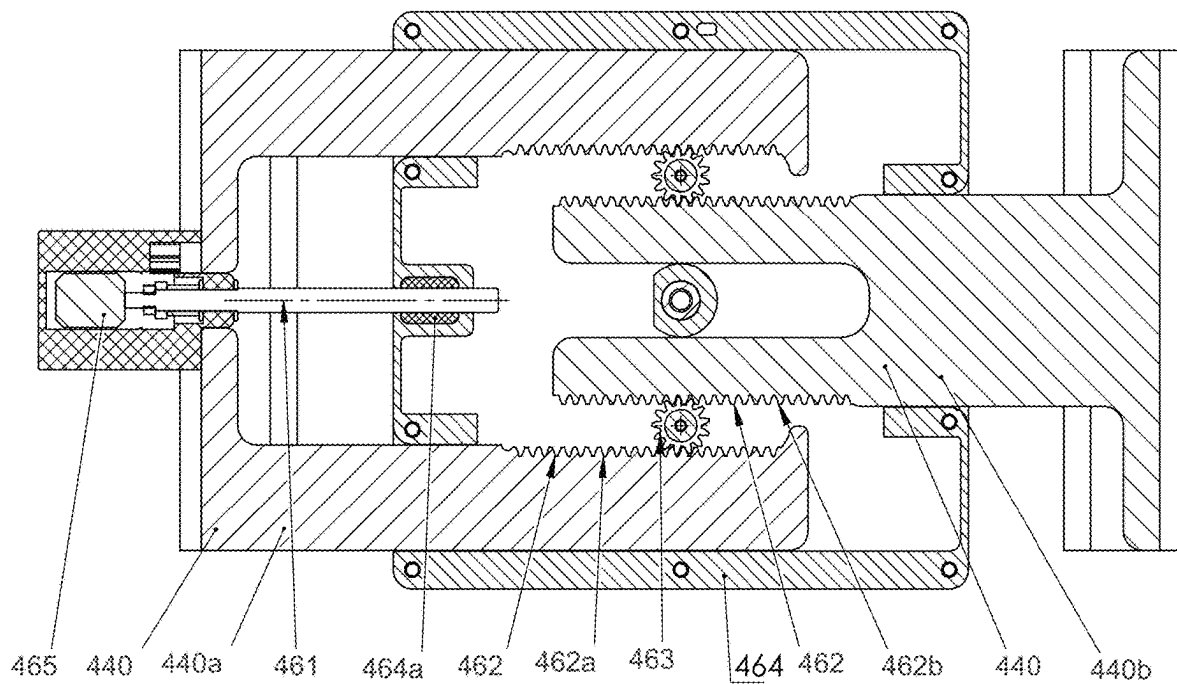


FIG. 119



SECTION E-E

FIG. 120

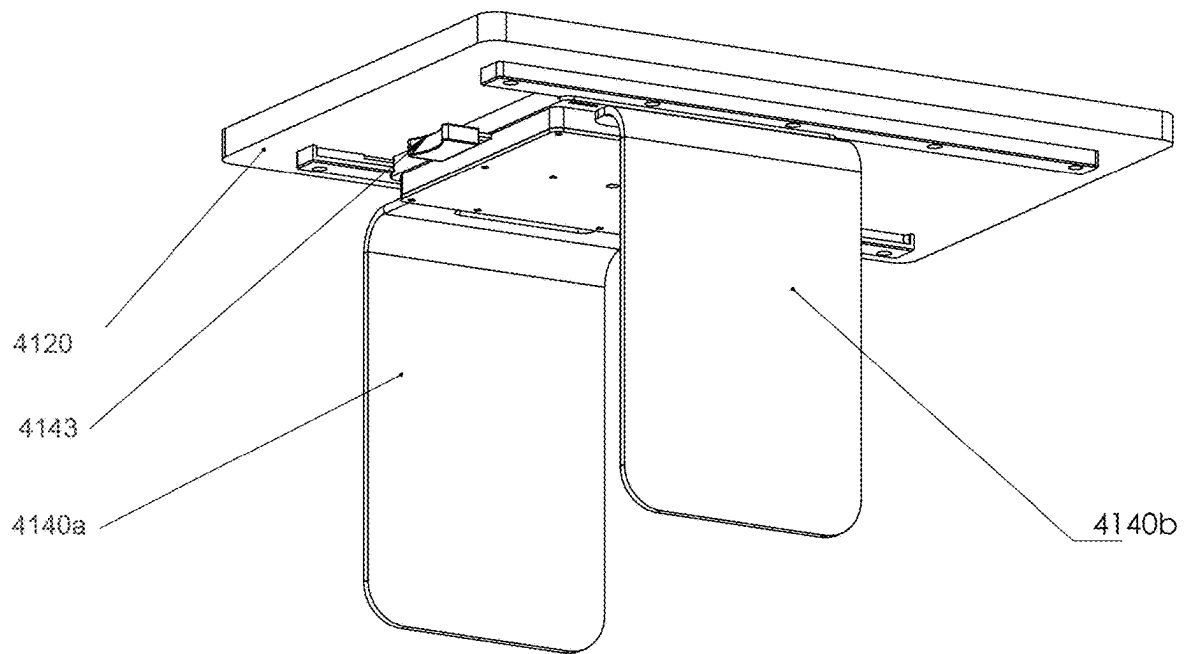


FIG. 121

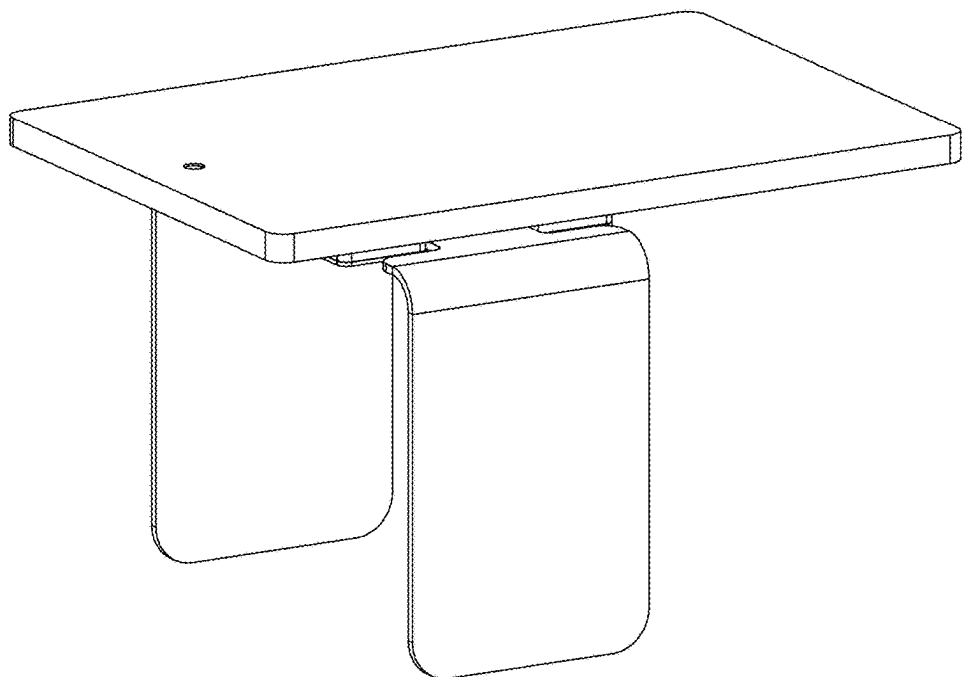


FIG. 122

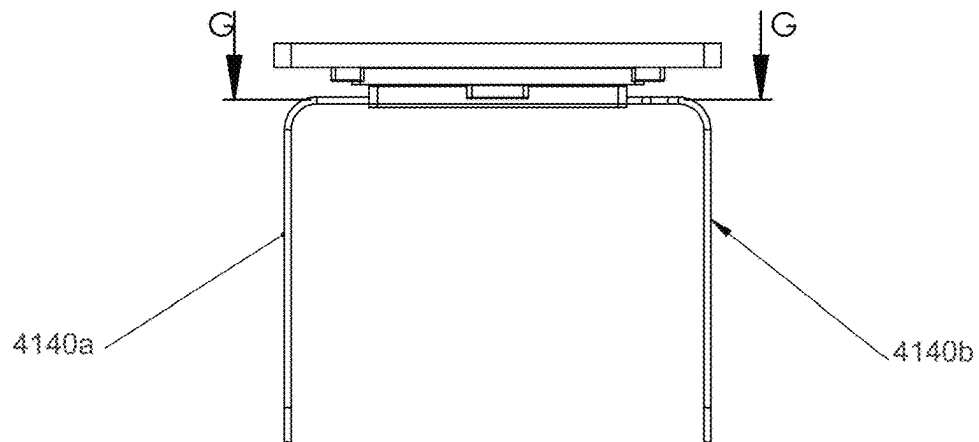
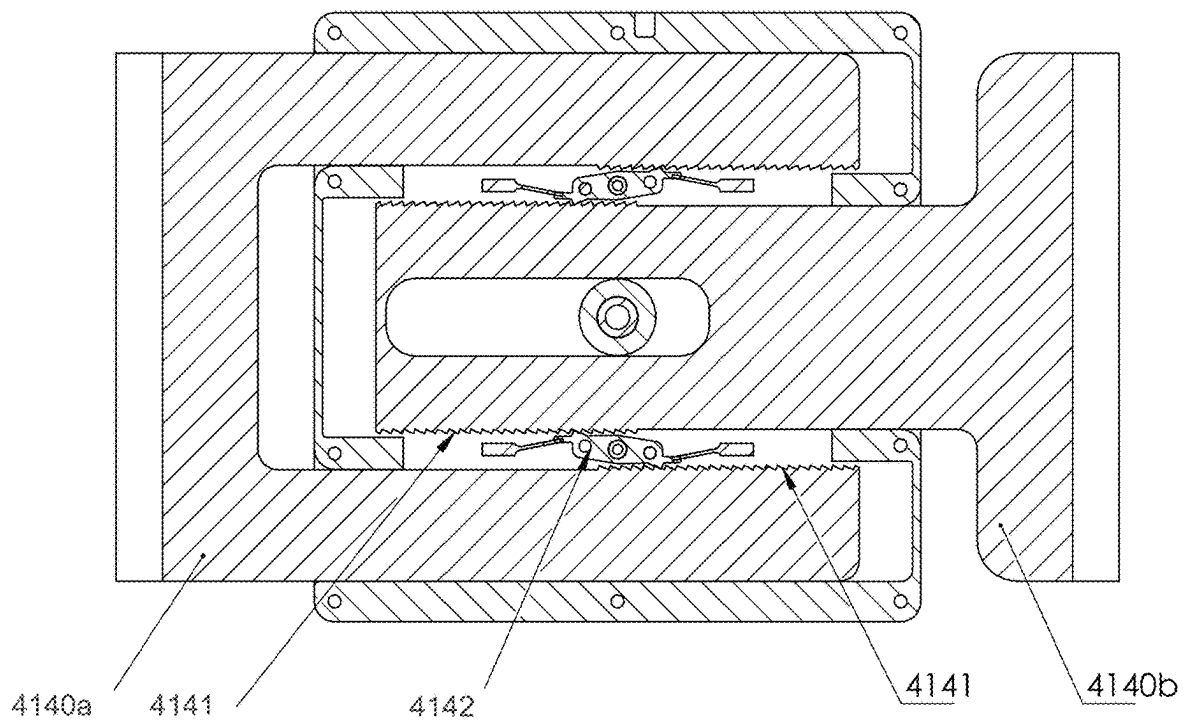


FIG. 123



SECTION G-G

FIG. 124

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AUXILIARY TABLE DEVICE AND METHOD THEREOF**PRIORITY**

This application is a Continuation in Part Application of U.S. application Ser. No. 18/342,278, which was filed in the U.S. Patent and Trademark Office (USPTO) on Jun. 27, 2023, which is a Continuation in Part Application of U.S. application Ser. No. 17/850,445, which was filed in the USPTO on Jun. 27, 2022, claims priority under 35 U.S.C. § 119 to Polish Patent Application No. 438,515, filed in the Polish Patent Office on Jul. 16, 2021, and issued as U.S. Pat. No. 11,689,068 on Jun. 27, 2023, which is a Continuation in Part Application of U.S. application Ser. No. 16/681,487, which was filed in the USPTO on Nov. 12, 2019, issued as U.S. Pat. No. 11,374,417 on Jun. 28, 2022, and claims priority under 35 U.S.C. § 119 to Provisional Application No. 62/758,239, which was filed in the USPTO on Nov. 9, 2018, the entire content of each of which are incorporated herein by reference in their entireties.

BACKGROUND OF THE DISCLOSURE**1. Field of the Disclosure**

The present disclosure relates generally to charging electronic devices.

2. Description of the Related Art

Due to the increasing use of electronic devices (i.e., mobile phones, tablets, and personal computers (PCs)), more and more charging locations are needed. Recently, many electronic devices are capable of wired or wireless charging, however, convenient charging locations may not exist.

Users of electronic devices frequently use their electronic devices while sitting on or around furniture (i.e., sofas). Thus, there is a need for charging locations to be situated on or around furniture, and not simply at a desk or table.

Therefore, there is a need to place a charger on or around a piece of furniture that can be characterized by various forms or different sizes. If the user is not interested in obtaining a piece of furniture with a charger built into it, it may be necessary to use a separate device for charging.

Further, wireless charging is increasing in popularity. Wireless charging is beneficial to users due to convenience. Users can place their electronic device in a wireless charging zone to start charging the device without plugging it, as in the case of wall chargers and other traditional charging devices. However, ordinarily a wireless charging induction coil is stationary. Therefore, wireless chargers may be limited since they require the electronic device to be placed in a specific area for charging. Frequently, the charging area is small and a user must be careful when placing the electronic device in the appropriate charging area.

In addition, traditional chargers may be plugged into wall outlets, and the charging locations are limited by the length of the charging cable extending out from the wall outlet.

Thus, there is a need to manufacture more convenient charging devices that are adaptable to various locations, and can change positions to a resting position when not in use.

SUMMARY

The present disclosure has been made to address the above-mentioned problems and disadvantages, and to provide at least the advantages described below.

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An auxiliary table device is provided and includes a horizontal surface comprising a usable surface; a first surface vertically positioned below the horizontal surface and configured to engage with a first side of a portion of furniture; and a second surface positioned below the horizontal surface and configured to adjustably engage with a second side of the portion of furniture while in a vertical position, wherein at least one edge of the horizontal surface extends beyond an edge of the first or second side of the portion of furniture when the first surface and the second surface are engaged with their respective sides of the furniture.

A method of using an auxiliary table device is provided. The method includes engaging a first surface of the auxiliary table device vertically positioned below a horizontal surface with a first side of a portion of furniture; and adjustably engaging a second surface, positioned below the horizontal surface, with a second side of the portion of furniture while in a vertical position, wherein at least one edge of the horizontal surface, which comprises a usable surface, extends beyond an edge of the first or second side of the portion of furniture when the first surface and the second surface are engaged with their respective sides of the furniture.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects, features and advantages of the present disclosure will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings in which:

FIG. 1 shows a diagram illustrating a top perspective view of a charger shelf movable along X, Y, and Z axes, according to an embodiment of the present disclosure;

FIG. 2 shows a diagram illustrating a profile view of a charger shelf movable along X, Y, and Z axes showing angles or rotations α , β , and γ , respectively, according to an embodiment of the present disclosure;

FIG. 3 shows a diagram illustrating a top perspective view of a charger shelf with a charging/energy transfer zone, according to an embodiment of the present disclosure;

FIG. 4 shows a diagram illustrating a top perspective view of a charger shelf with multiple stationary charging/energy transfer zones, according to an embodiment of the present disclosure;

FIG. 5 shows a diagram illustrating a top perspective view of a charger shelf with a movable charging/energy transfer zone, according to an embodiment of the present disclosure;

FIG. 6 shows a diagram illustrating a top perspective view of a charger shelf with an internal drive assembly, according to an embodiment of the present disclosure;

FIG. 7 shows a diagram illustrating a top view of a mobile coil module capable of locating an electronic device to be charged, according to an embodiment of the present disclosure;

FIG. 8 shows a diagram illustrating a top perspective view of a charger shelf capable of tilting in an operating position, according to an embodiment of the present disclosure;

FIG. 9 shows a diagram illustrating a top perspective view of a charger shelf capable of tilting in a resting position, according to an embodiment of the present disclosure;

FIG. 10 shows a diagram illustrating a top perspective view of a charger shelf capable of tilting in an operating position, according to an embodiment of the present disclosure;

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FIG. 11 shows a diagram illustrating a profile view of a charger shelf capable of tilting in an operating position, according to an embodiment of the present disclosure;

FIG. 12 shows a diagram illustrating a top perspective view of a charger shelf capable of tilting in an angled tilted position, according to an embodiment of the present disclosure;

FIG. 13 shows a diagram illustrating a profile view of a charger shelf capable of tilting in an angled tilted position, according to an embodiment of the present disclosure;

FIG. 14 shows a diagram illustrating a top perspective view of a charger shelf capable of tilting in an angled tilted position located next to a seat, according to an embodiment of the present disclosure;

FIG. 15 shows a diagram illustrating a profile view of a charger shelf capable of tilting in an angled tilted position, according to an embodiment of the present disclosure;

FIG. 16 shows a diagram illustrating a top perspective view of a charger shelf capable of moving vertically along a z axis, according to an embodiment of the present disclosure;

FIG. 17 shows a profile view of a charger shelf capable of moving vertically along a z axis, according to an embodiment of the present disclosure;

FIG. 18 shows a diagram illustrating a top perspective view of a charger shelf capable of rotating around a vertical axis, according to an embodiment of the present disclosure;

FIG. 19 shows a bird eye view of a charger shelf capable of rotating around a vertical axis, according to an embodiment of the present disclosure;

FIG. 20 shows a diagram illustrating a top perspective view of a charger shelf capable of rotating along 3 axes and positioned next to a seat, according to an embodiment of the present disclosure;

FIG. 21 shows a diagram illustrating a profile view of a charger shelf capable of rotating along a vertical axis and a horizontal axis, according to an embodiment of the present disclosure;

FIG. 22 shows a diagram illustrating a top perspective view of a charger shelf capable of sliding along an x axis, according to an embodiment of the present disclosure;

FIG. 23 shows a diagram illustrating a bottom perspective view of a charger shelf capable of sliding along an x axis, according to an embodiment of the present disclosure;

FIG. 24 shows a diagram illustrating a top perspective view of a charger shelf mounted with a magnetic holder, according to an embodiment of the present disclosure;

FIG. 25 shows a diagram illustrating a profile view of a charger shelf mounted with a magnetic holder, according to an embodiment of the present disclosure;

FIG. 26 shows a diagram illustrating a bottom perspective view of a charger shelf with a battery, according to an embodiment of the present disclosure;

FIG. 27 shows a diagram illustrating a top perspective view of a charger shelf with photo-voltaic solar cells, according to an embodiment of the present disclosure;

FIG. 28 shows a diagram illustrating a top perspective view of a charger shelf charging a smartphone, according to an embodiment of the present disclosure;

FIG. 29 shows a diagram illustrating a top perspective view of a charger shelf charging a smartwatch, according to an embodiment of the present disclosure;

FIG. 30 shows a diagram illustrating a top perspective view of a charger shelf charging a laptop computer, according to an embodiment of the present disclosure;

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FIG. 31 shows a diagram illustrating a top perspective view of a charger shelf with a display monitor, according to an embodiment of the present disclosure;

FIG. 32 shows a diagram illustrating a top perspective view of a charger shelf powering a heating device, according to an embodiment of the present disclosure;

FIG. 33 shows a diagram illustrating a top perspective view of a charger shelf powering an air conditioning device, according to an embodiment of the present disclosure;

FIG. 34 shows a diagram illustrating a top perspective view of a charger shelf charging a climate control device, according to an embodiment of the present disclosure;

FIG. 35 shows a diagram illustrating a top perspective view of a charger shelf charging an inductively powered appliance, according to an embodiment of the present disclosure;

FIG. 36 shows a diagram illustrating a top perspective view of a charger shelf including perimeter illumination, according to an embodiment of the present disclosure;

FIG. 37 shows a diagram illustrating a top perspective view of a charger shelf including illumination directed in an upward direction, according to an embodiment of the present disclosure;

FIG. 38 shows a diagram illustrating a top perspective view of a charger shelf including illumination outwardly rotatable from a side of the panel device, according to an embodiment of the present disclosure;

FIG. 39 shows a diagram illustrating a top perspective view of a charger shelf including multiple illumination modes with a motion detector, according to an embodiment of the present disclosure;

FIG. 40 shows a diagram illustrating a configuration of a wireless charger, according to an embodiment of the present disclosure;

FIG. 41 shows a diagram illustrating a sofa armrest with a wireless charger embedded therein, according to an embodiment of the present disclosure;

FIG. 42 shows a diagram illustrating a transparent view of a configuration of a wireless charger embedded in a sofa armrest, according to an embodiment of the present disclosure;

FIG. 43 shows a diagram illustrating accessing an interior of a sofa armrest via a service access point, according to an embodiment of the present disclosure;

FIG. 44 shows a diagram illustrating a sofa cushion with a wireless charger embedded therein, according to an embodiment of the present disclosure;

FIG. 45 shows a diagram illustrating a configuration of a charger, according to an embodiment of the present disclosure;

FIG. 46 shows a diagram illustrating a configuration of a charger embedded in a sofa cushion with a USB extension, according to an embodiment of the present disclosure;

FIG. 47 shows a diagram illustrating a configuration of a wireless charger and a configuration of a wired ball USB charger, according to an embodiment of the present disclosure;

FIG. 48 shows a diagram illustrating a configuration of a charger embedded in a middle section with additional power outlets and a wired ball USB splitter, according to an embodiment of the present disclosure;

FIG. 49 shows a diagram illustrating an interior configuration of a charger embedded in a middle section with additional power outlets and a wired ball USB splitter, according to an embodiment of the present disclosure;

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FIG. 50 shows a diagram illustrating a wired ball USB splitter, according to an embodiment of the present disclosure;

FIG. 51 shows a diagram illustrating a configuration of a wireless charger, according to an embodiment of the present disclosure;

FIG. 52 shows a diagram illustrating a configuration of a wireless charging pad placed in a drink holder of an armrest, according to an embodiment of the present disclosure;

FIG. 53 shows a diagram illustrating a configuration of a wireless charger with a USB port for wired charging, according to an embodiment of the present disclosure;

FIG. 54 shows a diagram illustrating a sofa cushion with a wireless charger positioned on a flip-down portion of a cushion, according to an embodiment of the present disclosure;

FIG. 55 shows a diagram illustrating a sofa armrest with a wireless charger with a USB port, according to an embodiment of the present disclosure;

FIG. 56 shows a perspective view of a charging system through a furniture panel that is non-transparent to visible and infrared light, according to an embodiment of the present disclosure;

FIG. 57 shows an exploded perspective view of a charging system with a charging module, according to an embodiment of the present disclosure;

FIG. 58 shows an exploded perspective view of a loading system with a loading module shown, according to an embodiment of the present disclosure;

FIG. 59 shows a perspective view of a charging system that is performing charging through a furniture panel with details of the charging displayed on the energy receiver, according to an embodiment of the present disclosure;

FIG. 60 shows a view illustrating a function of a positioning system, according to an embodiment of the present disclosure;

FIG. 61 shows a perspective view of a charging system performing charging through a furniture panel and a transparent decorative element, according to an embodiment of the present disclosure;

FIG. 62 shows an exploded perspective view of a charging system with a charging module, according to an embodiment of the present disclosure;

FIG. 63 shows an exploded perspective view of a loading system with a loading module shown, according to an embodiment of the present disclosure;

FIG. 64 shows a perspective view of a charging system performing charging through a furniture panel and a transparent decorative element, according to an embodiment of the present disclosure;

FIG. 65 shows a view illustrating a function of a positioning system, according to an embodiment of the present disclosure;

FIG. 66 shows a near navigation system for determining a position of a device, according to an embodiment of the present disclosure;

FIG. 67 shows a near navigation system for determining a position of a device, according to an embodiment of the present disclosure;

FIG. 68 shows a near navigation system for determining a position of a device, according to an embodiment of the present disclosure;

FIG. 69 shows a near navigation system for determining a position of a device, according to an embodiment of the present disclosure;

FIG. 70 shows a sensor of a near navigation system, according to an embodiment of the present disclosure;

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FIG. 71 shows a sensor of a near navigation system, according to an embodiment of the present disclosure;

FIG. 72 shows a sensor of a near navigation system, according to an embodiment of the present disclosure;

FIG. 73 shows a sensor of a near navigation system, according to an embodiment of the present disclosure;

FIG. 74 shows an inductive sensor, according to an embodiment of the present disclosure;

FIG. 75 shows a pointer of a near navigation system, according to an embodiment of the present disclosure;

FIG. 76 shows a pointer of a near navigation system, according to an embodiment of the present disclosure;

FIG. 77 shows a pointer of a near navigation system, according to an embodiment of the present disclosure;

FIG. 78 shows a pointer of a near navigation system, according to an embodiment of the present disclosure;

FIG. 79 shows a pointer of a near navigation system, according to an embodiment of the present disclosure;

FIG. 80 shows a far navigation system for determining a position of a device, according to an embodiment of the present disclosure;

FIG. 81 shows a far navigation system for determining a position of a device, according to an embodiment of the present disclosure;

FIG. 82 shows a navigation system in which two or more navigation systems are used, according to an embodiment of the present disclosure;

FIG. 83 shows a charging system in which functions are controlled by making the surface vibrate, according to an embodiment of the present disclosure;

FIG. 84 shows a perspective view of a device with a manual adjustment and ratchet assembly, according to an embodiment of the present disclosure;

FIG. 85 shows a front view of a device with a manual adjustment and ratchet assembly, according to an embodiment of the present disclosure;

FIG. 86 shows a cross-sectional view of a device with a manual adjustment and ratchet assembly, according to an embodiment of the present disclosure;

FIG. 87 shows a cross-sectional view of a ratchet assembly, according to an embodiment of the present disclosure;

FIG. 88 shows a manual adjustment and ratchet assembly mounted on an armrest, according to an embodiment of the present disclosure;

FIG. 89 shows a perspective view of a device with a worm and worm wheel, according to an embodiment of the present disclosure;

FIG. 90 shows a front view of a device with a worm and worm wheel, according to an embodiment of the present disclosure;

FIG. 91 shows a cross-section of a device with a worm and worm wheel, according to an embodiment of the present disclosure;

FIG. 92 shows a cross-sectional view of a worm and worm wheel assembly, according to an embodiment of the present disclosure;

FIG. 93 shows a device equipped with a worm and worm wheel mounted on an armrest, according to an embodiment of the present disclosure;

FIG. 94 shows a perspective view of a motorized device, according to an embodiment of the present disclosure;

FIG. 95 shows a front view of a motorized device, according to an embodiment of the present disclosure;

FIG. 96 shows a cross-sectional view of a motorized device, according to an embodiment of the present disclosure;

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FIG. 97 shows a sectional view of a motor unit, according to an embodiment of the present disclosure;

FIG. 98 shows a motorized device mounted on an armrest, according to an embodiment of the present disclosure;

FIG. 99 shows a top perspective view of a device with an additional surface, according to an embodiment of the present disclosure;

FIG. 100 shows a bottom perspective view of a device with an additional surface, according to an embodiment of the present disclosure;

FIG. 101 shows a top perspective view of a device with an additional surface, according to an embodiment of the present disclosure;

FIG. 102 shows a bottom perspective view of a device with an additional surface, according to an embodiment of the present disclosure;

FIG. 103 shows a device with an additional surface mounted on an armrest, according to an embodiment of the present disclosure;

FIG. 104 shows a device with an additional surface mounted on an armrest, according to an embodiment of the present disclosure;

FIG. 105 shows a device with movable clamps mounted on an armrest, according to an embodiment of the present disclosure;

FIG. 106 shows a device with movable clamps mounted on an armrest, according to an embodiment of the present disclosure;

FIG. 107 shows a device with movable clamps mounted on an armrest with a rotatable surface, according to an embodiment of the present disclosure;

FIG. 108 shows a device with movable clamps mounted on an armrest with a rotatable surface, according to an embodiment of the present disclosure;

FIG. 109 shows a bottom perspective view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 110 shows top perspective view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 111 shows a front view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 112 shows a cross-sectional view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 113 shows a bottom perspective view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 114 shows top perspective view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 115 shows a front view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 116 shows a cross-sectional view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 117 shows a bottom perspective view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 118 shows top perspective view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 119 shows a front view of a device with movable clamps, according to an embodiment of the present disclosure;

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FIG. 120 shows a cross-sectional view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 121 shows a bottom perspective view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 122 shows top perspective view of a device with movable clamps, according to an embodiment of the present disclosure;

FIG. 123 shows a front view of a device with movable clamps, according to an embodiment of the present disclosure; and

FIG. 124 shows a cross-sectional view of a device with movable clamps, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

The present disclosure is directed to providing a charging device for an electronic device and is described in detail with reference to the accompanying drawings. Detailed descriptions of constructions or processes known in the art may be omitted to avoid obscuring the subject matter of the present disclosure.

The descriptions of similar elements are omitted when those elements have already been described with respect to a previous figure.

The present disclosure provides a wireless charging device that may be detachably mounted in various rooms of, for example, a house, vehicle, airplane, or boat, and/or on any piece of furniture. The wireless charging device may be an intelligent device, with automatic detection of the position of a device to be charged and movement of an inductive coil towards the device to be charged.

Further, the present disclosure provides a charger that may be mounted on a bracket-member that attaches to a wall of a room, vehicle, or a piece of furniture. The charger (also referred to as “charging module”) may include at least one coil for supplying power to an electronic device, and have a designated area on a surface of the charger in which charging may occur. Additionally, the charger may be connected to a power source for charging.

Additionally, the charger may include a shelf and can be configured to move along or within at least one plane of the shelf, from a resting position to an optimal charging position. The expression “charger shelf” may refer to the charger and the shelf.

The charger can be moved along, at least, one axis from among X, Y, and/or Z axes corresponding to multiple planes. For example, the charger can be moved from a rest position to an optimal charging position. Further, the charger can rotate around each axis by an angle α , β , or γ . Accordingly, a shelf may be equipped with an internal drive unit to move and rotate the charger and/or shelf.

The shelf may be configured to search for an electronic device to be charged that is located on a surface of the shelf. The shelf may include at least one power coil movable on the shelf plane, and provided with a locator that adjusts the position of the power coil to the position of the electronic device to be charged. For example, the power coil may be adjusted vertically (e.g., towards or away from the electronic device to be charged) or along a plane that is parallel to the surface on which the electronic device to be charged is located.

Accordingly, an induction coil may be displaced (moved) to a charging area where an induction process (e.g., wireless charging) may occur. A locator may include a base, a first

motor place on the base, a base unit embedded in the base containing the wireless charging coil and a second motor.

The second motor may be configured to move the coil that is attached to a base unit (also referred to as the “coil base unit”) from a first position to a second position. The first motor may be configured to move the coil base unit in a first direction, and the second motor may be configured to move the coil base unit in a second direction. The first direction may be perpendicular to the second direction.

The charger can have many variants. For example, the charger can be a shelf that is raised and lowered as a hinged pivot member mounted on a hinge handle, with the charging area located at any angle between a horizontal or vertical position. Additionally, the charger can be a shelf rotatable around a vertical axis by a desired angle, fixed on a rotary handle. Further, the charger can be a shelf rising and falling vertically along a guide holder. In addition, the charger can be a shelf simultaneously movable in three dimensions on a hinged handle. Also, the charger can be a shelf pulling out and sliding in as a drawer along parallel guides. Additionally, the charger can be a shelf fixed in a magnetic holder.

The shelf may be equipped with a clamp for mounting mobile devices.

The charger may be powered by the mains, a battery, a rechargeable battery or a solar cell.

A device to be charged may be a smartphone, smartwatch or computer. The shelf may also be a display monitor. A device to be charged may be any electronic or electrical device equipped with a power receiver cooperating with the coil to supply the device with the electric current. The shelf of the charger can also be a heating station using radiation energy for heating.

The charger can also include a ventilating, humidifying or air-conditioning device that is mountable on the shelf.

The shelf of the charger can be equipped with a perimeter illumination, downwardly directed illumination, upwardly directed illumination, downwardly and upwardly directed illumination, or sideward directionally directed illumination along a periphery, with selectable illumination modes that allow a user to control the direction of the light output. The shelf can also be equipped with a motion sensor cooperating with the illumination.

An advantage of the solution is that the charger can be mounted on any support member, in any room and in any vehicle. The angle at which the charger is mounted, or a location of the charger may be selected freely by a user. The charger may simultaneously offer many additional functions, being a universal wireless charging station for any electronic or electrical device equipped with a receiver adapted to receive wireless power in the inductive charging area.

The charger may assume a shape of a separate shelf movable along, at least, one axis and within, at least, one plane from the rest position to the charging position, and may be provided with an attachment unit to the carrier in the form of a handle to enable such a movement.

FIG. 1 shows a diagram illustrating a top perspective view of a charger shelf movable along X, Y, and Z axes, according to an embodiment of the present disclosure. FIG. 2 shows a diagram illustrating a profile view of a charger shelf movable along X, Y, and Z axes showing angles or rotations α , β , and γ , respectively, according to an embodiment of the present disclosure.

Referring to FIGS. 1-2, the shelf 1 of the charger is attached to a holder 2 and can be moved along one or more of the axes X, Y, Z. Additionally, the shelf 1 can be rotated around each of the axes by an angle α , β , γ , respectively.

FIG. 3 shows a diagram illustrating a top perspective view of a charger shelf with a charging/energy transfer zone, according to an embodiment of the present disclosure.

Referring to FIG. 3, the charger includes a shelf 1 mounted to a holder 2, enabling the charger to be displaced by bringing the charger from a rest position to a charging operating position and enabling linear and angular positioning of a charging area with respect to a mounting area of the charger. The charger may be mounted to a piece of furniture, a wall of a room, a wall of the interior of a car body, or a wall of a water vehicle or an air vehicle.

An internal drive unit DRV, shown in FIG. 6, may be used to displace or rotate the charger.

FIG. 4 shows a diagram illustrating a top perspective view of a charger shelf with multiple stationary charging/energy transfer zones, according to an embodiment of the present disclosure.

Referring to FIG. 4, several inductive coils 3A, 3B, and 3C are arranged in the shelf 1, so as to increase the area to charge electronic device by induction (the charging area).

FIG. 5 shows a diagram illustrating a top perspective view of a charger shelf with a movable charging/energy transfer zone, according to an embodiment of the present disclosure.

Referring to FIG. 5, a charger including an inductive coil 33 is located inside the shelf 1, movable along the two axes of the shelf 1. The induction coil 33 is movable, and thus may expand the charging area for charging a device. The induction coil may be positioned on a wireless base with a motor to move the wireless base and the induction coil along one or more of the axes.

FIG. 6 shows a diagram illustrating a top perspective view of a charger shelf with an internal drive assembly, according to an embodiment of the present disclosure.

Referring to FIG. 6, the shelf 1 is provided with an internal drive unit DRV. In such a case, the shelf 1 may be attached to the holder and configured to move in the direction of, at least, one axis among the X, Y, and Z axes and, at least, from a rest position (e.g., a vertical folded-down position) to an optimal charging position (e.g., a horizontal foldout-out position). In addition, the charger 1 may be rotated around the X, Y, and Z axes along angles α , β , and γ , respectively.

FIG. 7 shows a diagram illustrating a top view of a mobile coil module capable of locating an electronic device to be charged, according to an embodiment of the present disclosure. The mobile coil module may be included in the shelf 1 to enable the charger to be move from a first position to a second position to charge a device.

Referring to FIG. 7, the base 4 may include the second motor 46 configured to move the coil base unit 33 from a first position to a second position, along a first guide 47, perpendicular to a second guide 45. The second motor 46 may be attached to the coil base unit or may be positioned somewhere else within the base 4 without being attached to the base unit of the coil 33. Additionally, the first motor 44 may be configured to move the base unit of the coil 33 in a first direction, and the second motor 46 may be configured to move the base unit of the coil 33 in a second direction, perpendicular to the first direction.

In addition, a power stage controller connected to the wireless charging coil may be configured to determine frequency characteristics of the wireless charging coil 33 and to control the power stage driver to provide a specific current to the wireless charging coil 33 based on the frequency characteristics of the coil and an input power value.

Thus, an electronic device may be intelligently searched for and located on the panel device 1 for charging. Accord-

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ingly, the induction coil 33 may be moved to redefine a charging zone (or charging area) to be within a predetermined distance of a device to be charged.

FIG. 8 shows a diagram illustrating a top perspective view of a charger shelf capable of tilting in an operating position, according to an embodiment of the present disclosure.

Referring to FIG. 8, a handle 21 of a shelf 1 is of a hinged form, and may include a locking device. The shelf 1 is in a horizontal position and may be locked in this position via the locking device of the handle 21.

FIG. 9 shows a diagram illustrating a top perspective view of a charger shelf capable of tilting in a resting position, according to an embodiment of the present disclosure.

Referring to FIG. 9, the shelf 1 is in a vertical position and may be locked in this position via the locking device of the handle 21. The shelf 1 may be rotated 90 degrees from the vertical position to the horizontal position.

FIG. 10 shows a diagram illustrating a top perspective view of a charger shelf capable of tilting in an operating position, according to an embodiment of the present disclosure. FIG. 11 shows a diagram illustrating a profile view of a charger shelf capable of tilting in an operating position, according to an embodiment of the present disclosure.

Referring to FIGS. 10-11, the charger shelf is mounted via the handle 21 to a piece of furniture. The shelf 1 may be rotated from a horizontal position to a vertical position.

FIG. 12 shows a diagram illustrating a top perspective view of a charger shelf capable of tilting in an angled tilted position, according to an embodiment of the present disclosure. FIG. 13 shows a diagram illustrating a profile view of a charger shelf capable of tilting in an angled tilted position, according to an embodiment of the present disclosure.

Referring to FIGS. 12-13, the shelf 1 is shown in a tilted position. The shelf 1 may be moved to any position within a range of 0 degrees to 180 degrees. The shelf may be locked using the locking device of the handle 21 at any position within the range.

FIG. 14 shows a diagram illustrating a top perspective view of a charger shelf capable of tilting in an angled tilted position located next to a seat, according to an embodiment of the present disclosure. FIG. 15 shows a diagram illustrating a profile view of a charger shelf capable of tilting in an angled tilted position, according to an embodiment of the present disclosure.

Referring to FIGS. 14-15, the shelf 1 is mounted to a wall next to a seat. The shelf 1 is in a tilted angular position. The shelf 1 includes a clamp 5 for holding an electronic device to be charged. The clamp 5 is particularly useful when the shelf 1 is in the tilted angular position, because it allows the electronic device to be held in a position for charging.

FIG. 16 shows a diagram illustrating a top perspective view of a charger shelf capable of moving vertically along a z axis, according to an embodiment of the present disclosure. FIG. 17 shows a profile view of a charger shelf capable of moving vertically along a z axis, according to an embodiment of the present disclosure.

Referring to FIGS. 16-17, the shelf 1 is movable along a vertical axis Z to adjust a height of the shelf 1. The handle 22 may include a vertical guide with a locking device. The locking device may lock the shelf 1 at a particular height along the vertical axis Z.

FIG. 18 shows a diagram illustrating a top perspective view of a charger shelf capable of rotating around a vertical axis, according to an embodiment of the present disclosure. FIG. 19 shows a bird eye view of a charger shelf capable of rotating around a vertical axis, according to an embodiment of the present disclosure.

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Referring to FIGS. 18-19, the shelf 1 is rotatable around a vertical axis. A handle 23 includes a pin 24 for enabling the shelf 1 to rotate around the vertical axis.

FIG. 20 shows a diagram illustrating a top perspective view of a charger shelf capable of rotating along 3 axes and positioned next to a seat, according to an embodiment of the present disclosure. FIG. 21 shows a diagram illustrating a profile view of a charger shelf capable of rotating along a vertical axis and a horizontal axis, according to an embodiment of the present disclosure.

Referring to FIGS. 20-21, the shelf 1 is rotatable around three axis at angles α , β , and γ . The handle may include a physical joint 25 (articulating handle) for enabling the shelf 1 to rotate around the three axis at angles α , β , and γ .

FIG. 22 shows a diagram illustrating a top perspective view of a charger shelf capable of sliding along an x axis, according to an embodiment of the present disclosure. FIG. 23 shows a diagram illustrating a bottom perspective view of a charger shelf capable of sliding along an x axis, according to an embodiment of the present disclosure.

Referring to FIGS. 22-23, the shelf 1 may be extended outwardly from a piece of furniture, a wall, or another object. The shelf 1 may act as a drawer and be movable into and out from the piece of furniture along an X axis. The bottom side of the shelf 1 includes guides 26 for allowing the shelf to extend outwardly or inwardly along a predefined path.

FIG. 24 shows a diagram illustrating a top perspective view of a charger shelf mounted with a magnetic holder, according to an embodiment of the present disclosure. FIG. 25 shows a diagram illustrating a profile view of a charger shelf mounted with a magnetic holder, according to an embodiment of the present disclosure.

Referring to FIGS. 24-25, the charger shelf is fastened to the furniture using a magnetic holder 27. The charger shelf can be mounted to a door frame, a desk leg, a chair leg, or a room structural element.

FIG. 26 shows a diagram illustrating a bottom perspective view of a charger shelf with a battery, according to an embodiment of the present disclosure.

Referring to FIG. 26, the shelf 1 is equipped with its own power source in the form of a shelf-mounted battery BAT that may provide power to an induction coil for charging a device.

FIG. 27 shows a diagram illustrating a top perspective view of a charger shelf with photo-voltaic solar cells, according to an embodiment of the present disclosure.

Referring to FIG. 27, the shelf 1 is equipped with at least one photo-voltaic cell SOL that may provide power to the coils and/or the battery BAT.

FIG. 28 shows a diagram illustrating a top perspective view of a charger shelf charging a smartphone, according to an embodiment of the present disclosure.

Referring to FIG. 28, the shelf 1 may wirelessly charge a smartphone SPH placed on a top surface of the shelf 1.

FIG. 29 shows a diagram illustrating a top perspective view of a charger shelf charging a smartwatch, according to an embodiment of the present disclosure.

Referring to FIG. 29, the shelf 1 may wirelessly charge a smartwatch SWT placed on a top surface of the shelf 1.

FIG. 30 shows a diagram illustrating a top perspective view of a charger shelf charging a laptop computer, according to an embodiment of the present disclosure.

Referring to FIG. 30, the shelf 1 may wirelessly charge a laptop LTP placed on a top surface of the shelf 1.

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FIG. 31 shows a diagram illustrating a top perspective view of a charger shelf with a display monitor, according to an embodiment of the present disclosure.

Referring to FIG. 31, the shelf 1 may include a monitor MON on a top surface of the shelf 1. Alternatively, the monitor MON can be built into other surfaces of the shelf 1. In addition, the shelf 1 may include a built-in modem, which when used with the monitor MON, may display media obtained over the Internet.

FIG. 32 shows a diagram illustrating a top perspective view of a charger shelf powering a heating device, according to an embodiment of the present disclosure.

Referring to FIG. 32, the shelf 1 may wirelessly provide power to a heating station to heat a mug that is placed on a top surface of the shelf 1.

FIG. 33 shows a diagram illustrating a top perspective view of a charger shelf powering an air conditioning device, according to an embodiment of the present disclosure.

Referring to FIG. 33, the shelf 1 may wirelessly provide power to an air conditioning device VEN that is placed on a top surface of the shelf 1.

FIG. 34 shows a diagram illustrating a top perspective view of a charger shelf charging a climate control device, according to an embodiment of the present disclosure.

Referring to FIG. 34, the shelf 1 may wirelessly provide power to a climate control device CLI that is placed on a top surface of the shelf 1. The climate control device CLI may provide hot or cold air, may filter air, or may humidify air.

FIG. 35 shows a diagram illustrating a top perspective view of a charger shelf charging an inductively powered appliance, according to an embodiment of the present disclosure.

Referring to FIG. 35, the inductively powered appliance ALL may be placed on or around a charging coil 33 located within the shelf 1. Once in place, the inductively powered appliance ALL may receive power. The inductively powered appliance may be any appliance capable of receiving power from the shelf 1.

FIG. 36 shows a diagram illustrating a top perspective view of a charger shelf including perimeter illumination, according to an embodiment of the present disclosure.

Referring to FIG. 36, the shelf 1 includes a perimeter backlight LGH along edges of the shelf 1 that upwardly projects light.

FIG. 37 shows a diagram illustrating a top perspective view of a charger shelf including illumination directed in an upward direction, according to an embodiment of the present disclosure.

Referring to FIG. 37, the shelf 1 includes a backlight LUP positioned on one side of the top surface of the shelf 1 that upwardly projects light.

FIG. 38 shows a diagram illustrating a top perspective view of a charger shelf including illumination outwardly rotatable from a side of the panel device, according to an embodiment of the present disclosure.

Referring to FIG. 38, the shelf 1 includes a backlight LSD extending from a side of the shelf 1 and is rotatable with respect to a surface of the shelf 1.

FIG. 39 shows a diagram illustrating a top perspective view of a charger shelf including multiple illumination modes with a motion detector, according to an embodiment of the present disclosure.

Referring to FIG. 39, the shelf 1 is equipped with a motion sensor SEN capable of being used to activate an illumination state of the shelf 1. For example, an illumination mode of the

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panel device may be in an inactive state (e.g., a sleep state), but may be switched to an active state when motion is detected.

FIG. 40 is a diagram illustrating a configuration of a wireless charger, according to an embodiment of the present disclosure.

The wireless charger of FIG. 40 includes a charging pad 101. The charging pad 101 may include a coil capable of producing an electromagnetic field that extends outward from the charging pad 101. The coil may be embedded inside the charging pad 101 so that an electronic device can be placed in an electromagnetic field by being placed on top of or near a surface of the charging pad 101. The charging pad 101 may be attached to a first cable (i.e., wire) which connects to a first electrical connector 102.

The first electrical connector 102 is connected to a second cable which is connected to a second electrical connector 103. The second electrical connector 103 is connected to a third cable which connects to the wall power supply 104. The first electrical connector 102 and the second electrical connector 103 are detachable such that each may be used to connect or disconnect the charging pad 101 from the wall power supply 104.

The wall power supply 104 may connect to a standard United States power supply of 120 volts (V) at 60 hertz (Hz), or may be connected to other power supply voltages and/or frequencies (i.e., alternating current and/or direct current power).

The first electrical connector 102 and the second electrical connector 103 are detachable, therefore the charging pad 101 may be detached from the second cable, the third cable, and the wall power supply 104. Likewise, the wall power supply 104 may be detached from the second cable, the first cable, and the wireless charger.

An electrical connector may be any device capable of forming an electrical connection among two or more electronic circuits. That is, when an electrical connector is in a disconnected state, the two or more electronic circuits may not be electronically coupled to each other. When an electrical connector is in a connected state, the two or more electronic circuits may be electronically coupled to each other.

FIG. 41 shows a diagram illustrating a sofa armrest with a wireless charger embedded therein, according to an embodiment of the present disclosure.

Referring to FIG. 41, the wireless charger is embedded into the armrest of the sofa such that an electronic device 105 may be wirelessly charged by being placed on top of the armrest near the embedded wireless charger.

By embedding the wireless charger inside of the armrest of the sofa, many of the wires of the wireless charger are hidden from view such that the charging configuration is aesthetically pleasing.

FIG. 42 shows a diagram illustrating a transparent view of a configuration of a wireless charger embedded in a sofa armrest, according to an embodiment of the present disclosure.

In particular, the positioning of the charging pad 101, the first electrical connector 102, and the second electrical connector 103 within the armrest is revealed by the transparent armrest.

The charging pad 101 is located a predetermined distance (i.e., 2.7 to 3.3 millimeter) below the top surface of the armrest. The first electrical connector 102 is located inside of a top portion of the armrest and the second electrical connector 103 is located outside of the armrest such that the second electrical connector 103 can easily be accessed

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without having to disassemble, reconfigure, or open the sofa armrest. Thus, the charging pad **101**, which is embedded in the sofa armrest, may be disconnected from the wall power source by adjusting the second electrical connector **103** to be in a disconnected state (i.e., opening the second electrical **103**).

FIG. **43** shows a diagram illustrating accessing an interior of a sofa armrest via a service access point, according to an embodiment of the present disclosure.

The service access point enables a top portion **401** of the armrest to be separated from a lower portion **402**. The service access point may be a zipper or other means (i.e., a magnet or a locking mechanism) for attaching/detaching the top portion **401** to the lower portion **402**. Alternatively, rather than completely separating a top portion **401** of the armrest from a lower portion **402** of the armrest, a service access point may be accessed through a pocket, hole, or window, which may be opened using a zipper, magnet, or a locking mechanism.

The service access point provides access to the charging pad **101**, the first electrical connector **102**, and/or the second electrical connector **103** when they are located inside of the armrest.

According to the present disclosure, a power charger can easily be replaced by using the service access point. For example, a user may notice that his or her electronic device (i.e. mobile phone) is not properly charging, and he or she may be inclined to attempt to fix the wireless charger. If the charger continues to not be working properly, then the user may decide that it is necessary to gain access to the charging pad **101** to troubleshoot it and/or replace it. Alternatively, if the user cannot gain access to the charging pad **101**, then the user may decide it is necessary to replace the entire piece of furniture which houses the charging pad **101**.

The service access point provides access to the interior of the sofa armrest, thereby allowing the user to troubleshoot and/or replace a portion of the wireless charger. For example, the first electrical connector **102** and/or the second electrical connector **103** may be switched to a disconnected state so that the top portion **401** of the armrest can be separated from the lower portion **402** of the armrest without wires limiting physical access to the service access point.

Thus, the service access point may allow a user to replace the wireless charger without having to replace the sofa armrest. In other words, if the wireless charger is built into the sofa armrest and cannot easily be accessed, then a user may be inclined to replace the sofa armrest and/or the sofa if the wireless charger is not operating properly.

FIG. **44** shows a diagram illustrating a configuration of a wireless charger embedded in a sofa cushion, according to an embodiment of the present disclosure.

Referring to FIG. **44**, the sofa cushion may flip down to provide access to the embedded wireless charger. That is, the charging pad **101** of the wireless charger can be located in a sofa cushion that may be opened from a first position to a second position. The first position may be a regular sofa orientation which a sofa cushion can be ordinary used (i.e., by allowing the user to sit on or against it). The second position may be a flipped down position of a portion of the sofa cushion which may provide an area for an electronic device **105** to be placed such that a wireless charger embedded in the flipped down portion of the sofa provides wireless charging to the electronic device **105**.

FIG. **45** shows a diagram illustrating a configuration of a charger, according to an embodiment of the present disclosure.

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Referring to FIG. **45**, a wireless charging pad (i.e., a coil for electromagnetic induction) may be located inside of a locking mechanism **601**. The coil may be wound around a central point for generating an electromagnetic field. The locking mechanism **601** may snugly fit inside of a pre-bored hole **602**. The locking mechanism **601** including the wireless charging pad can then be placed into a section of the sofa.

Sofas and other furniture for use with a wireless charging pad may contain pre-bored holes, which may be custom-made by a furniture factory. The pre-bored holes may be located on the bottom of, on the side of, on the top of, inside of, or outside of furniture to accommodate the wireless charging device. For example, a wireless charging device having a locking mechanism **601** may be placed into one of the pre-bored holes **602** and placed into a locked position such that an edge of the furniture is flush. Additionally, a furniture cover may be placed over the locking mechanism **601** in the pre-bored hole **602**.

When the locking mechanism **601** is placed into a pre-bored hole, a user may lock the locking mechanism **601** into place to affix a charger to the furniture. When it is desirable to remove the charger, a customer may unlock the locking mechanism **601** to remove it from the furniture. The locking mechanism **601** may include part or all of the charger. The charger may be a wired charger or a wireless charger.

The locking mechanism **601** may be connected by a first wire to an electrical connector **603**, which is connected to a wall power supply **604**. A USB power extension may be connected to the wall power supply **604** such that power can simultaneously be output to the wireless charging pad in the locking mechanism **601** through the electrical connector **603**, and to the USB power extension **605**. An electronic device **606** may be connected to the USB power extension **605** to receive a charge.

In order to attach the locking mechanism **601** to the furniture, the locking mechanism **601** may be placed into a pre-bored (previously drilled) mounting hole in the furniture. Next, a twist-lock turning part of the locking mechanism **601** may be twisted in a predetermined direction (e.g., clockwise). When the locking mechanism **601** is twisted in the predetermined direction, elastic latches may protrude in a direction away from the center of the locking mechanism **601**, causing the diameter of the locking mechanism **601** to increase. In this manner, the diameter of the locking mechanism **601** in the unlocked position may be less than the diameter of the locking mechanism **601** in the locked position.

By increasing the diameter of the locking mechanism **601** in the locked position, the locking mechanism **601** may be affixed to the interior walls of the pre-bored hole in the furniture. In this way, the locking mechanism **601** may be securely attached to the furniture when placed in the locked position.

Alternatively, the locking mechanism **601** may be attached to the pre-bored hole in the furniture using adhesive, latches, sliders, or clamps to fasten the wireless charger mounting mechanism into an intended position.

Accordingly, a user may make an installation decision regarding whether or not to install the locking mechanism **601** with a charging device in his/her furniture. This solution of reduces the risk of returning expensive furniture due to unsatisfactory performance of charging devices because a user may replace and/or reinstall part or all of a charging device associated with the furniture.

FIG. **46** shows a diagram illustrating a configuration of a charger embedded in a sofa cushion with a USB extension, according to an embodiment of the present disclosure.

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Referring to FIG. 46, the locking mechanism 601 may be locked into place inside of a portion of a sofa cushion. As illustrated in FIG. 46, a portion of the sofa cushion can be flipped down from a first position (i.e., closed position) to a second position (open position). The locking mechanism 601 may be in a locked position in a pre-bored hole. The pre-bored hole may be a part of a cover 701 which may be opened or closed to access an inside section of the portion of the flipped-down sofa cushion. The inside section of the portion of the sofa cushion may be used to place wires (i.e., USB power extension 605) when it is not being used.

Thus, a user may flip down a portion of a sofa cushion, thereby gaining access to the locking mechanism 601 having an embedded wireless charging pad. A user may charge a first electronic device by placing it on or near the locking mechanism 601 having the embedded wireless charging pad. The user may simultaneously charge a second electronic device by connecting it to the USB power extension 605.

FIG. 47 shows a diagram illustrating a configuration of a wireless charger and a configuration of a wired ball USB charger, according to an embodiment of the present disclosure.

Referring to FIG. 47, a locking mechanism 601, including a wireless charging pad embedded therein, is connected to the power supply 801 through the electrical 603. The power supply 801 connects directly to the electrical 603 to provide power to the wireless charging pad embedded in the locking mechanism 601.

Additionally, a second power supply 802 may provide power to a USB ball charger 803. The USB ball charger 803 may be detachable from the second power supply 802 and may have two USB ports 804 for simultaneously charging two electronic devices through the two USB ports 804. In this manner, the USB ball charger 803 can easily be moved and placed in the best suited location for charging electronic devices.

FIG. 48 shows a diagram illustrating a configuration of a charger embedded in a middle section with additional power outlets and a wired ball USB splitter, according to an embodiment of the present disclosure.

Referring to FIG. 48, three electronic devices may simultaneously be charged. A first electronic device 901 may wirelessly be charged through a wireless charging pad embedded into a middle section of a sofa. The first electronic device 901 may be placed on a portion of the middle section of the sofa that has a wireless charging pad embedded thereunder. For example, a wireless charging pad may be embedded into a locking mechanism, which can be inserted into a pre-bored hole. The locking mechanism may be hidden from view when the middle section of the sofa is in the closed position. That is, the pre-bored hole may be located on an inner surface of a door and/or flap such that the side of the door and/or flap opposite the locking mechanism provides a flush surface on which an electronic device 901 can wirelessly be charged.

Additionally, a second electronic device 902 and a third electronic device 903 can simultaneously be charged through the ball charger 803. The second electronic device 902 and the third electronic device 903 can be simultaneously charged while the first electronic device 901 is wirelessly being charged.

FIG. 49 shows a diagram illustrating an interior configuration of a charger embedded in a middle section with additional power outlets and a wired ball USB splitter, according to an embodiment of the present disclosure.

The middle section of the sofa may include hidden power outlets 1003 which are accessible by flipping open a small

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door 1001. The middle section may also include a wireless charger 1002 embedded therein.

FIG. 50 shows a diagram illustrating a wired ball USB splitter, according to an embodiment of the present disclosure.

Referring to FIG. 50, the USB splitter may be a USB ball charger 803 (i.e., a half-moon charger) capable of providing different power outputs to a first USB output socket 1101 and a second USB output socket 1102 according to what type of electronic device is plugged into each output socket.

The USB ball charger may include one or more custom integrated circuits (ICs) which may include a large variety of standards and protocols to communicate with many different types of electronic devices. For instance, when a first type of electronic device is connected to the USB output socket 1101, a custom IC may identify the type of electronic device by communicating with an external IC of the first electronic device and determine a first output power for charging. When a second type of electronic device is connected to the second USB output socket 1102, the custom IC may identify the second type of electronic device by communicating with an external IC of the second electronic device and determine a second output power for charging.

Additionally, the custom ICs may prioritize charging. For instance, when a first type of electronic device requires a relatively large power usage for charging when compared to a second type device, then the custom IC may distribute power in a way to ensure that both devices are charged according to their factory specifications. This may mean that the lower power usage device (i.e., the second type device) may first be charged completely and the higher power usage device may be charged after the lower power usage device is finished charging.

FIG. 51 shows a diagram illustrating a configuration of a wireless charger, according to an embodiment of the present disclosure.

The wireless charger of FIG. 51 includes a charging pad 1201. The charging pad 1201 includes a coil capable of producing an electromagnetic field that extends outward from the charging pad 1201. The coil may be embedded inside the charging pad 1201 so that an electronic device may be positioned in the electromagnetic field by being placed on top of the surface of the charging pad 1201. The charging pad 1201 may be attached to a first cable which connects to a first electrical connector 1202.

The first electrical connector 1202 is connected to a second cable which is connected to a second electrical connector 1203. The second electrical connector 1203 is connected to a third cable which connects to the wall power supply 1204. The first electrical connector 1202 and the second electrical connector 1203 are detachable such that either one may be used to connect or disconnect the charging pad 1201 from the wall power supply 1204. Additionally, the third cable may be detachable from the wall power supply 1204 such that the wall power supply 1204 may be disconnected from the second electrical connector 1203.

In particular, the wall power supply 1204 may include a USB port for the third cable to be connected to. The wall power supply 1204 may connect to a standard United States power supply of 120 volts (V) at 60 hertz (Hz), or may be connected to other power supply voltages and/or frequencies (i.e., alternating current and/or direct current power).

The first electrical connector 1202 and the second electrical connector 1203 are detachable, therefore the wireless charger may be detached from the second cable, the third cable, and the wall power supply 1204. Likewise, the wall

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power supply **1204** may be detached from the second cable, the first cable, and the wireless charger.

FIG. **52** shows a diagram illustrating a configuration of a wireless charging pad placed in a drink holder of an armrest, according to an embodiment of the present disclosure.

Referring to FIG. **52**, the wireless charging pad **1201** may be formed to fit into a cup holder on an armrest of the sofa. Specifically, the bottom side of the wireless charging pad **1201** may be shaped circularly so as to fit inside of the cup holder on the armrest of the sofa. In this manner, the wireless charging pad **1201** may appear to be attached to the armchair of the sofa because the wireless charging pad **1201** can be fitted to the cup holder. Additionally, the cable which attaches the wireless charging pad **1201** to the first electrical may run through the cup holder to access the interior of the sofa armchair. Therefore, the cable will not be visible when the wireless charger **1201** is placed into the cupholder.

FIG. **53** shows a diagram illustrating a configuration of a wireless charger with a USB port for wired charging, according to an embodiment of the present disclosure.

Referring to FIG. **53**, the wireless charger includes a charging pad **1401**. The charging pad **1401** includes a coil capable of producing an electromagnetic field that extends outward from the charging pad **1401**. The coil may be embedded inside the charging pad **1401** so that an electronic device may be positioned in the electromagnetic field by being placed on top of the surface of the charging pad **1401**.

In addition to including a wireless charger embedded therein, the charging pad **1401** may also include a USB port **1405** positioned on the exterior of the charging pad **1401**. This configuration may allow for a user to wirelessly charge a first electronic device on the charging pad **1401** and to simultaneously charge a second electronic device through the USB port **1405**.

The charging pad **1401** may be attached to a first cable which connects to a first electrical connector **1402**. The first electrical connector **1402** may be connected to a second cable which may be connected to a second electrical connector **1403**. The second electrical connector **1403** may be connected to a third cable which connects to the wall power supply **1404**. The first electrical connector **1402** and the second electrical connector **1403** may be detachable such that either one may be used to connect or disconnect the charging pad **1401** to the wall power supply **1404**.

The wall power supply **1404** may connect to a standard United States power supply of 120 volts (V) at 60 hertz (Hz), or may be connected to other power supply voltages and/or frequencies (i.e., alternating current and/or direct current power).

The first electrical connector **1402** and the second electrical connector **1403** may be detachable, therefore the wireless charger may be detached from the second cable, the third cable, and the wall power supply **1404**. Likewise, the wall power supply **1404** may be detached from the second cable, the first cable, and the wireless charger.

FIG. **54** shows a diagram illustrating a sofa cushion with a wireless charger positioned on a flip-down portion of a cushion, according to an embodiment of the present disclosure.

Referring to FIG. **54**, the sofa cushion may flip down to provide access to the wireless charger. That is, the charging pad **1405** of the wireless charger can be located in a sofa cushion that may be opened from a first position to a second position. The first position may be a regular sofa orientation which a user can sit and/or ordinarily use a cushion. The second position may provide an area for an electronic device to be placed such that a wireless charger embedded in or

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placed on the flipped down portion of the sofa provides wireless charging to the electronic device.

FIG. **55** shows a diagram illustrating a sofa armrest with a wireless charger with a USB port, according to an embodiment of the present disclosure.

Referring to FIG. **55**, the charging pad **1405** of the wireless charger can be positioned on top of the sofa armrest. The charging pad **1405** can wirelessly charge a first electronic device by placing it near or on top of the charging pad **1405**. Additionally, the charging pad **1405** can charge a second electronic device through a USB port located on the surface of the charging pad **1405**.

FIGS. **56-60** are directed to a structure and operation of a charging system that performs charging through a non-transparent furniture panel. For example, the non-transparent furniture panel may not allow visible light and/or infrared radiation to visibly pass through a surface thereof.

FIG. **56** shows a perspective view of a charging system through a furniture panel that is non-transparent to visible and infrared light, according to an embodiment of the present disclosure. FIG. **57** shows an exploded perspective view of a charging system with a charging module, according to an embodiment of the present disclosure. FIG. **58** shows an exploded perspective view of a loading system with a loading module shown, according to an embodiment of the present disclosure. FIG. **59** shows a perspective view of a charging system that is performing charging through a furniture panel with details of the charging displayed on the energy receiver, according to an embodiment of the present disclosure. FIG. **60** shows a view illustrating a function of a positioning system, according to an embodiment of the present disclosure.

Referring to FIGS. **56-60**, a charging system may include of a furniture board **151** (e.g., a non-transparent, wooden, and/or solid board) that provides a charging and mounting location for the charging module **152** (e.g., a portable wireless communication device). The charging module **152** may be mounted (e.g., positioned) on the bottom of the board **1** and equipped with an energy transmitter **154** (e.g., an induction coil for transmitting wireless power) and a matrix **156** of position sensors **155**. The position sensors **155** may be arranged in a predetermined pattern such that each position sensor **155** is spaced apart from each other position sensor **155** by a specified distance. In response to the board **151** not being transparent, the applied position sensors **155** operate according to capacitive signals, radio signals or other principle signals which permit transmission through a non-transparent board material (e.g., wood).

The wireless energy transmitter and wireless energy receiver most efficiently operate when placed at an ideal orientation at a center **159** of the transmitter (e.g., the board **151**) and a center **1512** of the receiving device **153**. The correct arrangement of the above-mentioned system elements is possible thanks to the installation of an internal smartphone application **157** in the receiving device **153**, which will enable a display on the receiving device **153** to display suggestions (e.g., direction arrows that suggest a movement and/or point towards the center **159**) **158** of the movement **1513** in relation to the board **151**.

The display of the receiving device **153** may display a position correction icon **1510** (e.g., a direction arrow that is illuminated or highlighted) when an energy receiver included in the device **153** is positioned within a predetermined distance along the surface of the board **151** in relation to the transmitter **154**. For example, if the device **153** moves **1513** in a direction towards the center **159** of the board **151**, and a horizontal distance from the transmitter **154** to the

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device **153** is less than a predetermined number (e.g., 5 centimeters), then the device **153** may display the position correction icon **1510** indicating a direction towards the center **159**. Accordingly, the application **157** may notify the user of the correct positioning of the transmitter means with respect to the power receiver. Additionally or alternatively, the application **157** may provide an audible notification.

FIGS. **61-65** are directed to a structure and operation of a charging system that performs charging through a transparent furniture panel. For example, the transparent furniture panel may allow visible light and/or infrared radiation to visibly pass through a surface thereof.

FIG. **61** shows a perspective view of a charging system performing charging through a furniture panel and a transparent decorative element, according to an embodiment of the present disclosure. FIG. **62** shows an exploded perspective view of a charging system with a charging module, according to an embodiment of the present disclosure. FIG. **63** shows an exploded perspective view of a loading system with a loading module shown, according to an embodiment of the present disclosure. FIG. **64** shows a perspective view of a charging system performing charging through a furniture panel and a transparent decorative element, according to an embodiment of the present disclosure. FIG. **65** shows a view illustrating a function of a positioning system, according to an embodiment of the present disclosure.

Referring to FIGS. **61-65**, a charging system may include of a furniture board **161** (e.g., a non-transparent, wooden, and/or solid board) that provides a charging and mounting location for the charging module **162** (e.g., a portable wireless communication device). In addition, the board **161** may include an additional central flat element **161a** that is transparent and may allow visible light and/or infrared radiation to visibly pass through a surface thereof. In other words, the furniture board **161** may not be transparent (e.g., the furniture board **161** may be made of wood), and the central flat element **161a** may be transparent (e.g., the central flat element **161a** may be made of a clear plastic material). The charging module **162** may be mounted (e.g., positioned) on the bottom of the board **161** and/or flat element **161a** and equipped with an energy transmitter **164** (e.g., an induction coil for transmitting wireless power) and a matrix **166** of position sensors **165**. The position sensors **165** may be arranged in a predetermined pattern such that each position sensor **165** is spaced apart from each other position sensor **165** by a specified distance. In response to the board **161** and/or flat element **161a** being transparent, visible light, infrared radiation, or other wireless transmission signals may be transmitted through the board **161**. That is, the board **161** and/or flat element **161a** may include a transparent surface and display information through the transparent surface.

The wireless energy transmitter and wireless energy receiver most efficiently operate when placed at an ideal orientation at a center **169** of the transmitter (e.g., the board **161**) and a center **1612** of the receiving device **163**. The correct arrangement of the above-mentioned system elements is possible without necessarily installing a smart-phone application in the receiving device **163**. Rather, a display may be included in the board **161** and/or flat element **161a** to display suggestions (e.g., direction arrows that suggest a movement and/or point towards the center **169**) **168** of the movement **1613** in relation to the board **161**.

The display of the board **161** and/or flat element **161a** may display a position correction icon **1610** (e.g., a direction arrow that is illuminated or highlighted) when an energy receiver included in the device **163** is positioned within a

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predetermined distance along the surface of the board **161** in relation to the transmitter **164**. For example, if the device **163** moves **1613** in a direction towards the center **169** of the board **161**, and a horizontal distance from the transmitter **164** to the device **163** is less than a predetermined number (e.g., 5 centimeters), then the board **161** and/or flat element **161a** may display the position correction icon **1610** indicating a direction towards the center **169**. Additionally or alternatively, the board **161** and/or flat element **161a** may project light signals from a light ring **1611** on the surface of the board **161** and/or flat element **161a**.

According to an embodiment, a wireless charging navigation system is disclosed that aims to enhance the charging experience by determining the optimal alignment (orientation) of a mobile device with a charging coil on a charger panel, thereby achieving efficient wireless charging.

FIGS. **66-69** show near navigation systems for determining a position of a device, according to embodiments of the present disclosure.

Referring to FIGS. **66-69**, the charger **201** features a usable surface **202** (e.g., a surface for wireless charging) where the user places the device **203** (e.g., a mobile device, tablet, computer, or other electronic device capable of wireless charging) for charging. The device **203** includes a display screen **204** and a receiving coil **205** that operate in conjunction with the transmitting coil **206** in the charger to enable wireless charging.

The near navigation system identifies the position of the device **203** using parameters determined based on the receiving coil **205** in relation to the transmitting coil **206** of the charger. The near navigation system may, therefore, provide information to assist a user to position the device **203** in a predetermined area based on the identified position. Precisely positioning the coils **205** and **206** maximizes charging efficiency and minimizes energy losses. Additionally, the near navigation system also determines the position of the device **203** in relation to an ideal charging position in the predetermined area when using a charger with a non-transparent usable surface **202**.

The near navigation system comprises various sensors **207** arranged on the surface **202**, including capacitive, inductive, radio, optical (visible light), and infrared sensors. These sensors are organized into two circles: a smaller circle **208** and a larger circle **209**. Both circles have sensors placed at equal angular distances, ensuring omnidirectional functionality.

Once the device **203** approaches a sensor, the sensor detects its presence and undergoes a state change (e.g., switching from an inactive state to an active state). A programmed CPU microprocessor **210** analyzes the activated sensors, constructs a map of the device's location on the plane where the wireless charging transmitting coil **206** is located, and calculates the displacement vector **211** relative to the charger's transmitting coil **206**. The displacement vector **211** is a parameter that provides a direction (e.g., an angle) and a distance (e.g., a length) of the device **203** (or the receiving coil **205** of the device **203**) relative to the transmitting coil **206**. In this manner, the displacement vector **211** "points to" the area considered to be the ideal charging position. Calculation of the displacement vector **211** occurs in real-time, multiple times per second, to provide up-to-date and actionable information for further processing.

The resulting displacement vector **211** is utilized to generate visual, audible, or other forms of feedback on a user interface. This feedback informs the user about the neces-

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sary direction **212** in which to move the device **203** for achieving the optimal alignment of the wireless charging system's coils **205** and **206**.

Depending on the material properties of the charger's surface, two types of planes can be distinguished: transparent planes **213** (e.g., a transparent surface) and non-transparent planes **214** (e.g., a non-transparent surface).

In the case of the charger's surface including transparent planes **213**, the feedback about the displacement vector **211** is presented through a segmented circumferentially illuminated indicator **215** (e.g., a charger indicator light) located centrally along the main axis of the wireless charging transmitting coil **206**. The segments **227** of the indicator **215** are selectively lit by the microprocessor circuit **210** to provide clear guidance on the required direction of movement for the device **203**.

In the case of non-transparent planes **214**, the screen **204** of the device **203** itself serves as the medium for displaying the displacement vector **211**. Utilizing one or more transceivers for wireless communication WC, the displacement vector **211** is transmitted from the navigation system to the device **203**, where it is displayed through an application APP on the device's screen **204**, offering clear instructions to the user regarding the required direction and/or distance for moving the device **203** to the optimal charging position.

In addition, the navigation system incorporates an earth's magnetic field sensor **216**, for determining the magnetic inclination angle α (e.g., an inclination of the charging system) and the relative direction of the navigation system's position vector with respect to the device **203**, which also includes a built-in earth magnetic field sensor **239**. By considering the inclination angle β (e.g., an inclination between field vectors) between the magnetic field vectors, the navigation system corrects the displacement vector **211** to provide accurate information about the spatial relationship between the two systems (e.g., the coils **205** and **206**). For example, the direction of the displacement vector **211** may be corrected based on an angle between the charging panel and the device **203**, as shown in Equation 1, below.

$$\begin{aligned} \text{angle between charging panel and device} &= \text{inclination} \\ \text{angle } \beta &= \text{inclination angle } \alpha \end{aligned} \quad \text{Equation 1}$$

In more detail, the present application relates to a near navigation system that utilizes a variety of sensors to accurately determine the position of a device **203** within a given space. The system may include two or more distinct groups of sensors, each tailored to different types of surfaces and operating principles (e.g., capacitive sensors, inductive sensors, optical sensors, sensors for transparent surfaces and/or for non-transparent surfaces).

FIGS. 70-73 show sensors of a near navigation system, according to various embodiments of the present disclosure.

Referring to FIGS. 70-73, a first group of sensors utilizes the phenomena of capacitance and inductance changes, as well as wave detection around radio antennas, to enable precise positioning. For example, capacitive sensors **217** are employed during system startup to measure the capacitance of an antenna **218** (e.g., a capacitive sensor antenna). A capacitance measured with the capacitive sensors **217** is stored as a reference value in the system memory. When a device **203** is moved into a position that modifies the capacitance of the antenna, the computing system analyzes the capacitance difference between the real-time measurement and the stored reference value. Based on a predetermined minimum repeatable difference (e.g., a difference

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between the real-time (variable) measurement and the stored (constant) reference value), the system determines an output signal as a logical 0 or 1.

Inductive sensors **219** operate based on changes in the inductance around a magnetic element **220**. Similar to the capacitive sensors **217**, the inductance of the magnetic element **220** is measured and stored as a reference value during system startup. When a device **203** is moved into a position that alters the inductance, the computing system compares the real-time inductance measurement with the stored reference value. Based on a predetermined minimum repeatable difference, the system determines an output signal as a logical 0 or 1.

Radio sensors **221** may also be employed and utilize electromagnetic waves with millimeter wavelengths (e.g., frequencies above 70 gigahertz (GHz)). Radio sensors **221** include at least one transmitting antenna **222** and at least one receiving antenna (designated as **223**). The transmitting antenna **222** emits a signal at a specific frequency, and the direction and nature of the radiation are determined by the antenna's construction. When a device **203** enters the operating range of the transmitted beam (e.g., an area in which the device **203** may receive and perform a designated function based on the transmitted beam), the device reflects the electromagnetic wave, which is subsequently received by the system's receiving antenna **223**. The computing system performs calculations to determine a distance between the tested object (e.g., the device **203**) and the antennas **222** and/or **223**.

The second group of sensors may include optical sensors **224**, which rely on the reflection of waves in the visible and infrared light range. These sensors are specifically designed to work with surfaces transparent to visible and infrared light waves. A light emitter **225** may emit a beam of light in a predetermined direction with specific characteristics, while a light detector **226** captures a reflected light beam. By analyzing the strength of the reflected light beam or a phase shift of the reflected light beam with respect to the transmitted light beam within the visible or infrared range, the computing system may accurately determine the distance between the object and the emitter.

Accordingly, the near navigation system described herein incorporates capacitive, inductive, radio, and optical sensors to enable precise positioning of mobile devices in various environments. The system's ability to adapt to different surface types and leverage different sensing principles ensures accurate and reliable near navigation capabilities.

FIG. 74 shows an inductive sensor, according to an embodiment of the present disclosure.

Referring to FIG. 74, the sensor **219** may be used to determine the position of the receiving coil **205**, not the device **203** (e.g., an electronic device or mobile device) itself. The device **203** is equipped with a built-in wireless charging receiving coil **205**. However, simpler circuits, for example, that analyze the mobile device's contour in relation to the transmitting coil **206** may lack the necessary precision to achieve efficient wireless charging. To address this, a specialized inductive sensor **219** is employed in the near navigation system to accurately detect the position of the receiving coil **205**.

To accomplish accurate detection of the receiving coil **205**, the inductive sensor **219** may be slid along the underside of the device **203**, allowing it to detect changes in inductance using its own inductive sensor coil **220** (including a magnetic element). When the sensor **219** is moved over an area of the device's **203** housing where there is no receiving coil **205** present (e.g., a position in which there is

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no receiving coil **205** present above or below the sensor **219**, the inductance readings (L) decrease compared to a reference level labeled as “Lo.” Conversely, as the inductive sensor coil **219** is glided close to (e.g., above or below a position in which there is a receiving coil **205** present) the windings of the wireless charging receiving coil **205**, the inductance readings (L) rise and surpass the “Lo” reference value. This behavior of inductance variations may be utilized to accurately determine the precise position of the built-in receiving coil **205**, regardless of the shape or contour of the device **203**.

The user interface of the device **203**, e.g., a smartphone, integrates advanced graphical capabilities through the installed application APP to visually guide the user towards the ideal charging position based on the device’s displacement vector **211**. The user may receive graphical notifications via pointers, described below.

FIGS. **75-79** show different types of pointers of a near navigation system, according to various embodiments of the present disclosure.

FIG. **75** shows a pointer of a near navigation system, according to an embodiment of the present disclosure.

Referring to FIG. **75**, an eye level pointer is illustrated. A fixed grid **228** comprised of numerous circles with varying diameters and a shared center serves as a map element that indicates shift directions, such as up-down and left-right. A mobile pointer **229**, in the form of a circle imitating a particle of air in water, is present on this map’s surface. The mobile pointer **229** moves across the map in tandem with the movements of the device **203** on the usable surface **202** of the charger **201**. The motion of the mobile pointer **229** indicates the suggested direction for moving the device **203** to an ideal charging location (e.g., towards a charging coil).

During a final positioning phase of the device **203** in near navigation, when the transmitting coil **206** and receiving coil **205** overlap but aren’t precisely aligned, the wireless charging process may be initiated, albeit with reduced efficiency. At this stage, the user may be informed through auditory, visual, or alternative means that the wireless charging process has commenced but with low efficiency. Thus, the device **203** may output a sound or graphic after wireless charging has commenced with a voltage and/or amplitude that is below a predefined threshold (e.g., charging efficiency is below a predefined threshold).

The additional end navigation feature enables precise alignment of the device **203** so that the transmitting coil **206** and receiving coil **205** are perfectly matched, resulting in optimal charging efficiency. The attainment of this ideal coil positioning is signified by a distinct sound, visual cue, or other means that differentiates it from an earlier notification, allowing the user to discern the different processes. In other words, a first sound or graphic may be output after wireless charging has commenced with a voltage and/or amplitude that is below a predefined threshold (e.g., charging efficiency is below a predefined threshold, and a second sound or graphic may be output when an ideal position of the device **203** is achieved (e.g., when a voltage, amplitude, and/or charging efficiency is equal to or greater than a predefined threshold). As illustrated in FIG. **10**, the mobile pointer **229** may change color in accordance with the navigation phase it is in.

FIG. **76** shows a pointer of a near navigation system, according to an embodiment of the present disclosure.

Referring to FIG. **76**, an arrow-vector pointer is illustrated. A vector pointer **230** is anchored at a stationary point **231** in the middle of the application screen. The length and direction of the vector pointer **230** may dynamically evolve,

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mirroring the movements of the device **203** on the usable surface **202** of the charger **201**. The varying length and direction of the vector pointer **230** indicate the recommended movement direction for the device **203** during the charging process.

FIG. **77** shows a pointer of a near navigation system, according to an embodiment of the present disclosure.

Referring to FIG. **77**, a reticle type indicator is illustrated. More particularly, a fixed point **231** is shown on the application screen. The reticle pointer **232**, functioning as a dynamic component within an application, exhibits time-dependent movements that are intricately tied to the motions of the device **203** on the usable surface **202** of the charger **201**. By observing the distance and direction of the crosshair’s center from the fixed point, the intended movement direction for the charging device **203** is indicated.

FIG. **78** shows a pointer of a near navigation system, according to an embodiment of the present disclosure.

Referring to FIG. **78**, a segment pointer is illustrated. More specifically, the application screen displays a centrally positioned segmented luminous pointer **233** (e.g., a mobile device pointer) along its axis. One or more edges of the segmented luminous pointer **233** are parallel to one or more edges of the device **203**. This luminous pointer **233** is divided into equally spaced segments **234** encircling its circumference. These segments **234** are actively and distinctly illuminated by the microprocessor **210**, following a characteristic pattern. Their illumination provides the user with valuable information regarding the recommended movement direction for the device **203** being charged.

FIG. **79** shows a pointer of a near navigation system, according to an embodiment of the present disclosure.

Referring to FIG. **79**, a hybrid pointer is illustrated. When employing a transparent usable surface, navigation information can be presented in two or more different types of display methods.

A first method involves utilizing a segmented light indicator **215**, positioned centrally along the main axis of the wireless charging transmit coil **206** on the charger’s usable surface **202** (e.g., a surface that may be used for wireless charging). This luminous indicator **215** is divided into equal segments **227** of the luminous indicator **215** along its circumference. These segments **227** are activated and illuminated in a characteristic manner by the microprocessor circuit **210**, which communicates precise information to the user regarding the movement direction for the device **203** necessary to improve charging efficiency.

A second method entails simultaneously displaying the indicator **215** on the charger’s usable surface **202** and displaying any of the aforementioned pointers (eye level, vector arrow, reticle, or segments) on the screen of a mobile device (such as a smartphone) through the installed controlling application APP. This allows for simultaneous presentation of navigational information on the charger’s usable surface **202** and on the device **203**. By employing this parallel and simultaneous approach, the functionality becomes more flexible, enabling enhanced utilization of the navigation features.

FIGS. **80-81** show far navigation systems for determining a position of a device, according to various embodiments of the present disclosure.

Referring to FIGS. **80-81**, the position of a mobile device relative to a charging transmitter may be determined.

A far navigation system (FNS) is designed to provide an approximate determination of the near navigation system (NNS) location within a functional space. By utilizing radio communication and the properties of electromagnetic

waves, such as wave propagation and phase shifts, the FNS enables long-distance navigation.

Once the FNS establishes radio communication with the device **203**, such as a smartphone, it continuously measures the strength of the radio signal RS in real-time. Concurrently, the FNS collects data from the mobile device's **203** built-in sensors, including the earth's magnetic field sensor **216**, accelerometer **235**, and gyroscope **236**. By utilizing the programmed microprocessor system **210**, the FNS calculates the mobile device's **203** position relative to the antenna **237** of the long-range navigation radio system in three-dimensional space using the collected data.

The desired displacement vector **211** is transmitted between the far navigation system and the mobile device via wireless communication WC. The mobile device's screen **204** serves as the medium for displaying the displacement vector information. The displacement vector **211** is displayed by the installed application APP on the device's screen **204**, providing a clear indication to the user regarding the intended movement direction of the device **203**.

By utilizing the FNS, the user is informed of the direction in which to move the device **203** to obtain information about the location of the NNS. Once the NNS is located, the NNS takes over the navigation tasks automatically, providing precise commands to the user regarding the accurate positioning for charging mobile device. Accordingly, the FNS may automatically deactivate when the NNS is activated.

A simplified variant of the long-distance navigation system involves calculating information for the user interface solely based on the strength of the radio signal RS. In this simplified approach, the system constructs a spherical signal strength map **238**. The signal strength map **238** is displayed on the mobile device's screen **204** via the installed application APP, providing clear information to the user about the approximate distance to the NNS antenna **237**. This enables the user to assess whether their current motion brings the device **203** closer or farther away from the central part of the NNS. Additionally, the simplified system relies on the user's intuitive behavior to locate the radio signal RS with the highest amplitude, thereby determining the precise position of the NNS within the functional space.

FIG. **82** shows a navigation system in which two or more navigation systems are used, according to an embodiment of the present disclosure.

Referring to FIG. **82**, three wireless charging systems are illustrated on one surface of the charger. More or less wireless charging systems may be employed.

In a large functional space, the installation of multiple navigation systems with wireless charging may be necessary. It is possible to have multiple charging systems situated within a single usable area **202**, which are not visibly apparent on the non-transparent surface **214**. In such scenarios, the user is informed through the installed device **203** application APP about the presence of additional navigation systems equipped with wireless charging functionality. The user can then choose the system that suits their needs at that moment.

When there are multiple radio signal RS sources available, triangulating the signals enables a more precise determination of the mobile device's position in space. The FNSs engage in communication with each other and with the device **203**. Through an intra-system negotiation process, the negotiation function identifies the main unit with the highest operational priority (e.g., the transmitting coil with the highest operational priority that will most efficiently provide charging). This main unit can be any navigation system or application installed on the mobile device. As a

result of these operations, the selected main unit receives data from individual external systems, performs navigation calculations, and provides feedback to the application APP installed on the device **203**.

FIG. **83** shows a charging system in which functions are controlled by making the surface vibrate, according to an embodiment of the present disclosure.

Referring to FIG. **83**, the wireless charging navigation system includes a built-in accelerometer **239**. The operating system functions gather acceleration values Acc from the accelerometer, capturing movement parameters in three-dimensional space. These acceleration values, along with their variations in different directions, determine the activation of specific additional functions controlled by changes in the acceleration experienced by the navigation system.

The described operational principle effectively expands the number of virtual buttons, denoted as Ax, Ay, Az, within the device. Due to the combinational nature of the generated acceleration changes over time, the number of different combinations of virtual buttons becomes virtually limitless in number. These virtual buttons, namely Ax, Ay, and Az, are utilized to control various functions, such as adjusting brightness, activating or deactivating red green blue white (RGBW) lighting, and other related operations. The changes in acceleration may be induced by user behavior, such as tapping, touching, or shifting the usable surface **202** of the device in different axes and directions. These actions create identifiable vibrations that are detected by the accelerometer **239**.

FIGS. **84-86** show a perspective view, a front view, and a cross-sectional view, respectively, of a device with a manual adjustment and ratchet assembly, according to various embodiments of the present disclosure. FIG. **87** shows a cross-sectional view of a ratchet assembly, according to an embodiment of the present disclosure. FIG. **88** shows a manual adjustment and ratchet assembly mounted on an armrest, according to an embodiment of the present disclosure.

Referring to FIGS. **84-88**, the device (e.g., an auxiliary table device or an auxiliary shelf) may be a shelf that can be mounted on the armrest **31** of a sofa or another part of furniture. The device comprises a clamping mechanism with a fixed horizontal shelf **32** above the clamping mechanism, that is connected to a vertical fixed part of the clamping mechanism. Under the shelf **32**, a displaceable pressure plate **34** is mounted (e.g., the displacement pressure plate may apply pressure and act as a "claw" or "clamp", as described below. Also, the term "clamp" or "claw" may be generally referred to as a "surface"), which acts as the second adjustable part of the clamp.

The pressure plate **34** is mounted on a joint **316**, where the arm **303** is pivoted. The other end of arm **303** is rotatably mounted in the holder **321** on the axis **317**. The holder **321**, including the ratchet mechanism (assembly) **37**, is attached to the base **36**, which is fastened by screws **39** to the lower inner surface of the shelf **32**.

The holder **321** houses a ratchet assembly **37** which includes internal teeth within the holder **321** and locking elements **322** pressed against them by springs **323**. This assembly ensures automatic engagement and locks the ratchet mechanism **37**. Predetermined angles (or a range of angles) on the teeth enable automatic locking when pressing the plate **34**.

Without manual unlocking, the arm **303** cannot move back. The interlocking point is marked with the reference numeral **324**. Simultaneous pressing of buttons **318** causes

the elements **322** to approach each other and decouple with the internal teeth of the holder **321**.

Disengaging the elements **322** and holder **321** causes simultaneous disengagement of the arm **303** and therefore the entire pressure plate **34**. The clamp allows for adjustment to the width of the armrest or furniture portion, supporting the dismantling and remounting of the device.

FIGS. **89-91** show a perspective view, a front view, and a cross-section, respectively, of a device with a worm and worm wheel, according to various embodiments of the present disclosure. FIG. **92** shows a cross-sectional view of a worm and worm wheel assembly, according to an embodiment of the present disclosure. FIG. **93** shows a device equipped with a worm and worm wheel mounted on an armrest, according to an embodiment of the present disclosure.

Some of the components shown in FIGS. **89-93** are similar to the embodiments described with reference to other figures. Therefore, previously stated descriptions of these components may be applied to FIGS. **89-93**.

Referring to FIGS. **89-93**, the pressure mechanism includes a worm **313**, a worm wheel **312**, and a knob **311** equipped with a torque limiter **314**.

The worm may be a gear in the form of a screw, which meshes with a worm wheel. The worm converts rotational motion into linear motion. In a worm drive, the worm can easily turn the gear, but the gear cannot turn the worm, leading to a locking mechanism unless the worm has a special.

The worm wheel may be a gear that meshes with the worm. It may be similar in appearance to a spur gear but designed to work with the spiral thread of a worm. The worm wheel rotates perpendicularly to the worm, translating the rotational motion of the worm into a different plane.

A knob may be a control element designed to be turned manually to adjust, control, or manipulate a mechanism. The knob may be used to adjust the engagement and disengagement of the worm mechanism.

A torque limiter may be a safety feature that prevents the knob from applying excessive force that could damage the mechanism or the furniture. The torque limiter may ensure that the force applied is consistent and prevents overtightening, thereby protecting both the device and the furniture from damage.

This setup forms a kinematic system that is self-locking, ensuring that the pressure arm **303** does not move back after the torque on the knob **311** is removed.

The worm mechanism permits blocking movement in both directions. To unlock the clamp, the knob **311** should be turned counterclockwise. Conversely, to lock the clamp, a clockwise movement of the knob **311** may be used. Other embodiments may include counterclockwise movement for locking and clockwise movement for unlocking.

The knob **311**, integrated with a torque limiter **314**, prevents excessive pressure against the surface of the furniture **31**, ensuring that the furniture is not damaged during the clamping process.

FIGS. **94-96** show a perspective view, a front view, and a cross-section, respectively, of a motorized device, according to various embodiments of the present disclosure. FIG. **97** shows a cross-sectional view of a motorized device, according to an embodiment of the present disclosure. FIG. **98** shows a motorized device mounted on an armrest, according to an embodiment of the present disclosure.

Some of the components shown in FIGS. **94-98** are similar to the embodiments described with reference to other

figures. Therefore, previously stated descriptions of these components may be applied to FIGS. **94-98**.

Referring to FIGS. **94-98**, the worm and worm wheel assembly is driven by a drive motor **320**, which may be electric. To release the clamp, the drive motor **320** may operate counterclockwise, and to lock the clamp, it may operate clockwise.

The drive motor **320** may include a built-in overload system that prevents too much pressure against the surface of the furniture **31**, safeguarding both the device and the furniture from potential damage due to excessive force. To achieve this, the drive motor **320** may include a sensor that measures the force that is applied against the furniture, and if the force is equal to or greater than a threshold, the drive motor **320** may reduce its pressure.

The pressure plate **34** may move along an arc. Also, the shelf (the horizontal portion of the self) may include an integrated electronic system **35** with many functions, for example wireless charging, a positioning system for locating a charging space for wirelessly charging, a weather indicator, lighting devices in many colors and with many lighting functions, as well as other functions described in this disclosure.

FIGS. **99-102** show a top perspective view, a bottom perspective view, a top perspective view, and a bottom perspective view, respectively, of a device with an additional surface, according to various embodiments of the present disclosure. FIGS. **103-104** show a device with an additional surface mounted on an armrest, according to various embodiments of the present disclosure.

Some of the components shown in FIGS. **99-104** are similar to the embodiments described with reference to other figures. Therefore, previously stated descriptions of these components may be applied to FIGS. **99-104**.

Referring to FIGS. **99-104**, the device features another additional usable surface **351** connected by a rotating arm **358**. Pivot points **359** and **3510** on the arm **358** enable rotational movement and shifting of the additional usable surface **351** in any direction in a plane parallel to the main (e.g., horizontal) plane of the L-shaped shelf **352**. For example, the additional usable surface **351** may be rotated up (vertically) or down (vertically), and maintain a substantially parallel orientation with respect to the horizontal shelf **352**.

The kinematic system of the additional usable surface mechanism utilizes a rigid arm **358** and rigid pivot points **359** and **3510** to define the movement of the additional surface **351** in such a way that the user using the sofa or furniture element **355** can ergonomically adjust the location and angle of the additional usable area **351** to his needs.

The shelf and the additional usable surface **351** can be used in the folded position as well as in any other position permitted by the kinematic system of the arm **358**.

The additional surface **351** and the electronic system **353** (corresponding to the electronic system **35**) contribute to a multifunctional utility aspect, enhancing the user's experience by providing multiple functionalities while being ergonomically adjustable to the user's preference.

FIGS. **105-106** show a device with movable clamps mounted on an armrest, according to various embodiments of the present disclosure. FIGS. **107-108** show a device with movable clamps mounted on an armrest with a rotatable surface, according to various embodiments of the present disclosure.

Some of the components shown in FIGS. **105-108** are similar to the embodiments described with reference to other figures. Therefore, previously stated descriptions of these

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components may be applied to FIGS. 105-108. In addition, the phrases “movable clamps” and “clamping claws” may be used interchangeably and may refer to the same component of the device.

Referring to FIGS. 105-108, the shelf 4120 with the double-sided clamping system can be mounted on the armrest of the sofa 401. Prior to installation, the movable clamps may be adjusted to be as far apart as possible (or to a setting that is wider than the armrest, or piece of furniture, that is desired for it to be mounted on). To secure the shelf 4120 in place, the knob 445 may be turned clockwise (or counterclockwise), which brings the surfaces of the pressing paws closer together and locks the shelf 4120 onto the armrest of the furniture.

The configuration allows a top surface of the shelf 4120, equipped with a double-sided pressure system, to be moved and rotated by a specific length and/or at any angle. This flexibility is facilitated by a rotation system integrated into the main body of the shelf, ensuring that users can adjust the position of the shelf 4120 to suit their ergonomic needs and preferences.

FIGS. 109-110 show a bottom perspective view and a top perspective view, respectively, of a device with movable clamps, according to various embodiments of the present disclosure. FIG. 111 shows a front view of a device with movable clamps, according to an embodiment of the present disclosure. FIG. 112 shows a cross-sectional view of a device with movable clamps, according to an embodiment of the present disclosure.

Some of the components shown in FIGS. 109-112 are similar to the embodiments described with reference to other figures. Therefore, previously stated descriptions of these components may be applied to FIGS. 109-112.

Referring to FIGS. 109-112, the shelf 420 includes two movable clamps 440, aligned such that their main surfaces are parallel to each other and perpendicular to the usable surface of the shelf 420. The movement of the clamps 440 is facilitated by a turnbuckle (e.g., a roman screw) 441. This turnbuckle 441 is equipped with a right-hand thread 441a interacting with a right-hand nut 442a, and a left-hand thread 441b interacting with a left-hand nut 442b. The central position of the turnbuckle 441 is secured within a bearing holder 443, which is firmly attached to the main body 444 of the device. This holder permits rotational movement while blocking longitudinal motion along the main axis of the turnbuckle 441.

Manual manipulation of the rotation knobs 445 controls the turnbuckle 441. Rotating the knob 445 clockwise (or counterclockwise) brings the surfaces of the clamps 440 closer together, effectively locking the shelf 420 onto the armrest of the furniture. Conversely, turning the knob 445 counterclockwise (or clockwise) causes the surfaces of the clamps 440 to separate, thereby unlocking the shelf 420 from the armrest. This mechanism allows for precise adjustment of the shelf's position and orientation relative to the furniture.

Accordingly, the auxiliary table device may include a turnbuckle mechanism for manually adjusting the position of the second surface beneath the horizontal surface. This mechanism features a turnbuckle comprising a threaded rod with opposing threads at each end. These threads may engage with matching threaded components fixed to the structure of the second surface. By rotating the turnbuckle, the user can precisely adjust the distance between the clamps attached to the second surface, allowing for a customizable fit to various furniture widths.

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FIGS. 113-114 show a bottom perspective view and a top perspective view, respectively, of a device with movable clamps, according to various embodiments of the present disclosure. FIG. 115 shows a front view of a device with movable clamps, according to an embodiment of the present disclosure. FIG. 116 shows a cross-sectional view of a device with movable clamps, according to an embodiment of the present disclosure.

Some of the components shown in FIGS. 113-116 are similar to the embodiments described with reference to other figures. Therefore, previously stated descriptions of these components may be applied to FIGS. 113-116.

Referring to FIGS. 113-116, the shelf 420 includes two movable clamps 440, with their main surfaces arranged parallel to each other and perpendicular to the usable surface of the shelf 420. The movement of the clamps 440 is driven by a screw 451 and a gear transmission built upon toothed racks 452 and a gear wheel 453. The pressure claw (clamp) of one of the clamps, 440a, is directly actuated by the screw 451, which is rotatably attached to the clamp 440a, with the nut 454a securely affixed to the main body 454.

The movement of the second clamp, 440b, is facilitated by a differential mechanism. A toothed rack 452a mounted directly on the clamp 440b is driven directly by the screw 451. This rack engages with a gear wheel 453, which is rotatably mounted in the main body 454, transforming the linear movement of the rack into a rotational movement of the gear wheel 453. A second gear rack 452b is coupled with the same gear 453 on the opposite side, resulting in counter-rotating movements of the racks when actuated.

Manual operation of the screw 451 is controlled by the rotation of the knob 445. Turning the knob 445 clockwise (or counterclockwise) causes the surfaces of the clamps 440 to draw closer together, thereby locking the shelf 420 onto the armrest of the furniture. Conversely, turning the knob 445 counterclockwise (or clockwise) causes the surfaces of the clamps 440 to separate, thereby unlocking the shelf 420 from the armrest. This setup allows for precise control over the positioning and securing of the shelf relative to the furniture.

Accordingly, the auxiliary table device may incorporate a clamp adjustment system operated via a screw and gear transmission. This system may include toothed racks and a central gear wheel that synchronizes the movement of clamps positioned on either side of the second surface. The central gear wheel may be driven by the rotation of a screw, converting rotational motion into linear motion of the toothed racks. As the screw is turned, the gear wheel moves the toothed racks in unison, causing the clamps to either converge or diverge. This mechanism provides an efficient means of securing the device to the furniture, with the capability to adjust to a wide range of furniture sizes while maintaining a firm hold.

FIGS. 117-118 show a bottom perspective view and a top perspective view, respectively, of a device with movable clamps, according to various embodiments of the present disclosure. FIG. 119 shows a front view of a device with movable clamps, according to an embodiment of the present disclosure. FIG. 120 shows a cross-sectional view of a device with movable clamps, according to an embodiment of the present disclosure.

Some of the components shown in FIGS. 117-120 are similar to the embodiments described with reference to other figures. Therefore, previously stated descriptions of these components may be applied to FIGS. 117-120.

Referring to FIGS. 117-120, the shelf 420 incorporates two movable clamps 440, with their main surfaces arranged

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parallel to each other and perpendicular to the usable surface of the shelf 420. The movement of the clamps 440 is actuated by an electric motor 465, a screw 461, a differential mechanism based on toothed strips 462, and a gear wheel 463. One of the clamps, 440a, is directly driven by the electric motor 465, the screw 461, and the nut 464a, which is fixedly attached to the main body 464.

The movement of the second clamp, 440b, is facilitated by a differential mechanism where a toothed rack 462a mounted directly on clamp 440a is driven by the screw 461. This rack engages with a gear wheel 463, which is rotatably mounted in the main body 464. The linear movement of rack 462a is converted into rotational movement of the gear wheel 463. A second gear rack 462b is coupled with the same gear wheel 463 on the opposite side, resulting in counter-rotating movements of the racks when actuated.

The rotation of the screw 461 is automatically controlled by the electric motor 465. Clockwise (or counterclockwise) rotation of the motor 465 brings the surfaces of the clamps 440 closer together, effectively locking the shelf 420 onto the armrest of the furniture. Conversely, counterclockwise (or clockwise) rotation of the motor 465 allows the surfaces of the clamps 440 to separate, thereby unlocking the shelf 420 from the armrest. This system provides automated control over the positioning and securing of the shelf relative to the furniture, enhancing usability and precision.

Alternatively, both clamps 440 may be driven by the motor 465, or one clamp may remain stationary while the other is driven by the motor 465.

FIGS. 121-122 show a bottom perspective view and a top perspective view, respectively, of a device with movable clamps, according to various embodiments of the present disclosure. FIG. 123 shows a front view of a device with movable clamps, according to an embodiment of the present disclosure. FIG. 124 shows a cross-sectional view of a device with movable clamps, according to an embodiment of the present disclosure.

Some of the components shown in FIGS. 121-124 are similar to the embodiments described with reference to other figures. Therefore, previously stated descriptions of these components may be applied to FIGS. 121-124.

Referring to FIGS. 121-124, the shelf 4120 features two independently movable clamps 4140, with main surfaces aligned parallel to each other and perpendicular to the usable surface of the shelf 4120. Movement of the clamps 4140a and 4140b is manually controlled by adjusting them from the inside or outside, allowing each clamp 4140a and 4140b to operate independently from the other. This configuration facilitates smooth adjustments and shifting along the adjustment plane of the clamp mechanism relative to the surface of the furniture armrest.

Each clamp 4140a is equipped with locking teeth 4141 and a rotary pawl 4142 that is released using a lever 4143. The design restricts free movement of the clamps 4140a and 4140b only towards each other, as the latch system prevents the surfaces of the clamps 4140a and 4140b from moving back and away from each other. This kinematic setup allows for manual locking of the shelf mechanism onto the armrest (or other portion) of the furniture. To disengage and remove the shelf from the furniture armrest, the pawl 4142 must be released by manipulating or rotating the lever 4143, which is mechanically connected to the rotary pawl 4142, facilitating the release and movement of the clamps 4140a and 4140b in a direction that enables the disassembly of the shelf.

While the disclosure has been shown and described with reference to a certain preferred embodiment thereof, it will

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be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure.

What is claimed is:

1. An auxiliary table device, comprising:
 - a horizontal surface comprising a usable surface;
 - a first surface vertically positioned below the horizontal surface and configured to engage with a first side of a portion of furniture; and
 - a second surface positioned below the horizontal surface and configured to adjustably engage with a second side of the portion of furniture while in a vertical position, wherein at least one edge of the horizontal surface includes a wireless charging coil affixed below a top surface of the at least one edge of the horizontal surface, and the at least one edge of the horizontal surface extends beyond an edge of the first or second side of the portion of furniture when the first surface and the second surface are engaged with their respective sides of the furniture.
2. The auxiliary table device of claim 1, further comprising a manual adjustment mechanism having a ratchet assembly configured to adjust the second surface.
3. The auxiliary table device of claim 2, wherein the ratchet assembly includes a locking feature that automatically engages when the second surface reaches a predetermined engagement position.
4. The auxiliary table device of claim 1, further comprising a motorized adjustment mechanism configured to move the second surface towards the second side of the portion of furniture to engage the second surface with the second side of the portion of furniture or to move the second surface away from the second side of the portion of furniture to disengage the second surface with the second side of the portion of furniture.
5. The auxiliary table device of claim 4, wherein the motorized adjustment mechanism includes a motor unit with a built-in overload protection system configured to prevent damage to the furniture.
6. The auxiliary table of claim 1, wherein the second surface comprises a worm and worm wheel assembly configured to adjust the engagement of the second surface with the second side of the portion of furniture.
7. The auxiliary table device of claim 6, wherein the worm and worm wheel assembly is configured to provide a self-locking feature that prevents the second surface from moving unless manually operated.
8. The auxiliary table device of claim 1, wherein the horizontal surface includes an integrated electronic system selected from a group including wireless charging modules, lighting devices, and electronic display units.
9. The auxiliary table device of claim 1, further comprising a manual lever associated with a locking mechanism configured to manually lock or unlock the engagement of the second surface with the second side of the portion of furniture.
10. The auxiliary table device of claim 1, wherein the second surface is configured to rotate from a horizontal position to a vertical position to engage with the second side of the portion of furniture.
11. The auxiliary table device of claim 1, wherein the second surface is movable relative to the horizontal surface using a turnbuckle mechanism to facilitate manual adjustment.
12. The auxiliary table device of claim 11, wherein the turnbuckle mechanism comprises a threaded rod with opposing threads on each end that interact with corresponding

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threaded components attached to the second surface, allowing for positioning by rotating the turnbuckle.

13. The auxiliary table device of claim **1**, wherein the second surface includes a clamp adjustment mechanism driven by a screw and a gear transmission.

14. The auxiliary table device of claim **13**, wherein the gear transmission comprises toothed racks and a gear wheel, enabling synchronized movement of clamps positioned on opposing sides of the second surface.

15. The auxiliary table device of claim **14**, wherein the movement of the gear wheel is actuated by rotation of the screw, which translates rotational motion into linear motion of the toothed racks, thereby driving the clamps.

16. A method of using an auxiliary table device, the method comprising:

engaging a first surface of the auxiliary table device vertically positioned below a horizontal surface with a first side of a portion of furniture; and

adjustably engaging a second surface, positioned below the horizontal surface, with a second side of the portion of furniture while in a vertical position,

wherein at least one edge of the horizontal surface, which comprises a usable surface and a wireless charging coil affixed below a top surface of the at least one edge of

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the horizontal surface, and the at least one edge of the horizontal surface, extends beyond an edge of the first or second side of the portion of furniture when the first surface and the second surface are engaged with their respective sides of the furniture.

17. The method of claim **16**, further comprising adjusting the second surface using a manual adjustment mechanism that includes a ratchet assembly.

18. The method of claim **16**, further comprising moving the second surface towards the second side of the portion of furniture to engage the second surface with the second side portion of furniture or to moving the second surface away from the second side of the portion of furniture to disengage the second surface with the second side of the portion of furniture.

19. The method of claim **16**, further comprising manually locking or unlocking the engagement of the second surface with the second side of the furniture using a manual lever associated with a locking mechanism.

20. The method of claim **16**, further comprising rotating the second surface from a horizontal position to a vertical position and engaging the second surface with the second side portion of furniture.

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